

Physiological State and Executive Function in ADHD

An integrated brain–body hypothesis of fluctuating attention, effort and recovery

Alex Yesilcimen

Illustrated research edition v34 · 6 September 2026

Hypothesis / Perspective · Not peer reviewed

shadewaterlabs.com

Research perspective | No original participant data | Not a clinical guideline

The idea in plain language

The same person can think clearly one day and struggle with ordinary tasks the next. I want to understand how much of that change comes from the body's state, alongside the task, the setting and the person's ADHD.

Sleep, pain, migraine, hormones, circulation, immune activity and metabolism may contribute. Changes in these systems could affect how reliably attention stays engaged, how much effort thinking takes and how long recovery takes. ADHD might change the practical consequences of a disturbance without necessarily making the disturbance itself larger. That possibility is not the same as saying that everyone with ADHD has less ability, inflammation, or a damaged nervous system.

The proposed routes differ. Histamine concerns both allergic signalling and wakefulness. Endometriosis lets us examine recurring pain, hormonal and immune changes. Migraine gives us an episodic neurological state; Long COVID adds acquired disruption and recovery. GLP-1 signalling and vagal stimulation offer ways to test communication between body systems. I want to know which connections can be measured, not assume they point to one hidden disease.

The test is whether a measured change in the body predicts a change in this person's attention, experience or everyday functioning. Can it help us anticipate the next difficult period? The answer must improve on simpler explanations, including sleep, medication timing and context. Otherwise, the wider theory has not earned its place. Mental health symptoms and the conditions people live in belong in those comparisons, too. A bodily explanation should not erase someone's history or circumstances.

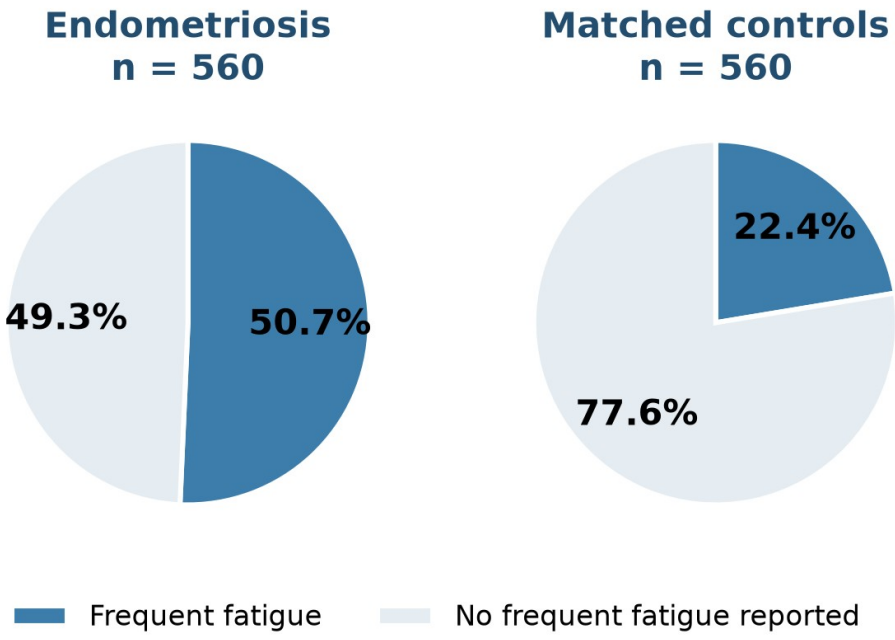
"Resilience" here is not willpower or a moral quality. It means a response and recovery pattern that a study would have to measure. The purpose is to explain fluctuation without blaming the person, and without replacing uncertainty with a fashionable molecule.

Medication is part of that changing state. A drug can make attention more reliable and change sleep, appetite or heart rate at the same time. A medicine taken for an overlapping condition can also affect thinking. We need to measure those effects separately before deciding what a better or worse day means.

The evidence in pictures

Fatigue is more common in endometriosis

In this matched study, about half of women with endometriosis reported frequent fatigue, compared with about one in five controls. That makes fatigue worth measuring alongside cognition. It does not identify the cause of either symptom.



Opening display A. Reported frequent fatigue: 50.7% of 560 women with confirmed endometriosis and 22.4% of 560 matched controls. Pale slices are the complementary proportions, calculated as 100% minus the reported percentage. Source: Ramin-Wright et al., 2018; see Section 6.6. [93]

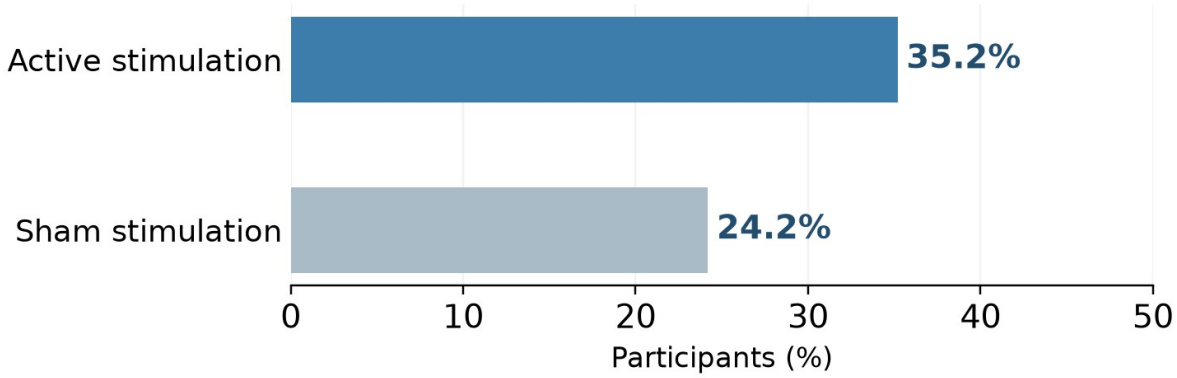
What this adds to the question

The observed difference is 28.3 percentage points. It is an association in this sample, **not a treatment effect**. Pain, insomnia, depression and occupational stress were also associated with fatigue within the endometriosis group. The study does not demonstrate that an immune pathway causes ADHD or that fatigue equals objective cognitive impairment. [93]

These opening displays preview selected results discussed in the paper. Each chart has its own population and endpoint; together they do not form a brain-health score. The numbered figures and fuller explanations remain in their original sections.

Two treatments with different outcomes

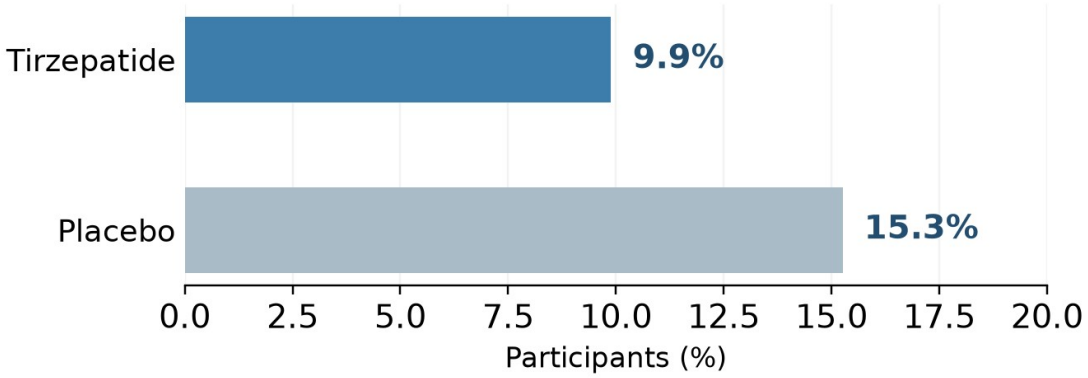
Implanted vagal stimulation and arthritis



Opening display B. RESET-RA: ACR20 arthritis response at three months in a randomized trial of 242 people with rheumatoid arthritis. ACR20 denotes at least 20% improvement under the trial's composite arthritis criteria. Active 35.2%; sham 24.2%. [57]

The raw difference is 11.0 percentage points. The published adjusted difference was 11.8 points (95% CI 0.6–23.1), using a prespecified one-sided primary test. This is evidence for an arthritis outcome, not cognitive improvement or equivalence with ear stimulation. See Section 10.1.

Tirzepatide and heart-failure events



Opening display C. SUMMIT: cardiovascular death or worsening heart failure in 731 people with preserved-ejection-fraction heart failure and obesity. Events: 36/364 on tirzepatide and 56/367 on placebo over a median 104 weeks. Percentages calculated from these counts. [115]

The hazard ratio was 0.62 (95% CI 0.41–0.95). Fewer worsening-heart-failure events drove the benefit; this does not establish a mortality benefit or ADHD efficacy. Bars show observed proportions, not annual risks or hazard ratios. See Section 9.4.

Reading the evidence

The opening question leads to Chapters 1–3, where I define what would count as an explanation. Chapters 4–8 examine changing states; Chapters 9–11 ask what treatment can and cannot tell us; Chapter 12 tests the idea against a wider health context. Chapters 13–16 bring those branches back to one research question. Use the links below to follow a particular thread. Underlined blue terms link to the glossary; the PDF also has a chapter outline.

The core idea and four quantities

Sleep and direct ADHD evidence

Histamine and immunity

Hormones endometriosis and fatigue

Migraine circulation and post-infection recovery

GLP-1 and other treatment evidence

Vagal regulation and ADHD medication

Mental health trauma gut and environment

Study design resilience and AI

What would weaken the hypothesis

Glossary and quantitative evidence

How to read the charts

A dot marks an estimate; a line shows its uncertainty. The caption gives the population, comparison and outcome. A line crossing the no-difference value leaves more than one interpretation open. It does not prove that two treatments are equivalent. Check each axis: a lower score is better in some figures and worse in others. In Figure 16, waffle grids show response proportions per 100, not actual participant counts; the printed intervals describe uncertainty in the treatment differences.

Most intervals are 95% confidence intervals. The sleep-surgery figure also labels adjusted 97% intervals. A percentage, a percentage-point difference, a standardized score and a risk ratio answer different questions; they cannot be combined into one brain-health score.

The 42 numbered figures include published results, calculations from reported counts and clearly labelled explanatory diagrams. They answer different questions. Chapters 4–8 cover sleep and overlapping physiological states; Chapters 9–12 examine treatment, neural regulation and wider health context. Chapter 14 sets out the proposed tests, including where AI could help.

Abstract

Attention-deficit/hyperactivity disorder (ADHD) varies within people as well as between them. Established state-regulation models already address arousal, effort and context. This narrative perspective asks whether measuring bodily state can improve prediction of fluctuating attention, perceived effort and everyday difficulty. It distinguishes functional executive margin, physiological reactivity, cognitive sensitivity and recovery. An equal performance loss can have different practical consequences; that is not the same as a larger bodily response, greater cognitive change per measured bodily change, or slower recovery.

Sleep intervention, circadian and waking-vigilance research motivate these distinctions without establishing universal ADHD hypersensitivity. Hormonal, allergic, migraine and post-infectious studies provide further test settings. Histamine, immune–neural communication, circulation, cellular energy, GLP-1 signalling and vagal pathways remain candidate mechanisms. Human associations, animal interventions and diagnosis-specific treatment effects are not interchangeable evidence for an ADHD mechanism. Established medication effects provide a direct ADHD benchmark; treatment-related sleep, autonomic and cognitive changes must be measured separately.

The proposed studies would measure cognition, subjective experience and physiology repeatedly, alongside a separate measure of everyday function. They would share a small set of measurements, then test predictions in later observations, new participants and different bodily states. Worked examples distinguish competing interpretations; a published sleep-trial figure illustrates why outcome choice matters. The framework is useful only if it improves on simpler accounts and survives explicit tests. It reports no original participant data, provides **no diagnostic criteria**, and does not recommend antihistamines, GLP-1 drugs or electrical stimulation for ADHD.

Keywords: ADHD; state regulation; executive function; vigilance; histamine; neuroimmune signalling; autonomic regulation; migraine; endometriosis; Long COVID; GLP-1; vagus nerve stimulation.

1. A broad question with a narrow standard of proof

ADHD is not a synonym for uniformly poor executive ability. Group-level differences coexist with substantial variation, and executive deficits are neither necessary nor sufficient to explain every case. This proposal concerns particular people, tasks and states—not a fixed intellectual ceiling. [1]

State dependence is also not a new discovery. Sergeant's cognitive-energetic model placed arousal, activation and effort alongside information processing and executive management. Intensive everyday assessment has since measured cognitive dynamics directly: one study

assessed 122 young adults five times daily for five days, finding that some inhibitory-control dynamics related to emotional difficulties where a baseline assessment did not. [2, 3, 4]

I want to measure the body's contribution more directly. Sleep, hormones, pain, autonomic regulation and immune activity might affect attention through different routes. Those routes need not operate in everyone. The question is whether a specified bodily change helps explain fluctuation within individuals better than the alternatives.

A model broad enough to explain every bad day after the fact would explain very little. This framework therefore has two levels. The core model concerns measurable relationships between physiological state, cognitive response and practical consequence. The mechanistic extensions ask whether particular neural, immune, vascular or metabolic processes explain those relationships. A successful core prediction would not automatically validate the histamine, GLP-1 or vagal extension.

1.1 Developmental risk and changing state

ADHD has a developmental background that this hypothesis needs to account for. A large genetic study compared 38,691 people with ADHD with 186,843 controls and identified 27 associated genomic regions. The results point to many small genetic contributions, with implicated genes enriched for expression during early brain development. They do not provide a genetic explanation for an individual's difficult afternoon. I see changing physiological state as a possible contributor to variation within that broader developmental picture. [150]

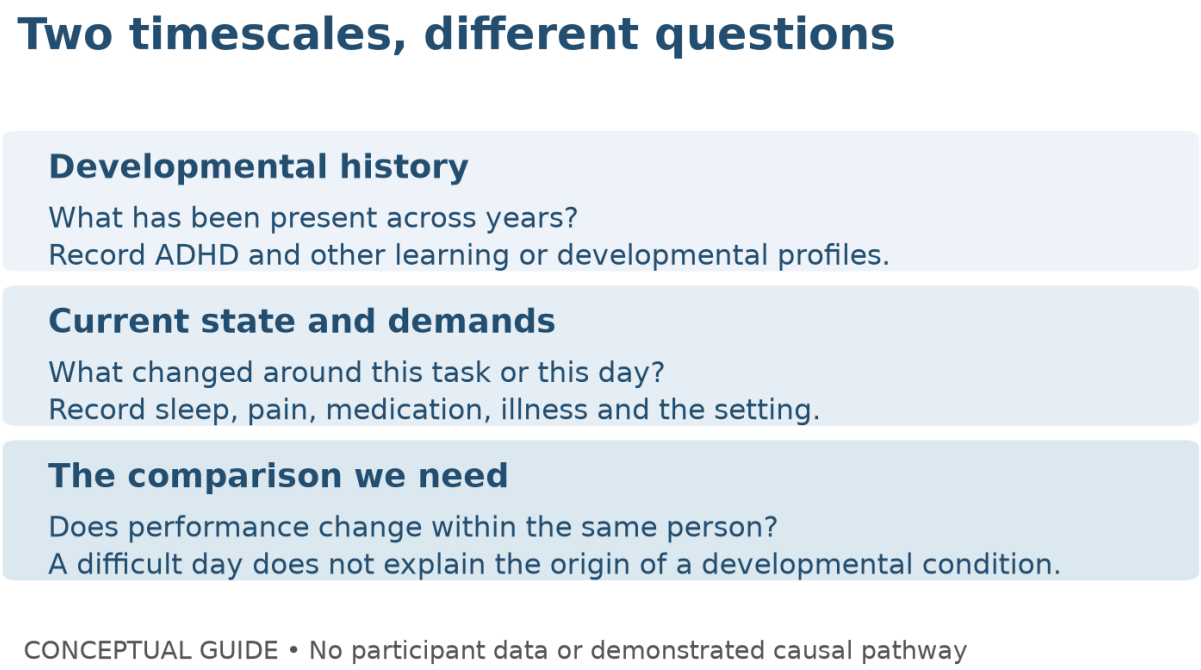
Symptoms also change over longer periods. In the Multimodal Treatment Study of ADHD, 558 participants with childhood ADHD were assessed eight times across 16 years. About 63.8% followed a fluctuating course; 9.1% met the study's definition of recovery, meaning sustained full remission through the end of follow-up. This is evidence about changes over years. It does not establish day-to-day fluctuation or identify a physiological cause. It also comes from the same research programme as the MTA evidence already discussed here, so it should not be counted as an **independent replication**. [151]

The timing of a diagnosis needs similar care. A repeated-assessment study followed 239 comparison participants without childhood ADHD. Many apparent late-onset cases identified by symptom checklists were excluded after closer review of impairment, timing, other psychiatric symptoms and substance use. That finding does not invalidate adult diagnosis or settle every case of late onset. For the proposed studies, I would distinguish longstanding difficulties recognized late from a newly acquired cognitive problem, and record how that distinction was assessed. [152]

Development, changes across years and changes across a day need different explanations. Our measurements should make clear which timescale they address.

Developmental history sets the starting point; a changing state asks what happens around it. Autism, dyslexia and other profiles also matter here, especially when a task places different

demands on different people. Section 12.9 returns to those differences. First, we need to separate four quantities that a single symptom score can hide.



Visual guide A. Developmental history and current state answer different questions. This is a conceptual reading aid for Chapter 1, not a model estimated from participant data.

2. What the hypothesis actually proposes

The Neurophysiological Resilience Hypothesis is the working name for this framework. It is built around explanations that can look the same in a symptom report.

2.1 Four explanations that must not be confused

The same difficulty can arise for different reasons. Someone may begin closer to a task's practical limit; their body may react more strongly; their cognition may change more for a comparable bodily shift; or recovery may take longer. These are separate claims.

Table 1. Four proposed quantities and the observations needed to distinguish them.

Proposed quantity	Plain-language question	What a study must distinguish
Functional executive margin	How much room is there before this task becomes unmanageable?	An equal absolute performance loss causes different functional consequences because starting margins differ.
Physiological reactivity	How strongly does the body respond?	The same challenge produces a larger measured bodily change.
Cognitive sensitivity	How much does thinking change for a comparable bodily shift?	The cognitive response is larger even when the measured physiological change is similar.
Recovery	How long does it take to return?	Cognitive recovery differs beyond what persistent physiology, medication, practice and context explain.

Functional executive margin is provisional and task-specific. It is not IQ and is not the established cognitive-reserve construct from aging research. A functional requirement must be defined independently of the group result. If “margin” merely renames a low baseline score or an arbitrary cutoff, it adds no explanatory value. Continuous performance and everyday functioning should then be reported without claiming that margin was measured. [5]

Nor does a lower starting point explain a larger absolute decline. It can explain why the same decline has a larger practical consequence. “Comparable physiology” also means comparable on the measurements taken, not proof that every relevant process in the two bodies was identical.

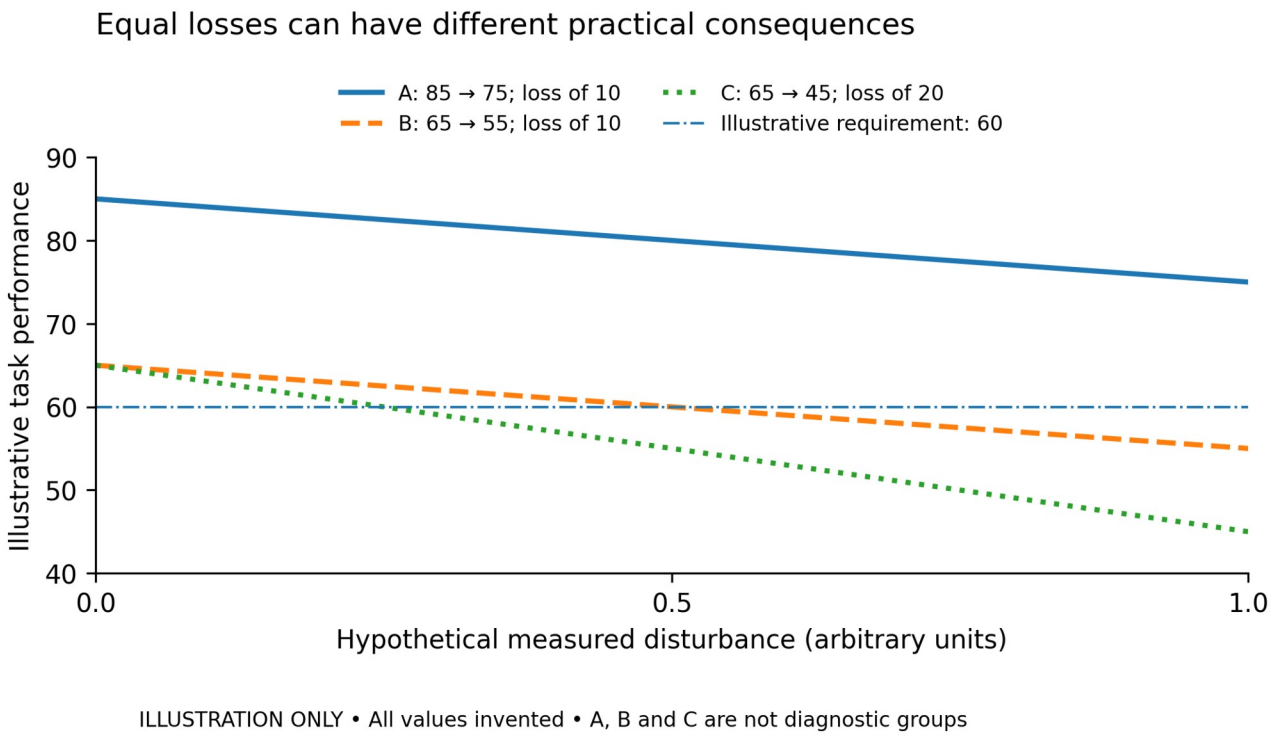


Figure 1. Margin and sensitivity are not the same claim. Hypothetical values, not research data. A and B each lose 10 points, but only B crosses the illustrative requirement of 60. C starts at B's value and loses 20: that illustrates greater sensitivity, not margin alone. No curve represents a diagnostic group or an IQ score.

Worked example 1 · A task requirement, not a ranking of people

In Figure 1, the invented requirement is fixed before comparing the curves. In a real study, it would need an independent functional anchor. Moving the cutoff afterwards to produce an ADHD difference would not test margin. A flexible deadline or external support could also change the practical requirement without changing the person's underlying task score.

2.2 Different outcomes deserve different measurements

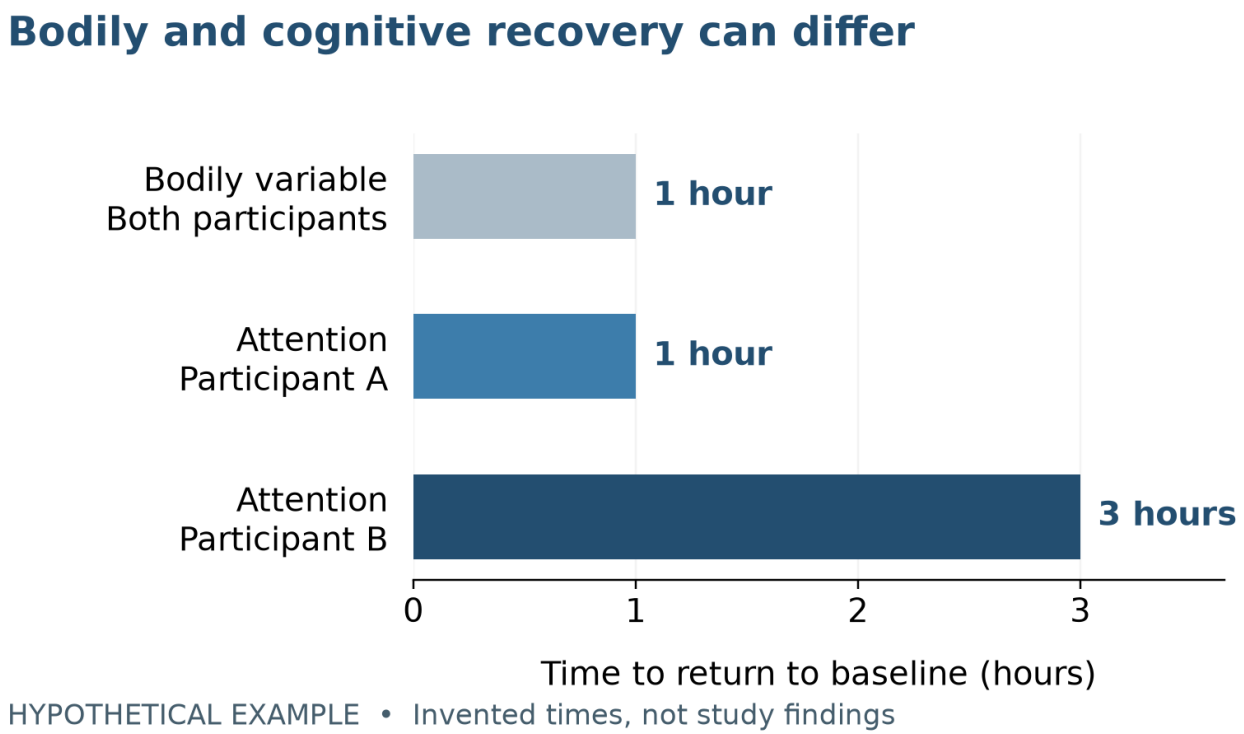
Brain fog is a report of reduced mental clarity. Sleepiness concerns a tendency to fall asleep. Fatigue concerns exhaustion or the experienced cost of continuing. A task lapse is a predefined behavioural event, such as missing a target. Everyday impairment concerns what a person can do in context. These working definitions guide measurement; they do not assign diagnoses.

Ratings and performance tests often capture different aspects of functioning. A normal short task does not invalidate a report that thinking has become exhausting. Conversely, a poor reaction-time result does not establish brain fog. Cognitive disengagement syndrome is a relevant research construct, but should not be used to relabel every foggy state or all ADHD inattention. [6, 7]

The model also needs context. In the MTA follow-up, remission periods were associated with greater environmental demands. Better functioning may have enabled greater responsibility; the study did not show that adding stress treats ADHD. It nevertheless cautions against treating structure, interest, urgency and reward as if they were simply more physiological burden. [8]

Worked example 2 · Recovery requires more than one follow-up

Suppose two hypothetical participants show the same initial change. A measured bodily variable returns to baseline for both after one hour, but attention returns after one hour in one participant and three hours in the other. That raises a recovery question. It does not yet show an ADHD-specific delay: unmeasured physiology, treatment timing and repeated-task fatigue remain alternatives. The times are invented to illustrate the design.



Worked example 2 chart. Both hypothetical participants have the same one-hour return of the measured bodily variable. Attention returns after one hour in A and three hours in B. Bars show only the invented return times; they do not show the depth or shape of either response.

The extra two hours are a question to investigate. They do not identify a mechanism or establish an ADHD-specific delay. Repeated measurements of physiology, attention and relevant context are needed to distinguish delayed recovery from an ongoing or unmeasured influence.

3. A map of the wider theory

Figure 2 shows the proposed connections and where they still need testing. It separates measured bodily states, candidate intermediate processes and three kinds of outcome. An immune, vascular or metabolic explanation must specify which connection it predicts; a shared molecule is not itself a shared cause.

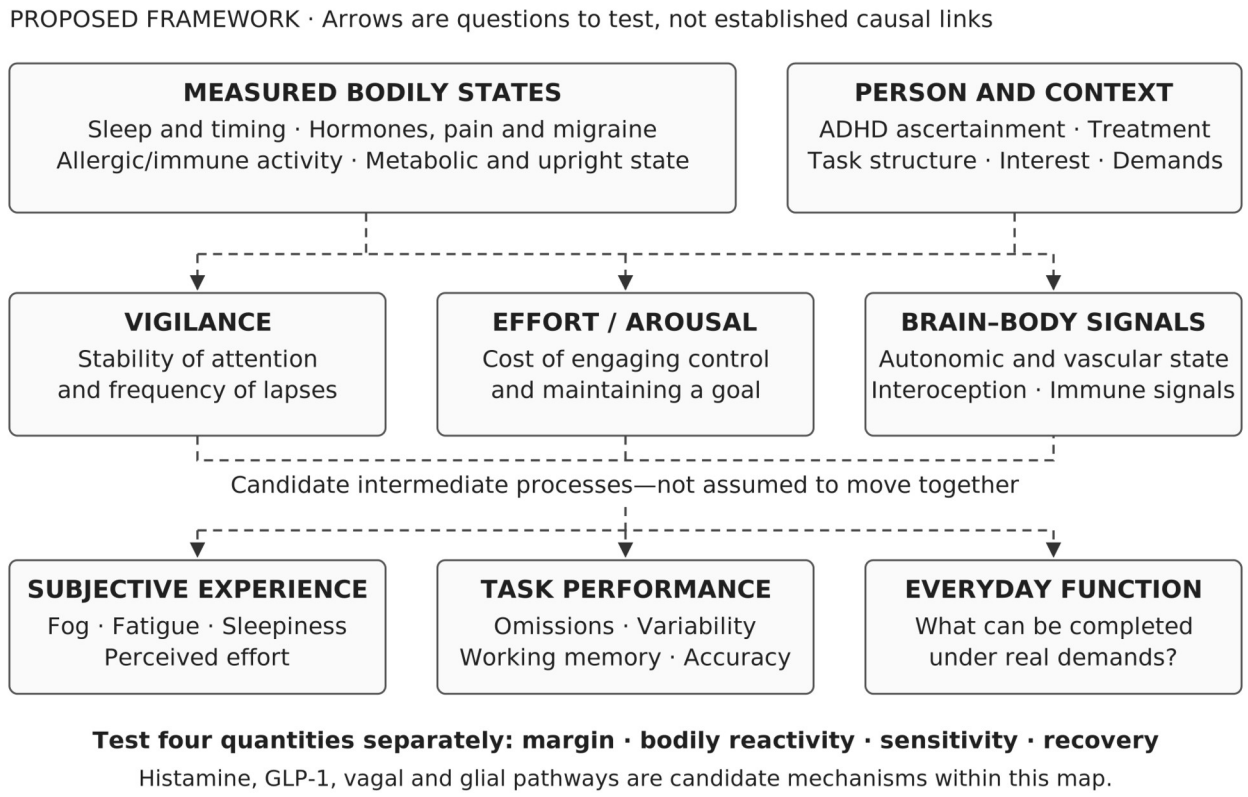


Figure 2. A map of the hypothesis, not a proven pathway. All connecting arrows are candidate relationships. “Person and context” may modify exposures, responses or consequences; the drawing is not a complete causal graph. Histamine, GLP-1, vagal and glial mechanisms belong within particular routes, not above them as universal explanations. The separate outcome boxes prevent subjective fog, test performance and everyday function from being treated as the same result.

Different conditions could affect the same measured process without being versions of one disease. To investigate that possibility, we need a proposed route from the condition to the outcome. Sections 4–10 distinguish evidence in ADHD from evidence elsewhere and from the proposed connections between them.

The map is a set of questions, not a completed chain of causes. I begin with sleep because it offers direct ADHD evidence, then move through immunity, hormones, migraine and post-infectious recovery. Treatment studies follow as tests of what can actually be changed. The wider health discussion then asks where the framework travels—and where it stops.

How evidence is used in this paper

An intervention in people with ADHD addresses a different question from an association in ADHD, an experiment in another human population, or a mechanism isolated in mice. A finding can justify a test without confirming the whole framework. The recovery example, Figures 1, 2, 7, 13, 30, 31 and 40, and Visual guides A, C and D are hypothetical or conceptual. The remaining charts show attributed study results or explicitly labeled calculations; some describe recruitment or follow-up rather than health outcomes.

3.1 Literature approach and limits

This is a targeted narrative synthesis, not a systematic review. Searches and source checks through 5–6 September 2026 covered the ADHD and cross-system questions discussed below. Studies were selected for relevance to cognition, competing explanations and predictions that could fail. Negative results were sought alongside positive findings. Coverage is selective: this paper does not review every diagnosis, exposure, peptide or treatment indication.

Primary reports and original evidence syntheses were prioritized. Numerical displays distinguish randomized, analyzed and imaging samples, clinical outcomes and biomarkers, and confidence intervals from Bayesian credible intervals. Verification used accessible full texts, tables and original abstracts; some methods and supplements remain inaccessible. This was not exhaustive screening, independent dual review, a formal risk-of-bias assessment or GRADE. Plots reproduce published estimates or explicitly labeled calculations. No original participant data or new meta-analysis is reported.

For this edition, I reread the manuscript for the order of its argument, clarity and overstatement. Selected high-stakes claims were checked again against original reports and official records; Appendix I records the scope. This was not a fresh independent verification of all 238 references. No new study or effect estimate is added, and the earlier source-access limits remain.

3.2 What the data show

The body matters to cognition. The harder question is which bodily change matters, for whom, and whether measuring it helps us predict anything. Several studies give us ways to test it. This evidence does not yet establish a single brain–body disorder behind ADHD and its overlaps.

Sleep provides a direct test of bodily-state effects. Changing sleep can change some ADHD symptom ratings, but the effect is not consistent across raters or cognitive tasks. Advancing the internal clock can also leave cognition unchanged. That is why target engagement, symptoms and everyday benefit need **separate measurements**. [9, 12, 19, 20, 65] Mild sleep-disordered breathing treatment also improved some symptoms without a clear gain on its primary cognitive outcomes. [92]

Medication has the strongest direct ADHD treatment evidence considered here. Short-term symptom benefit is established for several agents; cognitive tasks, learning, sleep and cardiovascular outcomes do not all move together. Medications for overlapping conditions can also affect thinking. Any state-based explanation needs treatment timing and co-medication in the model. [76, 77, 80, 82, 83]

Endometriosis, migraine and atopic dermatitis are associated with ADHD in large datasets. Those results tell us where to investigate. They do not tell us whether the route is immune signalling, pain, sleep, shared genetic liability, healthcare contact or some combination. A later recorded diagnosis is not necessarily a later onset of disease. [34, 35, 67, 68, 69] Endometriosis-related fatigue and iron status deserve separate assessment before a shared cognitive mechanism is inferred. [93, 94, 95]

Hormonal state deserves careful individual tracking. Small studies find changes in reported ADHD symptoms; a much larger general-population synthesis finds no robust overall menstrual-phase effect on objective cognition. Both findings can be true. Neither warrants assuming that all women, or all people with ADHD, lose cognitive ability at a particular phase. [32, 33, 66] Postmenopausal estradiol evidence also cautions against assuming that changing hormones improves cognition. [96]

Long COVID adds objective cognitive differences and smaller studies of brain barriers, autonomic responses and brain metabolites. Those findings came from different samples. They do not yet form a demonstrated sequence from immune activity to energy failure to brain fog. [44, 45, 70, 71]

Implanted vagal stimulation improved an arthritis outcome; semaglutide reduced cardiovascular events in a selected population; two large semaglutide trials did not slow symptomatic Alzheimer disease. None establishes ADHD efficacy. The useful next step is a defined prediction that can fail, not a longer list of plausible pathways. [56, 57, 73] Lifestyle and vascular trials add measured cognitive outcomes in older or hypertensive populations, with modest or outcome-specific effects. [97, 98]

PTSD and depression can involve measured cognitive difficulties; immune and infection histories predict some later diagnoses. Gut studies include promising and inconclusive trial results. None makes ADHD a hidden infection, autoimmune disorder or trauma response. Section 12 separates these claims and turns them into testable predictions. [99, 100, 104, 109, 110]

Treatment evidence gives the model more concrete tests. Chapter 9 groups the benefits reported in overlapping illnesses before examining the limits of cognitive and mechanistic interpretation.

Developmental evidence and long-term symptom change cannot explain a difficult day by themselves. Sections 1.1 and 5.4, followed by the later sections of Chapter 9, show why bodily markers, subjective experience and cognitive tests need **separate measurements**.

Mono overlaps with the proposed recovery question through prolonged fatigue and cognitive complaints after infection. A direct ADHD link, or a different recovery trajectory in people with ADHD, remains unestablished. Sections 8.4–8.9 separate these claims; Figures 37, 38, 39 describe fatigue-syndrome outcomes and follow-up, not ADHD risk.

Treatment findings sharpen the outcome question. A physical marker, a cognitive rating, an objective test and a personal goal can change differently. The adult sleep, ELAD, HISTORI and Long COVID trials below reinforce the need to specify which outcome improved before inferring a shared mechanism. [187, 188, 189, 190, 191, 192]

4. The direct ADHD foundation: sleep, vigilance and effort

4.1 Sleep can change symptoms without changing every task score

A crossover study enrolled 72 adolescents with ADHD; 48 adherent participants with complete actigraphy contributed to the main analyses. The protocols allowed 6.5 versus 9.5 hours in bed, producing an average 1.6-hour difference in measured sleep. Parent-rated inattention worsened during restriction (Vanderbilt scale, d = 0.60), whereas adolescent-rated inattention and continuous-performance testing did not show the same effect. Participants stopped medication for the study. This supports effects on specified symptom outcomes, not universal executive decline or an ADHD-versus-control interaction. [9]

A smaller study tested 16 young men with ADHD and 18 controls before and after 25 hours awake. Both groups deteriorated and were described as similarly affected. The study did not establish statistical equivalence or independently validate functional thresholds. Its result is compatible with an additive effect from different baselines; it does not prove the margin model. [10]

A newer experiment adds a more specific positive interaction: 19 young men with ADHD and 14 controls completed an emotional-face task before and after 25 hours awake. Target-face omissions showed a time-by-group interaction ($\chi^2 = 4.59$, $p = .032$), while several shape and later EEG outcomes did not. The small male sample and multiple endpoints require replication. This qualifies the earlier continuous-performance result: sensitivity may depend on the task and outcome, rather than being either absent everywhere or universally greater in ADHD. [75]

Hours asleep are not the only relevant measurement. In 62 medication-free adolescents—31 with ADHD and 31 controls—assessed with ambulatory polysomnography, sleep EEG features related to neurocognition despite no detected group differences in sleep duration, awakenings or latency. That supports measuring sleep physiology as well as duration, while leaving causality unresolved. [11]

4.2 Treating a sleep problem is a more direct test

Hiscock and colleagues randomized 244 children with ADHD and sleep problems to a brief behavioural sleep intervention or usual care. The adjusted parent-rated symptom differences were −2.9 points (95% CI −5.5 to −0.3) at three months and −3.7 (−6.1 to −1.2) at six months, using multiple imputation. Teacher-rated total ADHD scores did not show the same benefit. Most children were already taking stimulants; families were not blinded. This tests an adjunct in a selected clinical population, not the claim that sleep explains ADHD. [12]

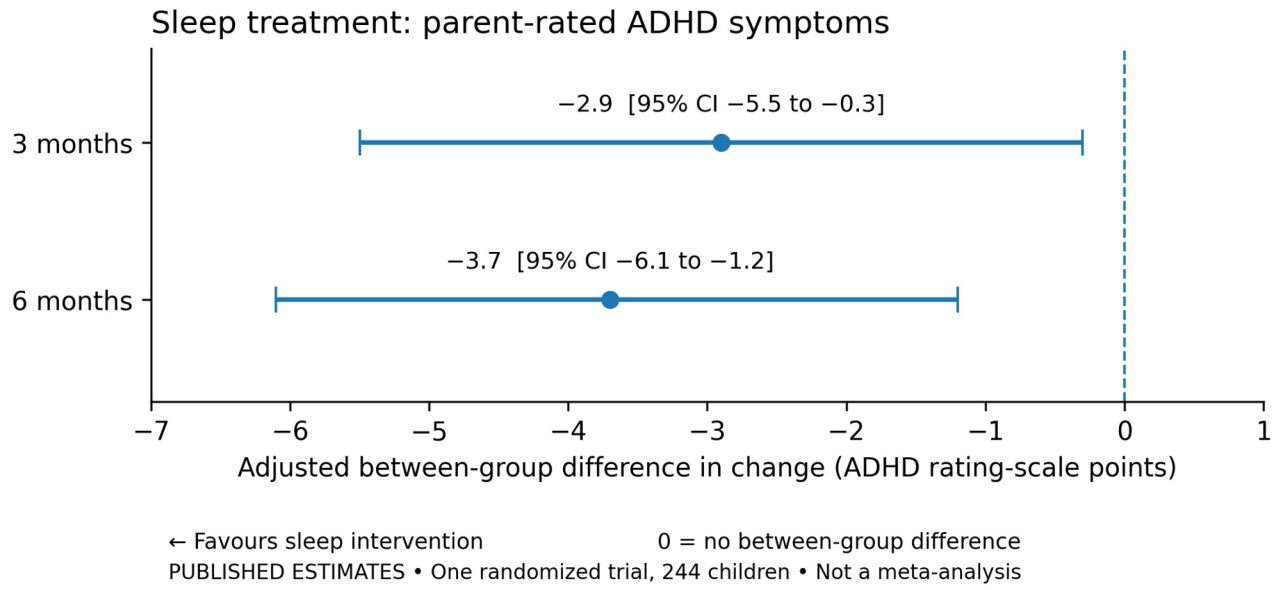


Figure 3. A defined intervention can help one outcome without improving every measure. Adjusted differences in parent-rated ADHD symptom change, with reported 95% confidence intervals, from Hiscock et al. The two follow-ups belong to the same randomized trial; they are not pooled independent studies. Estimates use the authors' imputed dataset. Families were not blinded, and teacher-rated total ADHD outcomes did not show the same benefit. [12]

Sleep-disordered breathing is another useful test. PATS randomized 459 children with habitual snoring and mild obstruction to surgery or watchful waiting; 458 entered analysis. At 12 months, neither primary executive-function nor attention outcome showed a clear between-group benefit. Sleepiness, symptoms and some health outcomes improved. Six children had a surgery-associated serious adverse event. This was not an ADHD trial. Improved sleep-related health cannot stand in for a measured cognitive gain. [92]

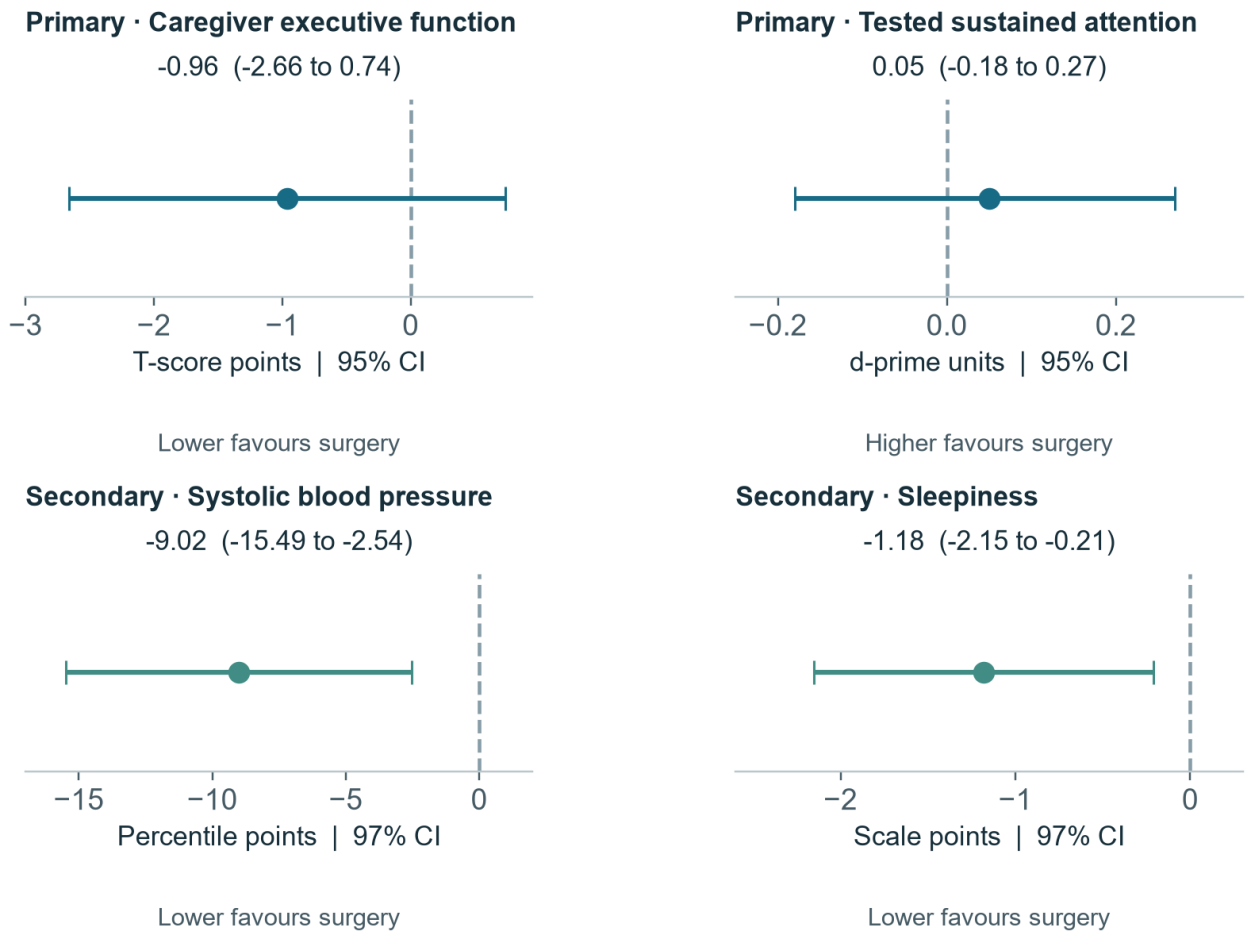


Figure 4. Sleep-related improvements can coexist with inconclusive cognitive outcomes. PATS, surgery minus watchful waiting: separate axes retain each outcome's units. Primary outcomes use 95% CIs; secondary outcomes use multiplicity-adjusted 97% CIs. Blood pressure is in percentile points, not mm Hg. Effect directions are labeled. These outcomes do not establish mediation or recommend surgery for ADHD. [92]

Adult evidence adds a useful limit. A preliminary open-label trial randomized 70 adults with ADHD and a positive sleep-disorder screen to three treatment conditions over 12 weeks. The primary comparison of usual ADHD care plus sleep treatment (22 participants) with usual care alone (25) did not show an additional subjective ADHD benefit: β −1.30 (95% CI −5.57 to 2.96). Sleep quality and fatigue improved more with the combined treatment; performance-based ADHD outcomes did not show an added benefit. Twenty percent missed the post-treatment assessment. The small trial leaves uncertainty, but improved sleep and reduced fatigue should not be relabeled as demonstrated ADHD improvement. [187]

4.3 Vigilance instability: more lapses, not less ability everywhere

The idea that poor performance can reflect intermittent lapses rather than uniform loss of ability has a long history. Doran and colleagues described sleep-deprivation state instability in 2001. Later, Andrillon and colleagues associated waking slow waves with attention lapses,

mind wandering and mind blanking in healthy adults. These are antecedents, not ADHD-specific diagnostic findings. [13, 14]

A 2026 waking-EEG study compared 32 adults with ADHD and 31 controls. Sleep-like slow-wave density was associated with omissions, slower and more variable responses, and sleepiness; cross-sectional mediation linked these signals to group differences. A pediatric study reported presentation- and region-dependent findings. The final adult report's complete methods remain an access gap; neither study establishes a diagnostic marker or causal pathway. [15, 16]

These findings motivate vigilance instability as an intermediate candidate: some impaired periods may involve more frequent interruptions in otherwise available performance. The proposed sequence—bodily state, lapse pattern, everyday consequence—contains three links that need testing. It should not be shortened to "ADHD is local sleep."

4.4 Circadian timing and demand

Among 43 evening-type students, including 19 with elevated ADHD symptoms, morning–evening testing produced a reaction-time interaction but not the same working-memory result. Morning always came first, leaving practice and order as alternatives. A larger study of 302 adolescents found that ADHD status did not strengthen associations between eveningness and poorer functioning. The designs differ, but together they argue against universal circadian hypersensitivity. [17, 18]

The PhASE trial randomized 51 adults with ADHD and delayed sleep phase to sleep education plus placebo, melatonin, or melatonin with morning bright light. The active regimens advanced dim-light melatonin onset by 88 and 118 minutes, respectively. Self-reported ADHD symptoms fell 14% in the melatonin arm; the combined arm did not show that symptom result. This is not evidence that the smaller clock shift was superior: a difference in within-arm significance is not a between-arm comparison. [19]

An exploratory sleep analysis of the same trial, using 49 baseline records, found no corresponding advance in sleep timing or general sleep improvement. This is a second analysis of the same participants. Target engagement, behaviour and symptoms still need separate outcomes. [20]

A separate double-blind trial in 105 medication-free children with ADHD and sleep-onset insomnia makes the distinction especially clear. After four weeks, sleep onset advanced by 26.9 minutes with melatonin but became 10.5 minutes later with placebo. Sleep duration also improved; behaviour, cognitive performance and quality of life did not show significant benefit. Better sleep is worthwhile, but its measurement cannot stand in for demonstrated cognitive improvement. [65]

A 2026 study monitored 54 adults with ADHD and 47 controls with actigraphy for ten days. Activity rhythms were less stable and later in the ADHD group. Several sleep measures initially correlated with cognitive performance, but **none of the cognitive correlations survived** correction for multiple comparisons. The groups also differed substantially in age

despite adjustment. The study gives us a reason to measure rhythms more carefully, while leaving their contribution to cognitive difficulty uncertain. [195]

Effort is another outcome, not a synonym for capacity. In a study of 20 adults with ADHD and 24 comparison participants, amphetamine medication increased willingness to invest cognitive and physical effort in the ADHD group. This identifies an experimentally tractable process; it does not show that immune activity caused the difficulty. [21]

A human immune-challenge experiment also linked inflammatory state to aspects of effort and reward allocation, while several proposed moderation effects were not supported. It did not compare people with and without ADHD. That comparison needs testing before similar motivational symptoms are assigned the same biological cause. [22]

4.5 A physiological challenge need not affect every domain

In a counterbalanced exposure study, 26 adults were analyzed after particulate-pollution and clean-air sessions. Four hours after exposure, selective attention and emotion-expression discrimination differed, while working memory and psychomotor vigilance did not show the same effect. This was not an ADHD study; inflammatory mediation was not established, and co-exposures limit attribution. It argues for a prespecified domain and time window, not a general “cognitive health” score. It is not a reason to add a pollution challenge to the proposed ADHD study. [23]

4.6 Does the association hold within an ordinary week?

A 28-day study followed 205 adolescents through two school weeks and two vacation weeks. It paired actigraphy and sleep diaries with morning and afternoon symptom ratings and a daily 3.2-minute vigilance task. Nights with more sleep or better sleep efficiency than a participant’s own average predicted better next-day task performance and less sleepiness and fatigue. The analysis separated these **within-person** associations from differences between adolescents. This was a convenience sample, not an ADHD study or a randomized sleep intervention; school always preceded vacation. Still, its design is much closer to the daily question in this paper than a single comparison between diagnostic groups. [173]

Our next step is to test whether that **within-person** relationship is different in ADHD and whether any physiological measure adds useful prediction beyond sleep, time of day and medication timing. A strong average association across people is not enough. We also need to know whether the signal is reliable enough to anticipate a difficult day for someone new.

A related design has already been used in type 1 diabetes. Researchers paired continuous glucose monitoring with cognitive tests over 15 days in 200 adults; 190 remained after data cleaning. Larger glucose excursions were associated with slower and less accurate processing speed, while sustained attention did not show the same association. Responses also varied between people. This is close to the kind of **within-person** test I have in mind. It remains observational evidence in diabetes, not proof of an ADHD mechanism or a reason to change glucose targets. The practical importance and stability of an individual response still need testing. [193]

An experimental glucose clamp gives a more direct, narrower result. In 94 adults with or without diabetes, researchers lowered glucose to about 2.8 mmol/L. Mean alertness reaction time increased by 32.1 milliseconds (SD 66.6), and working-memory errors and omissions increased by 2.0 (SD 5.5). These are **standard deviations** of the changes, not confidence intervals. The effects were largely similar across diabetes and hypoglycemia-awareness groups. Clinically low glucose can impair cognition; this fixed-sequence experiment does not show that ordinary meal-related fluctuations explain ADHD or that people with ADHD are more sensitive. [210]

5. Histamine and immunity: distinct compartments, possible links

Sleep establishes the first problem: a bodily change can matter, but symptoms, effort and task performance need not move together. Immune signalling is the next branch because it may intersect with sleep and alertness. The task is to identify the signal, its location and the outcome, rather than treating inflammation as one explanation.

5.1 There are at least two distinct histamine questions

Central neuronal histamine participates in arousal and cognitive regulation. Peripheral mast-cell histamine is part of immune signalling alongside other mediators. Receptor subtype and anatomical compartment matter. A proposed connection between them is not evidence that circulating histamine directly determines brain histamine or attention. [24, 25, 26]

In a circuit experiment, reducing H2-receptor signalling in specific parvalbumin-positive neurons produced ADHD-like behaviours in mice. The accompanying human tissue observations concerned patients described as having ADHD symptoms. The causal manipulation was in mice. It does not predict that blocking peripheral H1 receptors with an ordinary allergy medication treats ADHD. [24]

A pitolisant experiment tested central histamine signalling in humans. Sixty healthy participants were randomized; 58 formed the final sample and 52 contributed to key imaging/task analyses. An H3 inverse agonist altered learning-related network activity and working-memory computations. This is evidence that the intervention can change human cognitive processing, **not an ADHD treatment trial** or proof that its effects are exclusively histaminergic. [25]

The earlier ADHD trial was negative. In 72 adults, the H3 inverse agonist MK-0249 did not improve the primary ADHD outcome relative to placebo, whereas the active methylphenidate comparator did. This constrains that compound and protocol, not every histamine strategy. A molecule’s involvement in cognition is not sufficient evidence of treatment efficacy. [27]

5.2 Allergy overlap does not identify the route

An umbrella review supports ADHD associations with asthma and rhinitis, but diagnoses cannot separate immune effects from sleep disruption, symptoms, shared liability or ascertainment. In a daily time-series study, 21 children with ADHD and atopic disease were enrolled and 16 were analyzed; relationships between atopic and ADHD symptoms varied in timing and direction. These findings justify investigating individual relationships, not declaring stable allergy-sensitive ADHD subtypes. [28, 29]

A 2026 cytokine synthesis covered 74 studies across autism and ADHD together, including 990 ADHD participants, 4,331 autistic participants and 5,037 controls. Network analyses found lower TNF-α, IL-2 and MCP-1 in ADHD, rather than a broad elevated profile. These are pooled peripheral measurements, not a diagnostic signature or evidence that lowering inflammation would improve ADHD. They constrain a generic systemic-inflammation account without settling local brain immune activity. [30]

The question is whether allergic or immune changes affect cognition in some people through symptoms, sleep, nerve signalling or another measured route. A study must specify the route it is testing. Relief after an antihistamine could come from less discomfort or better sleep; it does not diagnose mast-cell activation syndrome or identify the cause of ADHD.

Atopic dermatitis studies put numbers on the overlap. A US survey analysis found adjusted ADHD odds ratios of 1.14 (95% CI 1.03–1.26) in children and 1.61 (1.25–2.06) in adults. A later synthesis estimated 1.28 (1.18–1.40). These observational estimates do not isolate histamine. Itch, disrupted sleep, treatment and diagnostic contact remain plausible contributors, and atopic dermatitis is not equivalent to mast-cell activation syndrome. [68, 69]

5.3 Immune communication is not the same as an autoimmune attack

The proposed immune model concerns communication and regulation. It does not require immune attack on brain tissue, and it should not use “neuroinflammation” as a catch-all explanation for pressure, dizziness or fog. In mice, Jin and colleagues identified vagal–brainstem circuitry that senses inflammatory signals and modifies peripheral responses. That establishes a specific experimental feedback route, not a diagnosis from a symptom cluster. [31]

Suppressing inflammation can also come at a cost. In the same circuit research, strong artificial activation that suppressed inflammation impaired protection in a bacterial-infection model. Appropriate regulation is a more defensible target than “the lowest possible inflammation.” Those mouse manipulations are not equivalent to ordinary vagal stimulation in people. [31]

5.4 What controlled human inflammation experiments show

Human immune challenges offer a more direct test than diagnostic overlap. In a randomized crossover study, 22 healthy volunteers received low-dose bacterial endotoxin or saline. After endotoxin, they rated their performance more poorly, but the researchers did not detect a

change in average reaction time or errors on two brief tasks. The small sample and three-minute tasks leave uncertainty. Still, the result is useful: feeling cognitively worse and showing a lower mean task score are outcomes we should measure separately. [153]

A newer experiment gets close to another distinction in this paper. Among 160 adults aged 60 or older, researchers randomized endotoxin or placebo within groups with and without insomnia. Of the endotoxin recipients, 26 had insomnia and 53 did not. The insomnia group had a larger self-rated depressed-mood response across the observation period, with a standardized difference of d = 0.64 (95% CI 0.16–1.12), despite similar measured rises in two inflammatory signals, IL-6 and TNF. Insomnia itself was not randomized. This was an acute mood outcome, **not a cognitive test**, a diagnosis of depression or an ADHD study. [154]

This gives us a concrete reason to separate the size of a bodily response from sensitivity to that response. Similar measured cytokine changes need not mean similar symptoms. Neither experiment identifies histamine, blood–brain barrier leakage or a particular pathway as the cause. I would use these findings to sharpen the measurements in our observational work; they are not a proposal to administer endotoxin in an ADHD pilot.

5.5 Inflammation needs more than a positive association

A 2025 analysis combined ALSPAC observations with genetic approaches. ADHD and C-reactive protein (CRP) had a genetic correlation of 0.27 (95% CI 0.19–0.35). That is evidence of shared common-variant influences, not proof that inflammation causes ADHD. The apparent CRP-to-ADHD effect was inconsistent across Mendelian-randomization sensitivity analyses, with evidence that variants could act through other pathways. Associations between childhood ADHD traits and later measured CRP were also not consistent across ages. Attrition, ancestry restriction and uncorrected multiple testing limit interpretation. [185]

Timing changes the question further. In Generation R, 6,555 mother–child pairs were followed with ADHD symptom ratings at ages 6, 10 and 14. Each additional reported prenatal fever exposure was associated with a 0.06-SD increase in symptoms (95% CI 0.01–0.11). Adjusted associations with the measured cytokine composite and CRP were not significant. The blood samples were taken in mid-pregnancy and could miss brief inflammatory spikes; fever was self-reported. The study neither proves an inflammatory mechanism nor rules out every prenatal immune effect. It also concerns developmental exposure, not the cause of an adult's difficult afternoon. [186]

5.6 Mast-cell disease and cognitive symptoms

Mastocytosis involves abnormal mast-cell accumulation. In one clinical sample of 57 patients, 42 reported cognitive complaints and 22 met the study's memory or attention impairment threshold. The study used standardized test norms, without a separately recruited healthy comparison group. Medication, referral selection and other symptoms complicate interpretation. It shows that cognitive difficulties deserve assessment in a defined mast-cell disease; it cannot establish their mediator or be treated as an MCAS or ADHD prevalence estimate. [198]

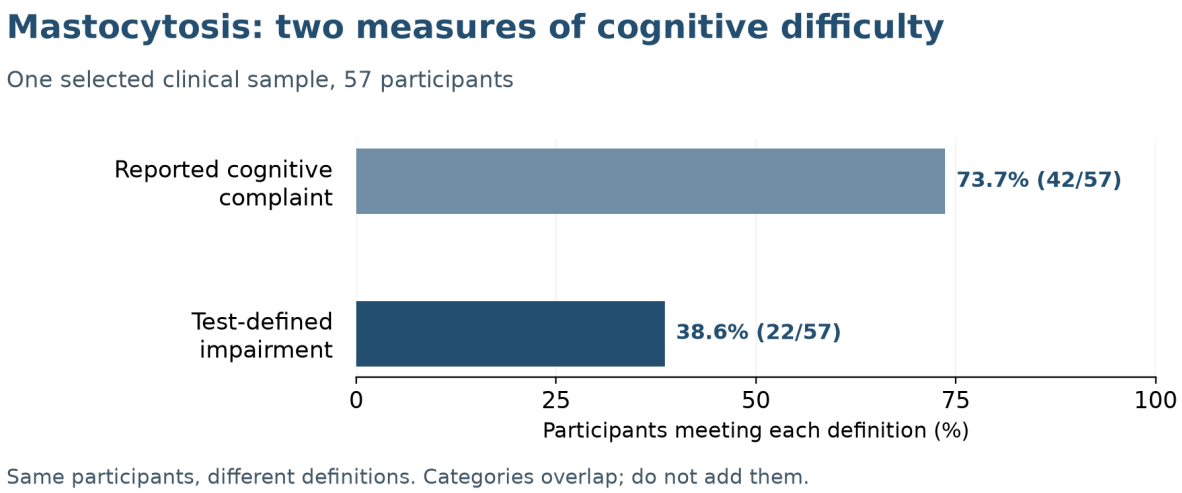


Figure 5. Two definitions in the same 57 patients. Percentages are calculated from reported counts and rounded. Symptoms and test thresholds are not interchangeable; the bars are neither treatment effects nor independent groups. [198]

Mast-cell activation and mast-cell activation syndrome (MCAS) also need separating. Clinical frameworks combine recurrent systemic symptoms, objective mediator evidence and response to directed treatment; tests and definitions still have limitations. A 2024 expert research report specifically identifies weak evidence for several neurological and psychiatric attributions. For our study, the diagnostic criteria, event timing and laboratory method must be recorded. Improvement after an antihistamine alone cannot identify the cause of brain fog. A negative test does not make the symptoms unreal, and a symptom cluster does not identify mast cells as its source. [199]

5.7 Brain barriers and fluid drainage

The blood–brain barrier regulates exchange between blood and brain tissue. Blood flow describes delivery through vessels; cerebrospinal fluid (CSF) movement is another process. A finding in one cannot substitute for a measurement of the others. The small Long COVID imaging comparison in Section 8.1 is human evidence about barrier permeability. It does not establish mast-cell causation or an ADHD abnormality. [44]

A 2025 Cell study adds a specific mechanism at the brain's outer coverings. In mice, dural mast-cell activation released histamine, widened bridging veins and restricted CSF passage at arachnoid cuff exit points. BBB permeability was unchanged at the measured one-hour time point in the activation experiment. Mast-cell depletion increased brain bacterial burden in meningitis models, showing a protective role in that setting. Human tissue observations established the presence of meningeal mast cells; the functional experiments were in mice. Fluid regulation and barrier leakage therefore need separate tests. Neither result demonstrates an explanation of human brain fog. [200]

A different study exposed cultured human brain endothelial and microglial cell lines to products from activated mast cells and found inflammatory responses and reduced tight-junction proteins. It also examined infected mice. Cell lines derived from humans are **not a**

clinical human experiment, and an acute infection model does not establish persistence in Long COVID. Together, these studies identify routes worth testing while leaving their relevance to ADHD unresolved. [201]

Table 2. Different brain-related measurements and their limits

Measurement	What it addresses	What it does not establish
Barrier permeability	Movement across a specified barrier, using an appropriate imaging or laboratory method.	The responsible cell, a cognitive mediator, or the cause of ADHD.
Cerebral blood flow	Blood delivery or a measured velocity proxy under specified conditions.	Barrier leakage or impaired CSF drainage.
CSF dynamics	Fluid movement around and through brain-associated spaces.	A general brain “detox” deficit or a proven human treatment target.
Cognitive outcome	A defined symptom report, task score or functional measure.	Which physiological route produced the difficulty.

6. Hormones PMDD and endometriosis

Hormones and endometriosis bring timing into sharper focus. Recurring symptoms offer a way to study the same person across changing conditions, while pain and sleep provide competing explanations we can measure. Shared diagnoses alone cannot choose between those routes.

6.1 The endocrine signal should be measured, not assumed

Roberts and colleagues collected saliva and symptom reports for 35 days in 32 regularly cycling women. Hormonal combinations predicted next-day ADHD symptoms, with trait impulsivity moderating relationships. This was a community sample, not proof of a diagnosed-ADHD interaction or objective cognitive decline. A later 35-day pilot in 30 stimulant-treated females found phase-related symptom variation without establishing hormonal or immune mediation. [32, 33]

A counterweight is a 2025 meta-analysis of 102 articles, 3,943 participants and 730 comparisons: it found no systematic, robust menstrual-phase differences in objective cognitive performance. Its mainly nonclinical scope does not exclude symptomatic ADHD or PMDD subgroups, or increased effort to maintain a score. It argues against treating small symptom studies as proof of universal cycle-related cognitive decline. [66]

For the proposed study, I would measure endocrine state repeatedly where it is relevant to the question. Calendar phase alone is not a mechanism. Hormones, mood, pain, migraine, sleep and treatment can change together; measuring them together would help us distinguish their roles.

6.2 PMDD and the direct ADHD overlap

PMDD involves impairing symptoms that follow a recurring premenstrual pattern and ease after menstruation begins. In a UK survey of 715 young adults, **provisional PMDD** was more frequent among those reporting a clinical ADHD diagnosis. Figure 6 shows the comparison. Retrospective screening cannot confirm PMDD, which needs prospective daily symptom tracking across at least two cycles. ADHD was oversampled, and depression and anxiety were common. Selection, recall and symptom overlap limit interpretation; neither hormones nor a cognitive mechanism were measured. [204]

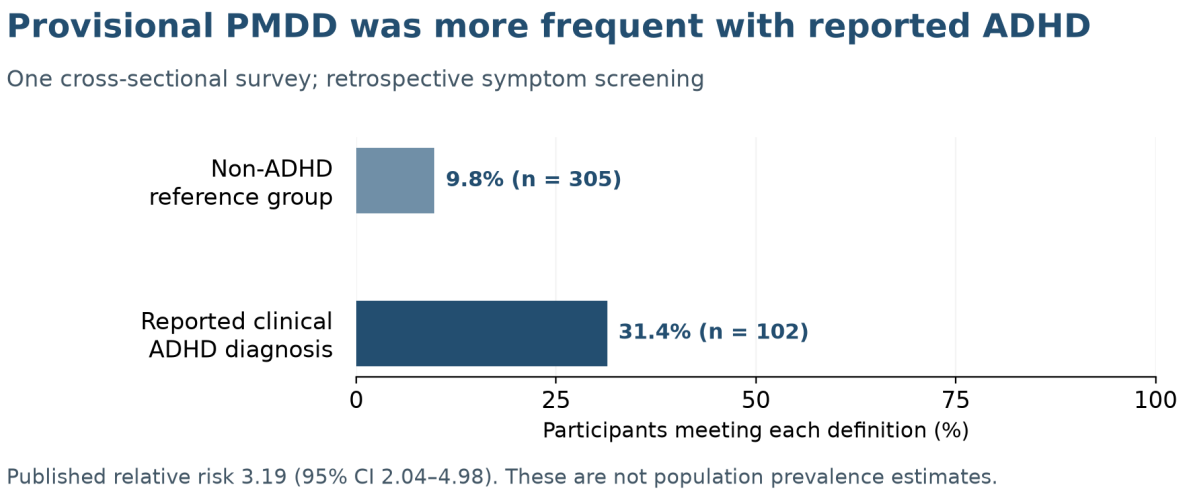


Figure 6. Reported proportions from Broughton et al. Bars show provisional screening results, not confirmed PMDD diagnoses. The ADHD group and reference group are selected subsets of the survey. The relative risk is an association, not evidence of shared causation. [204]

For the hypothesis, premenstrual exacerbation of ongoing ADHD or depression is a separate pattern from PMDD. Someone may have difficulties throughout the month and a premenstrual worsening, or a more restricted cyclical syndrome. Both deserve attention. Daily observations should distinguish baseline burden from a steeper response to the changing state.

Hormonal manipulation provides a more direct test. In a 2017 study, 22 women with PMDD were studied and 12 **selected responders** to ovarian suppression entered the relevant hormone add-back comparison. Symptoms rose during the first month of combined estradiol and progesterone exposure, then subsided during continued exposure in months two and three. This supports sensitivity to a change in hormones in that subgroup. The small, selected sample and combined treatment do not isolate estrogen, and cognition in ADHD was not tested. [205]

A 2025 extension compared 34 women with PMDD and 76 healthy participants during suppression and separate estradiol and progesterone add-back. Affective symptoms returned during add-back in the PMDD group but not the comparison group. Ten PMDD participants and 15 comparators came from the original cohort, so this is not a wholly independent

replication. The result strengthens the case for **differential sensitivity** while leaving the downstream mechanism open. [206]

6.3 Estrogen progesterone and metabolic signalling

Estrogen is a family of hormones; estradiol is the form studied in these experiments. Progesterone and its metabolites deserve their own place in the model. Allopregnanolone, a progesterone-derived neuroactive steroid, modulates GABA-A signalling. In two small dose studies, each with eight PMDD participants and eight controls, the higher-dose dutasteride experiment blocked the luteal allopregnanolone rise and reduced several symptoms; the lower-dose experiment did not. Several steroid pathways are affected, and these were mood outcomes. The experiment is a mechanistic clue, not an established PMDD treatment or an ADHD cognitive result. [208]

There is also a laboratory estrogen–mast-cell connection. Estradiol increased mediator release in cell lines and mouse-derived mast cells through estrogen-receptor-alpha-dependent mechanisms. The work included a human mast-cell line, not a trial in women with PMDD. It supports testing hormone–immune interactions; it does not establish an estrogen–histamine feedback loop as the cause of PMDD, BBB disruption or ADHD. Our model should leave those connections open until measurements in the same people connect them. [202]

The insulin connection has a small human experiment behind it. Eleven healthy, naturally cycling women completed insulin-versus-placebo nasal-spray comparisons during glucose clamps in follicular and luteal phases. Insulin increased the glucose infusion requirement in the follicular phase; the adjusted luteal result was not significant. A separate 15-person imaging study examined hypothalamic responses but lacked a contemporaneous placebo-spray comparison. Phase was verified with hormone measurements, yet estrogen and progesterone were not independently manipulated. This is a physiological interaction, not proof of cognitive impairment, PMDD or a pathological luteal state. [212]

6.4 Testosterone and estrogen treatment need cognitive outcomes

Menopause needs its own evidence. In ELITE-Cog, 567 healthy postmenopausal women contributed to a randomized estradiol comparison. The verbal-memory difference was −0.06 SD (95% CI −0.22 to 0.09) after a mean 57 months, with no clear advantage from starting nearer menopause. This does not test symptomatic perimenopause or ADHD, and does not negate relief of other menopausal symptoms. A hormonal hypothesis should not assume that replacement improves cognition. [96]

Testosterone belongs in the discussion on the same terms. In the Testosterone Trials, 493 men aged 65 or older with low testosterone and age-associated memory impairment formed the primary cognitive subgroup. One year of testosterone gel did not improve delayed paragraph recall versus placebo: the adjusted difference was −0.07 points (95% CI −0.92 to 0.79). This does not settle treatment for established endocrine indications, younger people or ADHD. It does show why a low hormone level and a plausible brain mechanism cannot promise a cognitive benefit. [207]

I would ask whether an endocrine change predicts a defined difficulty in the person being studied, and whether treating an established condition improves that difficulty. There is no single “better” estrogen or testosterone level supplied by these studies for attention or resilience. Reproductive stage, prescribed hormones, sleep, pain and mood all affect what a comparison can mean.

6.5 Why the ADHD–endometriosis overlap stays in the paper

In a Swedish register study of 854,361 women, 14,144 had an endometriosis diagnosis. The sibling comparison found a higher rate of subsequent recorded ADHD diagnosis (HR 1.98, 95% CI 1.34–2.93). Comparing sisters reduces some shared familial confounding; it does not remove differences in pain, sleep, medical contact or diagnostic delay. Later ADHD recognition cannot establish that endometriosis caused a developmental disorder. No cognitive task was measured. [34]

Cross-trait genetics provides another kind of evidence. One analysis estimated endometriosis –ADHD genetic correlations of 0.21 and 0.17 using two methods. These are population-level correlations in common-variant effects, not a 21% individual risk or a percentage of shared cause. The methods reuse source data, and the result does not identify an inflammatory pathway or establish causal ADHD-to-endometriosis effects. [35]

A human endometriosis study found higher lesion histidine decarboxylase expression and serum histamine, but no difference in peritoneal histamine or urinary methylhistamine. Histamine-related receptor transcripts were not uniformly increased either. Those null findings belong in the account: local tissue biology, circulating markers and excretion are not interchangeable compartments. [36]

A more specific 2025 study combined human lesion observations with cell and mouse experiments. It implicated the mast-cell receptor MRGPRX2 and a histamine–H1–TRPV1 route in sensory-neuron sensitization and pain. This is a candidate local immune-to-nerve mechanism. It is not a trial showing that an antihistamine treats endometriosis, and it did not measure ADHD cognition. [26]

My proposed connection is that recurring endocrine or local immune activity may increase pain, alter sleep or migraine burden, and thereby change cognitive experience or performance. A direct immune-to-cognition route is an additional hypothesis, not a necessary assumption. If pain and sleep explain the association, we should accept that answer rather than keep searching for a more exotic mechanism.

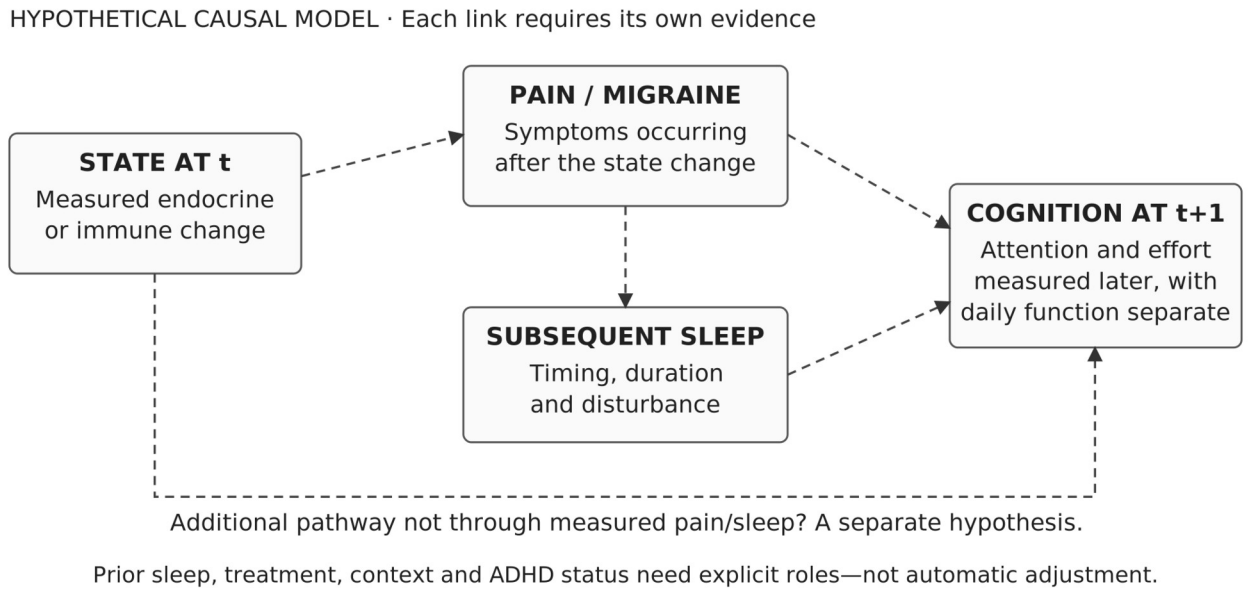


Figure 7. A plausible connection still contains competing explanations. Hypothetical time-ordered diagram for the endometriosis module; arrows do not assert demonstrated mediation. A total-effect question includes routes through subsequent pain and sleep. A question about a remaining association after those routes requires different assumptions. This is not a complete causal graph: prior exposures, treatment and confounders need explicit specification.

Worked example 3 · Do not adjust away the question

Imagine that attention worsens during a pain flare that has already disrupted the preceding night's sleep. A model including sleep might reduce the estimated flare–attention association. That could be consistent with sleep carrying part of the relationship, not proof that the original experience was unrelated to the flare. The study must state whether it is asking about the total relationship or a pathway not operating through sleep. This is an illustration, not a patient case or a causal conclusion.

6.6 Fatigue and iron status need their own measurements

In 560 women with confirmed endometriosis and 560 matched controls, frequent fatigue was reported by 50.7% versus 22.4%. Within the endometriosis group, fatigue was associated with insomnia, pain, depression and occupational stress. These are reasons to measure competing routes, not proof that they mediate ADHD cognition. [93]

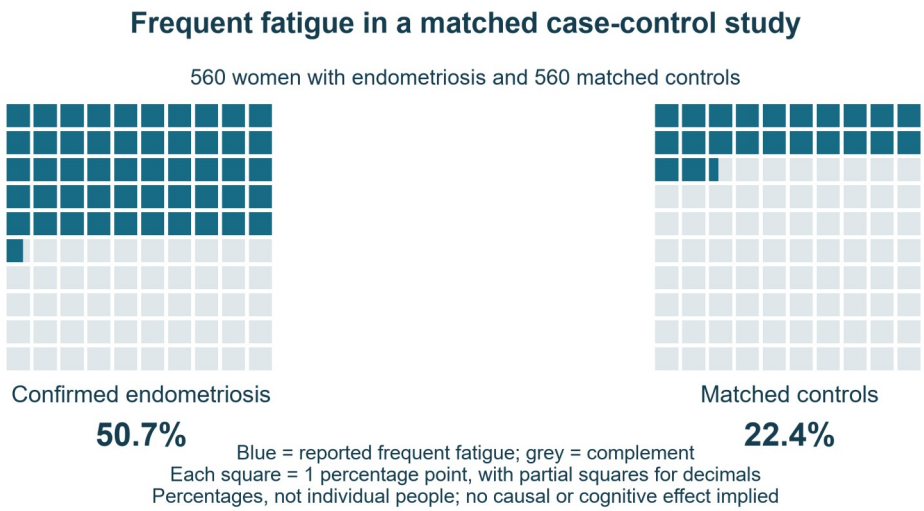


Figure 8. Fatigue is common in this endometriosis sample, but fatigue is not a cognitive test. Reported proportions from a matched case-control study; no confidence intervals are reconstructed. Selection, symptom reporting and unrecognized endometriosis in controls may affect the comparison. It cannot identify a shared cause with ADHD. [93]

Iron status also needs checking. In a selected clinic cohort of 251 people with symptomatic endometriosis, 134 (53.4%) had iron deficiency and 34 (13.5%) had iron-deficiency anemia. Anemia is included within iron deficiency; the percentages must not be added. This was a referral sample, not a prevalence estimate for everyone with endometriosis, and it did not measure objective cognition. [94]

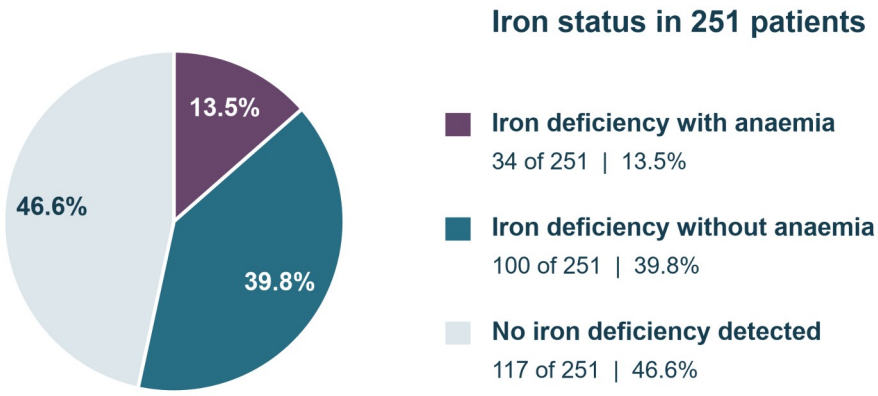


Figure 9. Iron-status categories in a selected endometriosis clinic cohort. Counts partition the same 251 patients: 34 with iron-deficiency anaemia, 100 with iron deficiency without anaemia (134 minus 34), and 117 without detected iron deficiency (251 minus 134). Percentages are calculated from these counts and rounded. This is not a general-population prevalence estimate or a cognitive outcome. [94]

A separate randomized trial enrolled 198 fatigued, nonanemic menstruating women with low ferritin. At 12 weeks, fatigue scores fell by 47.7% with iron and 28.8% with placebo; the reported contrast was −18.9 percentage points (95% CI −34.5 to −3.2). Those are mean score reductions, not the proportions cured. The trial did not test ADHD or cognition. Assessment guided by clinical history can identify a contributor without treating fatigue or fog

as proof of an immune–brain mechanism. These findings do not support routine supplementation without an indication. [95]

7. Migraine, circulation and body-signal processing

Recurring hormonal and pain states lead naturally to migraine: another setting in which a person can be compared during and between episodes. Circulation and posture extend that question to the conditions under which thinking is tested.

7.1 Migraine is more than a convenient analogy

In a Taiwan cohort, 81,441 people with ADHD were matched to 81,441 controls without a prior migraine diagnosis. During follow-up, 462 versus 212 received a migraine diagnosis; the adjusted hazard ratio was 1.92 (95% CI 1.64–2.34). The young-adult subgroup estimate was imprecise and not significant: 1.31 (0.58–2.98). These are newly coded diagnoses, not lifetime migraine prevalence. Figure 10 shows the observed counts on an absolute scale. [67]

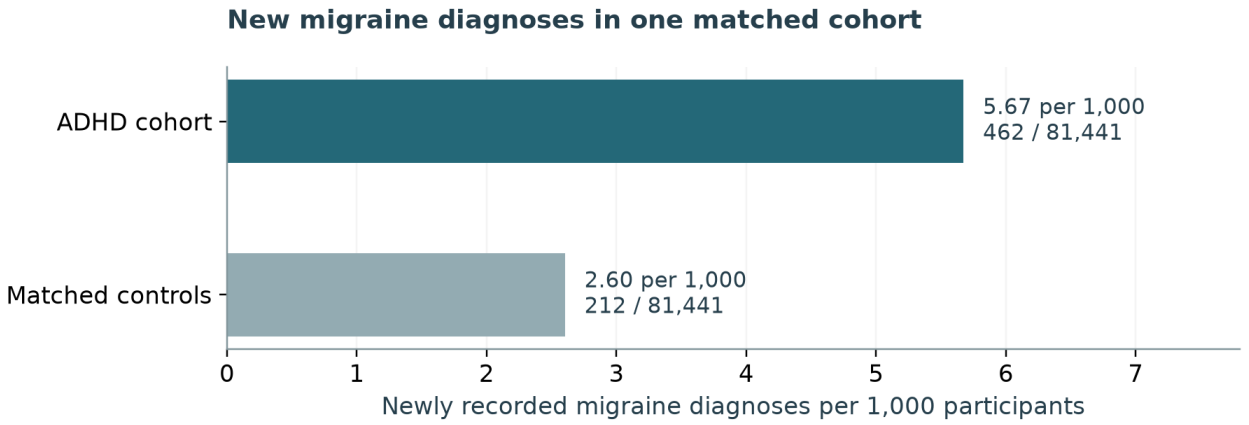


Figure 10. One cohort, two ways to read the result. Counts from Hsu et al.; bars are calculated as count / 81,441 × 1,000. Adjusted HR 1.92 (95% CI 1.64–2.34). Enrollment 2001–2009, follow-up through 2011: these descriptive proportions have variable follow-up and are not annual incidence rates or migraine prevalence. They cannot identify a shared cause. [67]

A migraine study included 39 patients, with 24 completing testing both during an untreated attack and between attacks. Processing speed and verbal-memory performance were worse during attacks. This supports **within-person** neurological-state variation; it does not establish a larger effect in ADHD or identify the entire mechanism. [37]

Migraine also supplies an example of a defined, clinically testable pathway. In the 955-participant STRIVE trial, blocking the calcitonin gene-related peptide (CGRP) receptor with erenumab reduced migraine frequency and interference with activities compared with placebo. That supports a specific migraine target, not a histamine explanation or evidence that ADHD improves when that target is treated. [38]

The relevant ADHD study would assess attack, interictal and recovery states, alongside pain, sensory symptoms, treatment and independent cognition. A treatment-related reduction in headache days and a reduction in cognitive fog are related but separate outcomes. Migraine medication itself is another possible contributor to cognitive change.

7.2 Orthostatic physiology offers a measurable challenge

Autonomic studies in adult ADHD are heterogeneous, and differences are more evident in some task/stress settings than as one resting marker. Interoception research—the detection and interpretation of bodily signals—is similarly limited and mixed. Neither literature justifies diagnosing dysautonomia from ADHD or a wearable HRV value. [39, 40]

POTSKog tested the relationship in POTS: 30 participants with postural orthostatic tachycardia syndrome (POTS) and 30 controls completed cognitive assessment across body positions in counterbalanced order. Attention and executive differences were found, with posture-related Stroop effects in the POTS group; leg crossing did not restore executive performance. It was not an ADHD study and did not establish that cerebral blood flow alone explained the results. [41]

That suggests a study we can actually run. Measure the change in posture, cardiovascular response, respiratory carbon dioxide and cognitive performance together. Transcranial Doppler measures blood-flow velocity, not absolute flow; vessel diameter and carbon dioxide affect interpretation. Cognitive effort can itself alter haemodynamic measurements, so a repeated-task control condition is essential. [42, 43]

Feeling light-headed or pressured does not establish impaired cerebral perfusion. Conversely, a symptom report should not be dismissed merely because one cardiovascular measurement is normal. Physiology, perception, symptoms and performance are different measurements whose relationships remain to be tested.

7.3 A normal pulse response does not answer every circulation question

In a specialist-clinic study, 429 people with ME/CFS and 44 healthy controls underwent head-up tilt with extracranial Doppler measurements of cerebral blood flow. Among the 247 patients whose heart rate and blood pressure responses were classified as normal, the mean end-tilt flow reduction was 24%, compared with 7% in controls. Referral bias may have enriched the patient sample for orthostatic problems. This was not an ADHD study and did not demonstrate that reduced flow caused cognitive symptoms. It gives a concrete reason to avoid treating pulse and blood pressure as complete substitutes for cerebral circulation. [177]

The useful follow-up would pair the circulation measure with an independent cognitive outcome and respiratory carbon dioxide, and assess reproducibility outside the original clinic. That would test a mechanism more directly than adding another diagnostic overlap. The existing safeguards around symptoms and exertion still apply.

8. Long COVID and ME/CFS: acquired disruption and delayed effects

So far, the emphasis has been on recurring states. Infection adds a different course: a recognizable illness followed by recovery that may be incomplete or uneven. Long COVID and mono help us ask about duration without assuming they share one mechanism with ADHD.

A large English community study measured cognition after COVID. Of 141,583 people who began an online battery, 112,964 completed it. Unresolved persistent symptoms were associated with a −0.42 SD adjusted global-cognition difference from the no-COVID group (95% CI −0.53 to −0.31). Smaller differences remained after reported symptom resolution. The study did not measure each person before infection; selection and confounding remain. These are group task-score differences, not observed individual IQ losses or evidence of irreversible decline. [70]

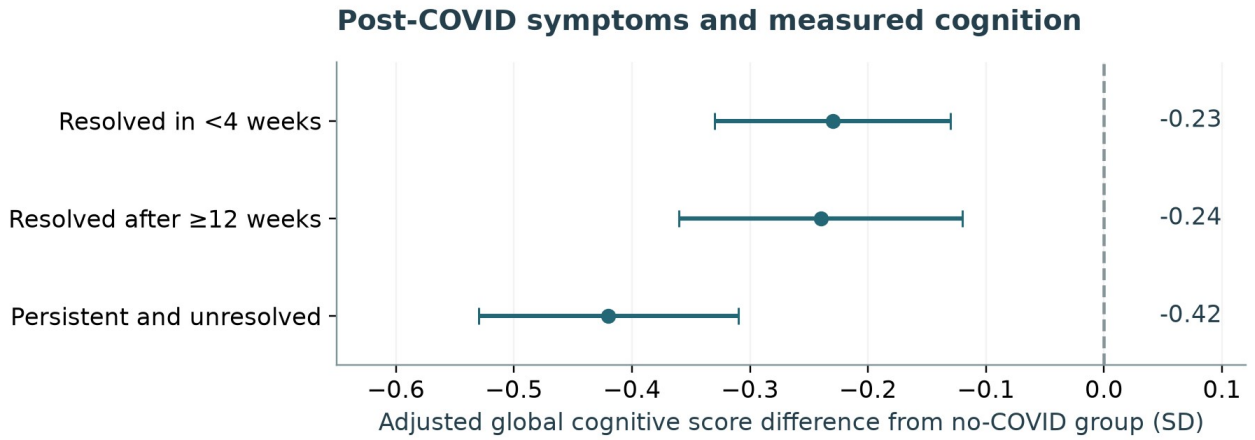


Figure 11. Published adjusted group differences with 95% confidence intervals, Hampshire et al. The reference group included people with no confirmed infection. Selected symptom groups from one community study are shown; they are not a pooled analysis or repeated stages in the same people. SD means standard-deviation units. No personal pre-infection cognitive baseline was available. [70]

8.1 Brain barriers and energy measures are distinct pieces of evidence

A Long COVID study associated cognitive symptoms with blood–brain barrier abnormalities and immune findings. Its dynamic-contrast MRI comparison included 10 recovered participants, 11 with Long COVID without brain fog and 11 with brain fog. This is a small mechanistic comparison, not evidence that ADHD shares its cause. [44]

A separate phosphorus spectroscopy study compared 27 post-COVID participants with 23 recovered controls. The authors reported lower ATP-to-phosphocreatine ratios and associations between anterior cingulate measures and cognition. A ratio is not a direct ATP-production measurement. The barrier and spectroscopy findings came from different studies; combining them does not establish one causal chain in the same participants. [45]

Long COVID and myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) also require a different sampling timescale. Post-exertional malaise can involve delayed worsening after physical or mental exertion; CDC guidance describes typical onset 12–48 hours later and persistence for days or longer. That is not adequately captured by an immediate “before and after” test. [46]

For this programme, post-infectious cohorts require symptom-informed limits, low-burden measures and delayed follow-up. Maximal exercise should not be introduced merely to create a strong signal. The module is intended to study naturally occurring change or a clinically justified protocol, not provoke avoidable deterioration.

Autonomic and muscle findings need equally careful denominators. In 24 selected patients referred for post-COVID symptoms, 23 had orthostatic intolerance on tilt, but only four showed POTS; 15 developed intolerance after nitroglycerin provocation. That cannot estimate population prevalence. A separate exercise study compared 25 people with Long COVID and post-exertional malaise with 21 recovered controls, finding skeletal-muscle abnormalities. Muscle biopsies are not brain-energy measurements, and neither study demonstrates mediation of ADHD cognition. [71, 74]

8.2 The mast-cell findings conflict—and that is part of the evidence

An exploratory report in 13 symptomatic post-acute COVID participants found elevations in selected mast-cell-derived proteases. A later case-control study of 24 Long COVID participants and 24 recovered controls found no significant differences in two mast-cell proteases and no corrected associations with the physical-function measures tested. [47, 48]

Timing and assays differed; the later Long COVID group was sampled a median of 60 weeks after diagnosis. These differences are reasons for prospective replication, not proof that mast-cell activity starts early and then disappears. Nor do two null blood markers rule out every local or episodic process. The defensible conclusion is narrower: persistent systemic mast-cell activation is not established as a general explanation, and a subgroup claim needs prespecified, reproducible evidence. [48]

8.3 There are two different antihistamine trial questions

The small blinded famotidine trial enrolled previously hospitalized adults with low cognitive-screen scores. Its full text reports 60 randomized, 50 analyzed after a first post-baseline assessment, and 45 completing 12 weeks. Mean time since symptom onset was about 29 days, so this is early post-COVID convalescence rather than a trial requiring established Long COVID. At week 12, endpoint differences favored famotidine by 2.44 MMSE points (95% CI 1.47–3.41) and 2.56 MoCA points (1.69–3.43). These are endpoint contrasts, not change-score contrasts. Excluding early withdrawals and carrying observations forward limit interpretation. The authors also state that the registry did not explicitly distinguish primary from secondary outcomes; the registry record itself remains to be independently checked. [49]

The larger STIMULATE-ICP trial asked a different question. It randomized 778 participants across four groups; 193 received famotidine–loratadine. Compared with no study drug, the adjusted fatigue difference at 12 weeks was −1.48 points (95% CI −2.88 to −0.08), with no significant between-group difference at 24 weeks. It was open-label, and the primary outcome was self-reported fatigue, not objective cognition. [50]

These trials are not replications: early recovery after hospitalization and established post-COVID condition differ, as do single-drug and combination treatment, masking, and cognitive-screen versus fatigue endpoints. Neither establishes a mast-cell mechanism. A positive screen result or a small fatigue benefit warrants a better-targeted trial; it does not justify a general antihistamine claim.

The vagal research offers another possible explanation for a drug effect. In a mouse endotoxin model, famotidine reduced selected cytokines even in mast-cell-deficient animals, while vagal interruption or deletion of an acetylcholine receptor abolished the effect. This is a possible neural anti-inflammatory route, not a demonstrated human Long COVID mechanism. The model and administration routes differ from oral clinical use. It illustrates why benefit from an H2 drug need not identify mast cells as the cause. [51]

8.4 Mono gives the recovery question a starting point

The overlap with this hypothesis is prolonged fatigue, cognitive complaints and recovery after infection. A direct causal overlap with ADHD has not been established, and the studies below do not show that pre-existing ADHD changes recovery after mono. Section 8.6 separates that unanswered question from the broader post-infectious evidence.

Mono is relevant because it gives us a recognizable illness to follow forward. Infectious mononucleosis is most often caused by Epstein–Barr virus, although other infections can produce the syndrome. Having had EBV is not the same as having had mono. Most people with mono improve within two to four weeks, but fatigue can last longer. The important question here is what happens when recovery takes months rather than weeks. [158]

In a prospective study of 301 adolescents with mono, 39 met the study's chronic fatigue syndrome criteria at six months, 22 at 12 months and 13 at 24 months. The published rounded proportions were 13%, 7% and 4%. Figure 37 shows these counts against the original cohort. Some participants were lost to follow-up, so everyone outside those counts cannot simply be labelled recovered. These were clinical fatigue-syndrome classifications, not ADHD diagnoses or cognitive-test scores. [160]

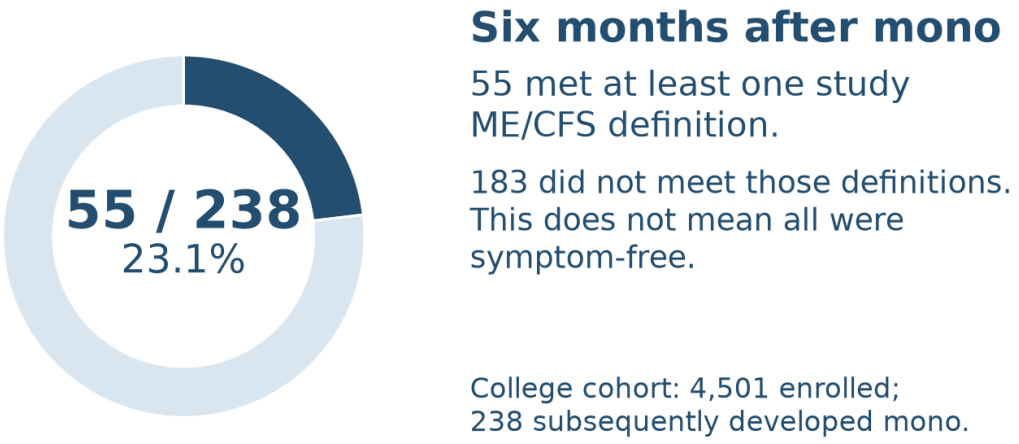


Figure 12. The denominator is the 238 students who developed mono, not all 4,501 enrolled students. Percentages are calculated from reported counts. This is a fatigue-syndrome classification, not a cognitive score or ADHD outcome. [161]

A college study adds something unusual: measurements from before the infection. Researchers enrolled 4,501 students; 238 subsequently developed mono, and 55 of those students—23%—met at least one of the study's ME/CFS definitions six months later. That percentage should not be averaged with the adolescent estimate. The samples and diagnostic definitions differ. What matters for this hypothesis is the design: a person's condition before infection can be compared with their later course, rather than reconstructed entirely from memory. [161]

A 2026 follow-up of that college cohort assessed 104 selected participants about seven years later. Among those who returned, 11 of 14 from the earlier "severe" group and 7 of 23 from the "moderate" group met ME/CFS criteria. Those labels originally meant meeting multiple case definitions versus one. The results and figure report 11/14, approximately 79%; the abstract's 81% is inconsistent with that count. Small groups, attrition and selected follow-up prevent treating these figures as the long-term risk for everyone with mono. They do show that persistent illness can extend well beyond two years. [172]

EBV is not the only infection that can precede prolonged illness. In the Dubbo cohort, which combined EBV, Ross River virus and Q fever, 28 of 253 participants met chronic fatigue syndrome criteria at six months. Acute illness severity predicted persistence. This supports studying recovery across infections, while leaving open whether the processes that maintain symptoms are shared, pathogen-specific, or different between people. [162]

8.5 Brain fog, test performance and autonomic symptoms

The cognitive evidence is more mixed than the phrase "brain fog" suggests. A Dubbo study assessed 107 people during acute infection and retested a subset after recovery. Some working-memory, planning and attention measures were poorer during illness. This was a mixed-infection study from the same research programme, not an independent EBV

replication. Recovery-selected retesting and practice effects also limit what the comparison can establish. [163]

At six months after EBV infection, another study compared 195 adolescents, grouped by fatigue status, with 70 healthy controls. The fatigued participants reported more cognitive difficulty, but the researchers did not detect group differences on the objective processing-speed, executive-function and memory measures used. That does not establish equivalent ability in every setting, or make the symptoms imaginary. It tells us that a brief test and the experience of getting through a day may capture different things. [164]

A related study from the same Norwegian programme examined 200 adolescents after EBV infection and 70 controls. Fatigued and non-fatigued participants had similar infection markers and mostly similar immune, hormonal and autonomic measurements. There were some small differences, but no strong relationship between symptoms and the measured biological changes. Persistent fatigue did not correspond to a higher measured viral load. These overlapping samples should not be counted as independent confirmations. [165]

For our hypothesis, this is a reason to measure symptoms, sustained performance, standing-related symptoms and recovery separately. A report of dizziness is not a diagnosis of POTS. A normal short task does not tell us how much effort it took, how long performance can be maintained, or what happens afterward. Those are questions for the proposed measurements, not conclusions supplied by the mono studies.

8.6 Is there a direct ADHD link?

The direct evidence located in this pass does not establish one. A US NHANES analysis had ADHD status available for 2,844 children and adolescents. EBV antibody positivity was not significantly associated with reported ADHD after adjustment: odds ratio 0.77, with a 95% confidence interval of 0.42–1.38. The same report found an association with reported learning disability, OR 2.76 (1.53–4.96). This was a cross-sectional survey using antibodies to past infection and reported diagnoses, with different available samples across outcomes. It did not measure mono severity, establish which came first, or test cognitive recovery in people with ADHD. [166]

I would not turn that null ADHD result into a protective effect. Nor would I use the learning-disability association to claim that EBV causes ADHD. The narrower possibility is still worth testing: a substantial infection might add fatigue or cognitive difficulty to someone who already has ADHD. We need a prospective ADHD-by-infection comparison to test whether their decline or recovery differs. I did not locate that direct test in this targeted search.

8.7 EBV and Long COVID: an unresolved connection

A study of 280 adults after COVID associated particular EBV antibody patterns with fatigue or subjective cognitive symptoms. It lacked pre-COVID samples, and antibody findings were indirect evidence about viral activity. The result generates a hypothesis; it does not show that EBV reactivation caused those symptoms or that treating EBV would improve them. [167]

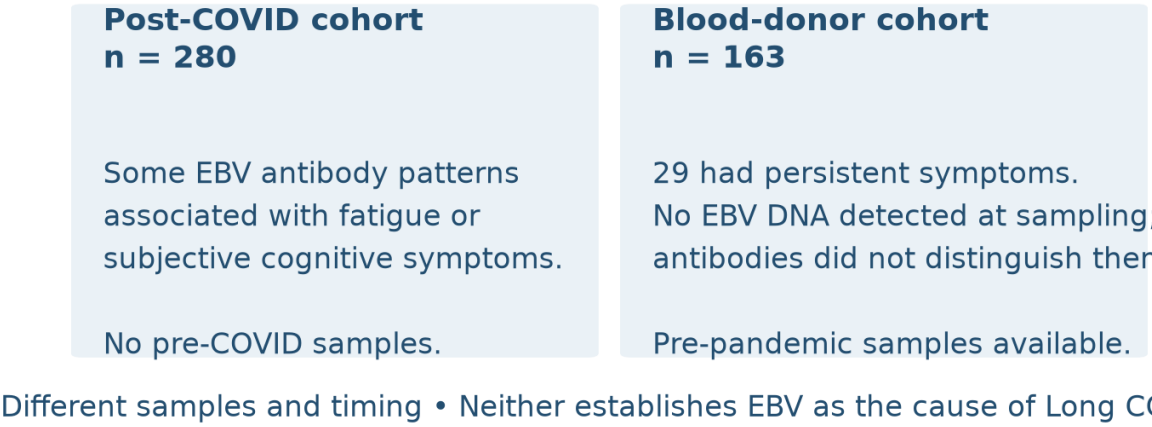


Figure 13. Evidence comparison, not a pooled estimate or demonstrated mechanism. Antibodies are indirect evidence of viral activity; the blood-donor study could miss transient or tissue-localized activity. The findings do not establish an ADHD link. [167, 168]

A different study could compare stored prepandemic samples in 163 previously EBV-infected blood donors. Among 29 with persistent post-COVID symptoms, EBV DNA was not detected at the sampled times, and antibody patterns did not distinguish the symptomatic group. Its selected, generally mild-infection population and sampling schedule leave room for transient or tissue-localized activity that blood testing could miss. Taken together, these studies do not support using EBV reactivation as a general explanation for Long COVID. [168]

The laboratory distinction matters. More than 90% of adults have evidence of past EBV infection. VCA IgG together with EBNA antibodies usually indicates past infection; high antibody levels can persist for years. Even early-antigen IgG can remain detectable in healthy people. A positive antibody result on its own cannot explain a person’s current cognitive symptoms. [159]

8.8 What mono changes in the hypothesis

Mono strengthens the case for treating recovery as something to measure. It also forces us to separate an initiating event from whatever maintains a later problem. An infection can precede persistent illness without a high blood viral load continuing to drive it. Immune changes, sleep disruption, pain, autonomic symptoms and altered activity may be candidates, but the evidence above does not identify one shared pathway or establish a histamine mechanism.

A useful study would record documented infection timing, the person’s previous level of function and ADHD history, then follow cognition, symptoms and everyday activity on separate tracks. Where feasible, it would recruit before infection and retain both recovered and persistently unwell participants. A prespecified interaction would test whether pre-existing ADHD predicts a larger change or slower recovery after accounting for baseline function and illness severity. If it does not, the claim of special ADHD sensitivity would lose support even if infection affects both groups.

This remains an observational proposal, with the existing safeguards around post-exertional symptom worsening. It does not justify provoking a crash or replacing assessment for other causes of persistent symptoms. Nothing in these studies establishes antiviral, GLP-1 or experimental-peptide treatment for ADHD or post-mono brain fog.

“Chronic active EBV disease” also needs its proper meaning. It is a distinct systemic inflammatory illness involving EBV-infected T or NK cells, with specific diagnostic requirements. It is not a synonym for prolonged fatigue after mono or a high IgG result. [171]

8.9 What a before-infection baseline and a comparison group add

UK Biobank provides an important addition to the post-COVID evidence. Researchers compared repeat brain measurements in 785 adults aged 51–81: 401 infected between visits and 384 controls. The second scan was, on average, 141 days after diagnosis in the infected group. Infection was associated with greater change in some brain measures and slower performance on the Trail Making Test between visits; findings remained after excluding 15 hospitalized cases. Pre-infection measurements reduce the chance that a pre-existing difference is mistaken for an illness effect. They do not eliminate selection or confounding, identify a viral mechanism, establish permanence, or make the findings applicable to ADHD. Imaging and cognitive analyses also have their own available samples. [174]

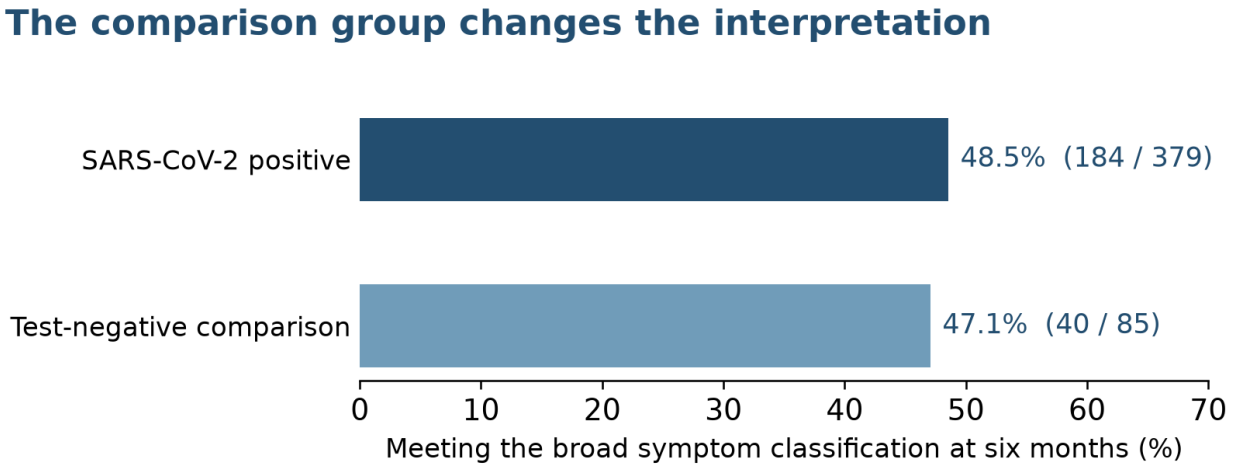


Figure 14. A broad persistent-symptom classification in nonhospitalized adolescents and young adults. The published difference was 1.5 percentage points (95% CI −10.2 to 13.1). Meeting symptom criteria does not give test-negative participants a confirmed Long COVID diagnosis. Selection, age and infection period limit generalization. [175]

A younger Norwegian cohort illustrates why comparison groups can change the answer. At six months, a broad persistent-symptom classification applied to 184 of 379 SARS-CoV-2-positive participants and 40 of 85 test-negative participants: 48.5% versus 47.1%. The published difference was 1.5 percentage points (95% CI −10.2 to 13.1). Applying symptom criteria to test-negative participants does not mean they had confirmed Long COVID. The selected controls may have had more symptoms than the general population; participants were young, nonhospitalized and mostly infected in the Alpha period. This result cannot

erase findings in other populations or prove a psychological cause. It shows how much an estimate can depend on the outcome definition and comparator. [175]

For a mono–ADHD study, I would therefore want both the person's previous function and a comparison group followed over the same period. The central question remains whether existing ADHD changes the response to infection. Similar symptoms after different illnesses make that question worth asking, but do not answer it.

8.10 Antiviral treatment is a test of a specific proposal

RECOVER-VITAL, reported in August 2026, randomized 964 adults with Long COVID; 959 entered the modified intention-to-treat analysis. Neither 15 nor 25 days of nirmatrelvir–ritonavir showed a significant primary benefit in the cognitive, autonomic or exercise symptom phenotypes. The cognitive analysis included 332 participants and used a patient-reported improvement threshold at day 90, not normalization of an objective cognitive test. Figure 42 shows its estimates and uncertainty. The comparator included ritonavir with nirmatrelvir-matching placebo. These results constrain the tested regimens; they do not disprove every viral-persistence mechanism or test EBV-directed treatment. [190]

8.11 Useful adaptation and cognitive restoration are different outcomes

RECOVER-NEURO randomized 328 adults with cognitive symptoms to five conditions over ten weeks. Computerized cognitive training, additional structured rehabilitation and active brain stimulation did not outperform their respective comparators on the primary self-reported cognitive-difficulty measure. For training versus its active comparator, the adjusted difference was 0.0 points (95% CI −0.2 to 0.2). Improvement within every group does not establish an intervention effect. More than half lacked performance-based impairment at entry, and an active comparator may itself help. [191]

CICERO asked a different question in 78 adults with objective cognitive impairment. Individualized rehabilitation improved attainment of trained personal goals versus usual care: an adjusted 2.88-point difference (95% CI 2.03–3.73) at three months and 1.72 (0.86–2.57) at six months. These were participant-reported goal ratings. Participants were unblinded, contact was not matched, and more than 40% of six-month cognitive tests were missing. Most cognitive and symptom outcomes did not differ. This supports a defined functional benefit, with limits; it does not establish general cognitive restoration. Different participants, comparators and endpoints prevent a direct ranking against RECOVER-NEURO. [192]

9. GLP-1 and other peptides in integrated health

The preceding chapters ask what changes alongside cognition. Treatment poses a harder question: if we change a target, does the relevant outcome improve? GLP-1 and other peptide studies are useful precisely because their results differ by condition and endpoint. The next three chapters move from these wider interventions to vagal stimulation and then back to established ADHD medication.

The chapter moves from proposed mechanisms to treatment results in defined illnesses, then asks what those results tell us about cognition. Dementia and trials with no cognitive benefit set useful limits. Chapter 11 covers medication used directly for ADHD.

Blood glucose, insulin signalling and HbA1c answer different questions. Glucose tracks circulating fuel availability; insulin helps regulate its use, while HbA1c summarizes glycemic exposure over preceding months. Insulin sensitivity describes responsiveness to that signal. Peripheral measures do not directly establish how the brain responds to insulin. Daily glucose excursions, long-term diabetes complications and a drug's receptor effects should therefore be tested as separate proposals. [209, 212]

9.1 How GLP-1 signalling could connect brain and immune state

GLP-1 is relevant because it allows a test of communication between metabolic signalling and other systems. Wong and colleagues found that anti-inflammatory effects in mouse Toll-like-receptor and sepsis models required central neuronal GLP-1 receptors rather than hematopoietic or endothelial receptors. Adrenergic and opioid signalling contributed to those effects. The experiment demonstrates a specific brain-to-immune route in those models; it does not establish that semaglutide improves human ADHD through that route. [52]

There is a shortcut to avoid here. An immune effect does not require assuming that every affected immune cell expresses the relevant receptor. Direct action on mast cells or microglia, indirect neural regulation, changes in body weight, and changes in other physiology are competing explanations that need different experiments.

Brain access is another distinction. In rodent tracing experiments, semaglutide reached selected brain regions through circumventricular and ventricular-adjacent sites, without broad passage across the BBB. Neural activity also changed in regions without direct drug access. A drug can therefore influence brain circuits through selective access and relays. This does not mean every GLP-1 drug has the same distribution, or that it repairs barrier leakage in humans. [211]

Immune effects can also involve cells other than mast cells. In an aspirin-exacerbated respiratory disease study, liraglutide reduced selected activation responses in platelets sampled from 31 affected participants and 11 controls. The drug was applied to samples outside the body; airway experiments were in mice. This is useful evidence of a possible peripheral route, but it is **not a clinical** asthma, MCAS or cognitive treatment result. [203]

9.2 Human studies measure different outcomes

A 47-person uncontrolled MCAS case series reported clinical benefit in 89% after GLP-1 receptor agonist treatment. That is the proportion reported to benefit in this selected series, not an 89% reduction in symptoms or a controlled response rate. Variable treatments, co-interventions, expectation and the absence of a comparison group prevent causal inference. The series is a lead for a controlled trial; it cannot demonstrate direct mast-cell stabilization, BBB repair or an ADHD benefit. [53]

A separate placebo-controlled study of 48 adults with alcohol use disorder found that semaglutide changed laboratory drinking, craving and some—but not all—drinking outcomes over nine weeks. This supplies human evidence of behavioural effects relevant to reward, but it did not directly measure dopamine or test executive function in ADHD. Reduced craving and improved attention are not interchangeable outcomes. [54] Larger and newer alcohol-use trials are discussed in Section 9.6.

The proposed ADHD link remains conditional: a metabolic intervention might affect cognition directly, through symptoms and sleep, through food intake or another physiological route, or not at all. Body-weight change should be measured rather than dismissed. A nonsignificant weight change or a statistical adjustment for weight alone does not establish a weight-independent neural mechanism.

The stronger human GLP-1 evidence is outcome-specific. SELECT randomized 17,604 adults with overweight or obesity and established cardiovascular disease, without diabetes. Major cardiovascular events occurred in 6.5% on semaglutide and 8.0% on placebo (HR 0.80, 95% CI 0.72–0.90). That is vascular benefit in a high-risk population; attention and immune mediation were not established. Adverse-event discontinuation was more frequent, 16.6% versus 8.2%. [73]

REWIND tested a closer cognitive question in type 2 diabetes. Among 8,828 participants with baseline and follow-up testing, the exploratory cognitive-impairment outcome gave HR 0.93 (95% CI 0.85–1.02). A post-hoc adjustment for baseline cognitive scores gave 0.86 (0.79–0.95). Reporting only “14% lower risk” would hide that analytic distinction. The endpoint was a low MoCA or processing-speed score, not an incident dementia diagnosis or ADHD improvement. [72]

GRADE provides a larger cognitive comparison in type 2 diabetes. Of 5,047 randomized participants, 3,721 had baseline and year-four cognitive assessments. Adding liraglutide, insulin glargine, glimepiride or sitagliptin to metformin produced **no significant between-group cognitive differences**. Higher average HbA1c was associated with poorer performance, but that association was observational within the trial; glucose targets were not randomized. Open-label treatment, follow-up selection, rescue insulin and modest glycemic separation limit the inference. The study did not demonstrate cognitive superiority for liraglutide in this population. [209]

The clinical benefits in the following sections remain valuable on their own endpoints. For this paper, a GLP-1 mechanism becomes more convincing if changes in the proposed mediator precede a cognitive change, the result survives appropriate comparisons, and simpler explanations are measured. Weight loss, improved sleep-disordered breathing, reduced symptoms and direct neural signalling are possibilities to distinguish, not competing stories we can settle from an individual treatment response.

9.3 Metabolic benefit during psychiatric treatment

HISTORI studied 154 antipsychotic-treated adults with schizophrenia, prediabetes and overweight or obesity. At 30 weeks, semaglutide reduced HbA1c by 0.46 percentage points (95% CI −0.56 to −0.36) and weight by 9.21 kg (−11.68 to −6.75) relative to placebo. Physical quality of life improved by 3.75 SF-36v2 points (1.52–5.98). Average schizophrenia symptom ratings and mental quality of life did not show significant benefit; cognition was not an efficacy endpoint. Gastrointestinal symptoms were more frequent, and repeated hospitalizations occurred in a few treated participants. Metabolic care can matter during psychiatric treatment without proving a cognitive or immune mechanism. [189]

9.4 Where treatment of overlapping illness has stronger evidence

For each treatment, we need to ask which illness improved, in whom and on what measure. GLP-1 receptor agonists such as semaglutide and the dual GIP/GLP-1 agonist tirzepatide now have substantial human evidence outside glucose control. These results are relevant because sleep, pain, circulation and illness burden could affect a person's available capacity. That last connection remains a hypothesis unless cognition and daily function are measured.

Sleep apnea. SURMOUNT-OSA comprised two placebo-controlled trials totaling 469 adults with obesity and moderate-to-severe obstructive sleep apnea. At 52 weeks, tirzepatide reduced breathing interruptions substantially more than placebo (Figure 15). One trial enrolled people without positive airway pressure treatment; the other enrolled people receiving it. For the latter, pressure treatment was withheld before endpoint assessment under the study protocol. This does not justify stopping prescribed treatment. Gastrointestinal adverse effects were common. Improved sleep-related reports are relevant, but the trials do not establish an ADHD cognitive effect. [113]

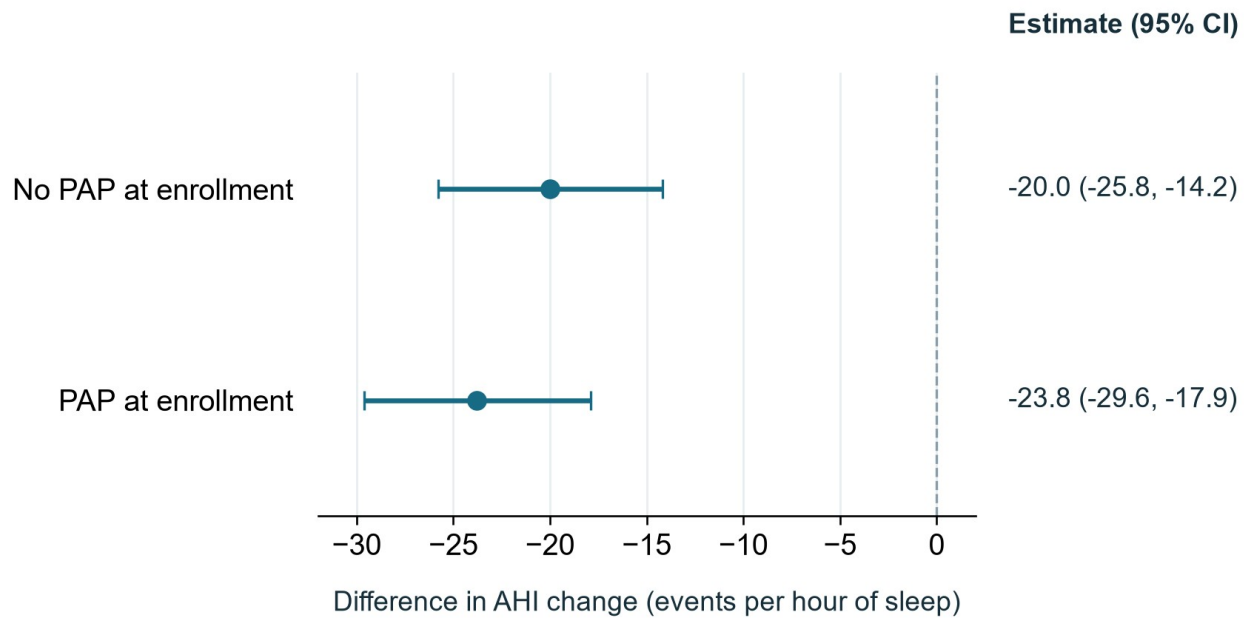


Figure 15. Tirzepatide minus placebo at 52 weeks, with 95% CIs. Lower favours tirzepatide. AHI counts breathing interruptions; PAP means positive airway pressure. No pooling. [113]

Heart failure. In STEP-HFpEF, 529 people with obesity and heart failure with preserved ejection fraction were randomized for 52 weeks. Semaglutide improved the Kansas City Cardiomyopathy Questionnaire clinical summary score by an additional 7.8 points (95% CI 4.8–10.9) on its 0–100 scale, and six-minute walking distance by 20.3 metres (8.6–32.1). These are symptoms, physical limitations and exercise capacity, not attention tests. [114]

SUMMIT tested a harder clinical outcome in 731 people with the same broad heart-failure phenotype and obesity. Over a median 104 weeks, cardiovascular death or worsening heart failure occurred in 36/364 participants on tirzepatide versus 56/367 on placebo: 9.9% versus 15.3%, HR 0.62 (95% CI 0.41–0.95). The advantage was driven by fewer worsening-heart-failure events; cardiovascular deaths were 8 versus 5, with wide uncertainty. Calling this a demonstrated mortality benefit would be wrong. [115]

Pain and mobility. STEP 9 randomized 407 people with obesity and knee osteoarthritis for 68 weeks. WOMAC pain scores, on a 0–100 scale, fell by 41.7 points with semaglutide versus 27.5 with placebo; weight fell by 13.7% versus 3.2%. A meaningful pain result can coexist with uncertainty about mechanism. These data do not demonstrate cartilage regeneration, an effect independent of weight loss, or improved executive function. [116]

Kidney disease. FLOW randomized 3,533 people with type 2 diabetes and chronic kidney disease. Over a median 3.4 years, its primary composite occurred in 331/1,767 on semaglutide and 410/1,766 on placebo, HR 0.76 (95% CI 0.66–0.88). The endpoint combined kidney failure, sustained major filtration loss, and kidney or cardiovascular death. It is not a 24-percentage-point reduction in kidney failure alone. This is strong organ-outcome evidence in a defined population, without a demonstrated cognitive mediator. [117]

9.5 Liver and immune disease sharpen the mechanism question

ESSENCE provides tissue-level evidence in metabolic dysfunction-associated steatohepatitis, or MASH: liver inflammation and injury associated with metabolic dysfunction. Its planned 72-week interim analysis covered the first 800 of 1,197 randomized participants with moderate or advanced fibrosis. Figure 16 shows two biopsy outcomes favouring semaglutide. These are histological responses, not yet proof of preventing cirrhosis-related death or improving cognition. Gastrointestinal events were more frequent, and the bodily-pain outcome did not show a significant between-group difference. A positive liver result is not a general symptom result. [118]

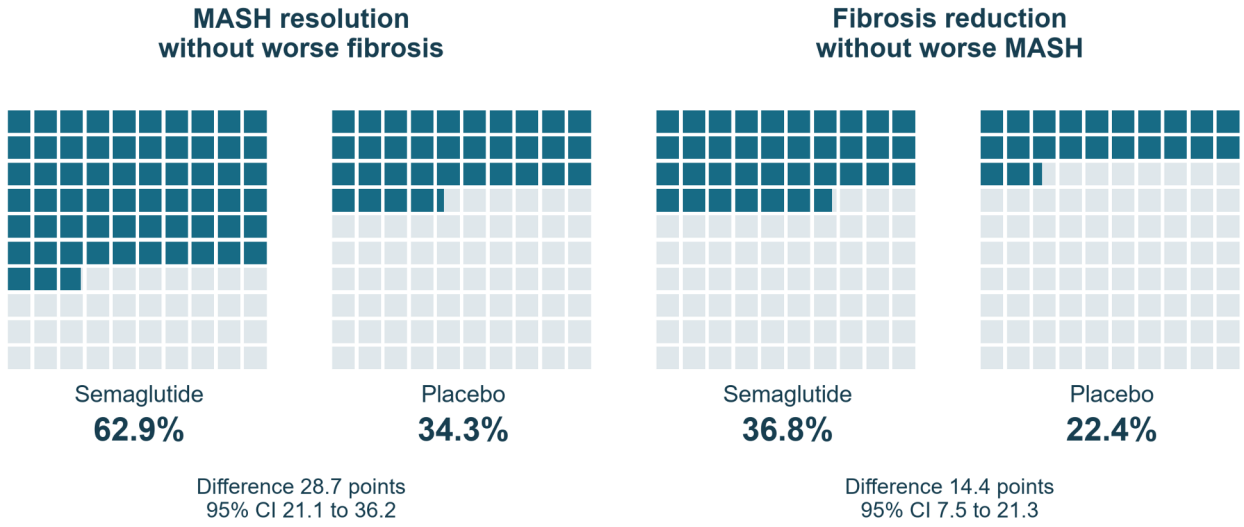


Figure 16. ESSENCE liver-biopsy responses at 72 weeks. Each grid represents 100%; one square is one percentage point, with partial squares for decimals. Blue shows the reported response proportion; grey is its complement. Squares are not individual participants. The two outcomes overlap and must not be added. First 800 randomized: semaglutide n=534, placebo n=266. Published differences and 95% CIs are retained. [118]

The smaller SYNERGY-NASH trial gives a related signal with tirzepatide. Among 190 randomized participants, 157 had evaluable week-52 biopsies; missing outcomes were handled by imputation. MASH resolution without worsening fibrosis occurred in 44–62% across the three tirzepatide groups versus 10% on placebo. This phase 2 study supports further testing. Its numbers cannot be used as a head-to-head comparison with ESSENCE because the trials differ. [119]

Psoriasis offers a direct overlap experiment. TOGETHER-PsO randomized 274 adults with psoriasis and overweight or obesity to the immune-targeted antibody ixekizumab, with or without tirzepatide. Treatment was open-label, but skin assessors were masked. At week 36, adding tirzepatide improved the joint skin-clearance/weight-loss primary endpoint and the key secondary skin-clearance endpoint (Figure 17). Nausea occurred in 24.6% versus 2.2%. This demonstrates added benefit in that treatment setting. It cannot separate immune signalling from weight change, drug exposure or other routes, and it did not test cognition. [141]

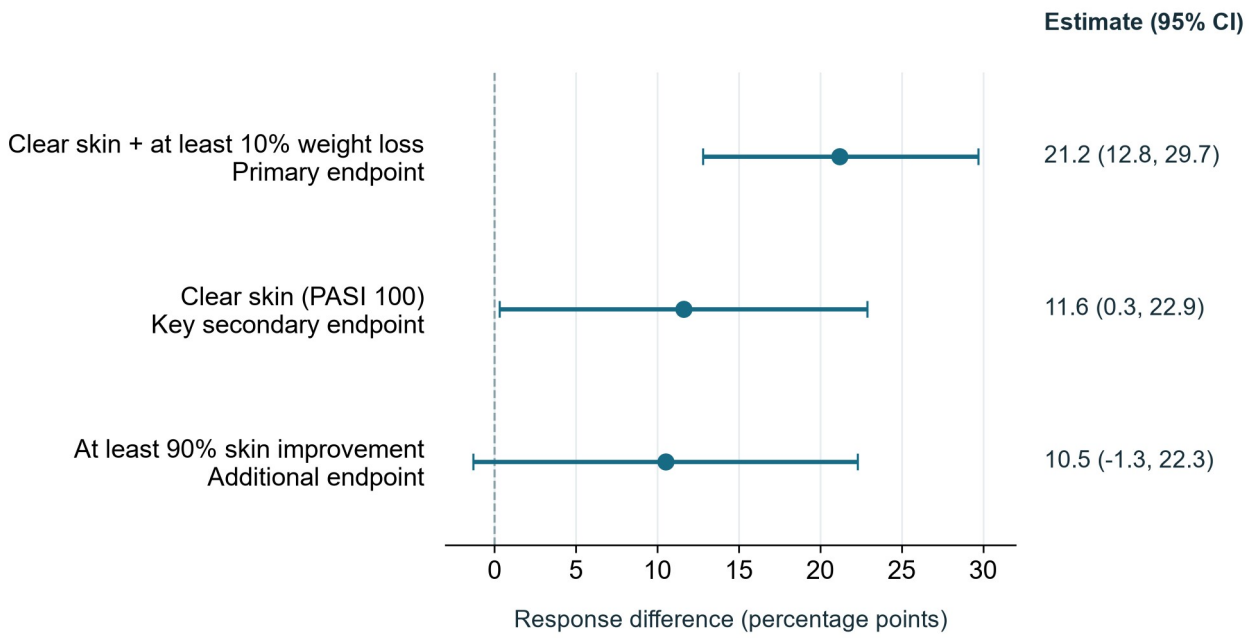


Figure 17. Combination minus ixekizumab alone; higher favours combination. Published 36-week contrasts and 95% CIs. The model estimates outcomes without discontinuation or prohibited treatment, imputing missing data. The additional endpoint lacks multiplicity control. [141]

Rheumatoid arthritis evidence is less secure. A retrospective study compared 173 people who took a GLP-1 drug with 42 who were prescribed one but did not start. Pain changed by −0.6 versus +1.3 cm on a visual scale. Access, adherence, other treatments and baseline differences can affect that comparison; roughly one-third of treated patients discontinued. A randomized trial would be needed to test whether the drug modifies the disease. [127]

The inflammatory bowel disease findings are also mixed. One electronic-record study reported lower five-year odds of surgery among GLP-1 users, OR 0.36 (95% CI 0.29–0.46). [128] A separate 2026 target-trial emulation in stable IBD found relapse rates of 7.9% in both groups in its first matched cohort, RR 1.01 (0.82–1.24). Its advanced-therapy cohort was also inconclusive, RR 0.80 (0.55–1.15). [129] The populations, outcomes and follow-up differ. An odds ratio for surgery and a risk ratio for relapse should not be pooled or treated as direct contradictions. Neither study randomized treatment; “emulation” describes the design, not removal of confounding.

9.6 Reward and headache outcomes need their own tests

The alcohol-use evidence has progressed beyond the small trial in Section 9.2. A 2026 trial randomized 108 adults with alcohol use disorder and obesity to semaglutide or placebo, with cognitive behavioural therapy in both groups. Over 26 weeks, the reported treatment difference in percentage of heavy-drinking days was −13.7 percentage points (95% CI −22.0 to −5.4); 88 participants completed treatment. Gastrointestinal effects were more common. This is a clinically relevant behavioural result in a specific population. It does not measure dopamine restoration, executive function or ADHD treatment response. [120]

A separate eight-week oral-semaglutide trial randomized 50 adults. Its primary laboratory cue-induced craving outcome was not significantly different, while the key secondary heavy-drinking-days outcome favoured treatment; drinks per day did not clearly differ. Reporting only the positive measure would hide the outcome hierarchy. Together, the studies support further alcohol-use treatment research without establishing one uniform effect across formulations, populations and outcomes. [121]

Headache evidence is earlier. In an uncontrolled 31-person pilot involving obesity and frequent or chronic migraine, headache days fell from 19.8 to 10.7 per month after 12 weeks of added liraglutide. Without a control group, expectation, other treatment and regression to the mean remain plausible contributors. A nonsignificant weight change does not establish a weight-independent mechanism. [122]

Idiopathic intracranial hypertension is a distinct condition. A small exenatide trial randomized 15 women and found lower intracranial pressure at 24 hours: difference −4.7 mm Hg (95% CI −8.8 to −0.5). At 12 weeks the interval crossed zero, −4.1 (−8.4 to 0.1); the study had prespecified a more permissive significance threshold of 0.10. This is an early pressure result, not a general migraine trial. [123] The related exploratory cognitive analysis found no significant between-group cognitive advantage. Improvements within the treated group alone cannot supply that missing comparison. [124]

9.7 Endometriosis and Long COVID remain open questions

There is a recent endometriosis signal, but its design matters. A February 2026 preprint analyzed 161 survey participants with confirmed or suspected endometriosis who had used GLP-1 therapy. Retrospective reports of symptom improvement are vulnerable to selective participation, recall, co-treatment and changes in weight or menstrual state. They do not establish lesion regression, a shared cause with ADHD, or cognitive benefit. A prospective controlled study should separate pelvic pain, bleeding, sleep, fatigue and attention rather than combine them into “feeling better.” [130]

For Long COVID, a tirzepatide trial had enrolled 1,058 participants in an August 2026 preprint. The report describes recruitment, baseline characteristics and the operation of a remote trial; it does not report treatment efficacy. Forecasts of improvement are not observed treatment effects. As of this review, that report cannot support a claim that tirzepatide treats Long COVID fatigue, autonomic dysfunction, blood–brain barrier disruption or mitochondrial impairment. Those are separate outcomes needing results. [131]

9.8 Other peptides are different treatments

“Peptide” describes how a molecule is built. It does not identify a shared therapeutic target, benefit or safety profile. An oral IL-23 receptor antagonist, a GLP-2 analogue and a GnRH agonist ask different clinical questions. Table 3 includes positive disease-specific examples alongside direct brain-related trials that did not confirm benefit. Antibodies such as ixekizumab are identified separately as protein biologics; sharing an immune target does not make treatments interchangeable.

Table 3. Different peptide targets and their measured outcomes

Treatment and target	Population and comparison	Measured result and boundary
Icotrokinra IL-23 receptor peptide [132]	Plaque psoriasis; 684 randomized; placebo comparison at 16 weeks.	Clear/almost-clear skin with ≥2-grade improvement: 65% vs 8%. ≥90% severity improvement: 50% vs 4%. Skin endpoints; no cognitive test.
Teduglutide GLP-2 analogue [134]	Short bowel with intestinal failure; 86 randomized; 24 weeks.	Weekly intravenous support volume reduced >20% at weeks 20 and 24: 27/43 (63%) vs 13/43 (30%). Not general gut repair.
Leuprolide GnRH agonist [140]	Endometriosis; 201 randomized to add-back regimens; everyone received leuprolide.	Pain improved from baseline. Randomization tests add-back, not leuprolide versus placebo. Bone-density trade-off discussed below.
Intranasal insulin [135]	Mild cognitive impairment/Alzheimer dementia; 289 randomized; primary device cohort 240; 12 months.	ADAS-cog12 cognitive score difference 0.026 points (95% CI −1.771 to 1.822): no primary advantage. Delivery-device problems limit generalization.
Intranasal oxytocin [136]	Autism; 290 enrolled, ages 3–17; placebo comparison; 24 weeks.	Modified social-withdrawal score difference −0.2 (95% CI −1.5 to 1.0). No primary benefit; this was not an ADHD trial.

Icotrokinra is particularly useful here because it is a targeted oral peptide with a positive immune-disease trial, rather than a general “immune support” claim. The FDA approved it in March 2026 for eligible adults and adolescents with moderate-to-severe plaque psoriasis. That is a US regulatory indication, not approval for ADHD or cognitive symptoms. [133]

The endometriosis example also makes the trade-off concrete. In the leuprolide study, everyone received the GnRH agonist and randomization concerned hormonal add-back. Pain improved from baseline across groups, but this was not a placebo-controlled estimate of leuprolide's pain effect. After a year, the group without add-back lost 6.3% of bone density; add-back groups preserved it. The hormone state created by treatment belongs in the model alongside any symptom relief. Cognition was not measured. [140]

9.9 Negative trials and research peptides set limits

Neurological plausibility has not produced a consistent GLP-1 treatment effect. A 156-person phase 2 Parkinson's trial of lixisenatide found a favourable motor difference at 12 months, 3.08 points (95% CI 0.86–5.30). [125] A separate 194-person phase 3 exenatide trial did not confirm benefit on its 96-week motor endpoint: adjusted difference +0.92 points (−1.56 to 3.39). [126] Different drugs and on/off-medication assessments prevent a direct numerical

comparison. Neither motor result is a cognitive outcome. The negative Alzheimer trials in Section 9.11 remain a further boundary.

The same standard applies to products sold as research peptides. FDA staff's 2026 BPC-157 review identified a small ulcerative-colitis trial reported only as a meeting abstract and several limited human reports. Human reports exist, but their designs and reporting do not establish the broad claims made for systemic treatment. [137]

For the nominated TB-500 fragment and MOTS-c products, the FDA reviews identified no clinical studies involving administration to humans. Evidence about full-length thymosin beta-4 cannot simply be assigned to the shorter TB-500 fragment. Measurements of naturally occurring MOTS-c, animal experiments and studies of modified analogues do not establish benefit from injected MOTS-c. These were staff evidence reviews for a compounding advisory process, not efficacy trials or final approval decisions. [138, 139]

9.10 Where mitochondria, mTOR, AMPK and glia belong

Immunometabolism means that cellular energy management and immune behaviour interact. A TREM2 study combined human Alzheimer tissue observations with cellular and mouse experiments, linking impaired microglial metabolic function to defective mTOR signalling and altered responses to pathology. It demonstrates a cell-specific immune–metabolic relationship; it does not establish the same defect in ADHD. [55]

The earlier mTOR/AMPK proposals fit here as candidate regulatory mechanisms, not as a master switch that should always be turned up or down. The study above is a reminder that reducing one signalling pathway is not inherently beneficial. Cell type, timing and the process being measured matter. Peripheral mitochondrial markers or muscle findings should not be relabelled as direct measurements of neuronal energy supply.

A metabolic measure needs to do more than look abnormal alongside the symptoms. It should predict the relevant cognitive or functional outcome and add something beyond simpler explanations.

9.11 Dementia is a different timescale, not the endpoint of ADHD fog

The TREM2 work makes glial and metabolic biology relevant to neurodegeneration, but that is not a disease progression from ADHD through Long COVID to Alzheimer’s. Mechanistic overlap does not establish that a fluctuating symptom is an early form of a progressive disease. [55]

In 3,808 participants across evoke and evoke+, oral semaglutide did not slow early symptomatic Alzheimer disease. At 104 weeks, differences in change on the Clinical Dementia Rating–Sum of Boxes were −0.08 points (95% CI −0.35 to 0.20) and +0.10 (−0.17 to 0.38), respectively. Lower values favor semaglutide; both intervals span zero. Biomarker improvement cannot replace this clinical outcome. These trials do not settle prevention in other populations, but their null results are not positive evidence for this hypothesis. [56]

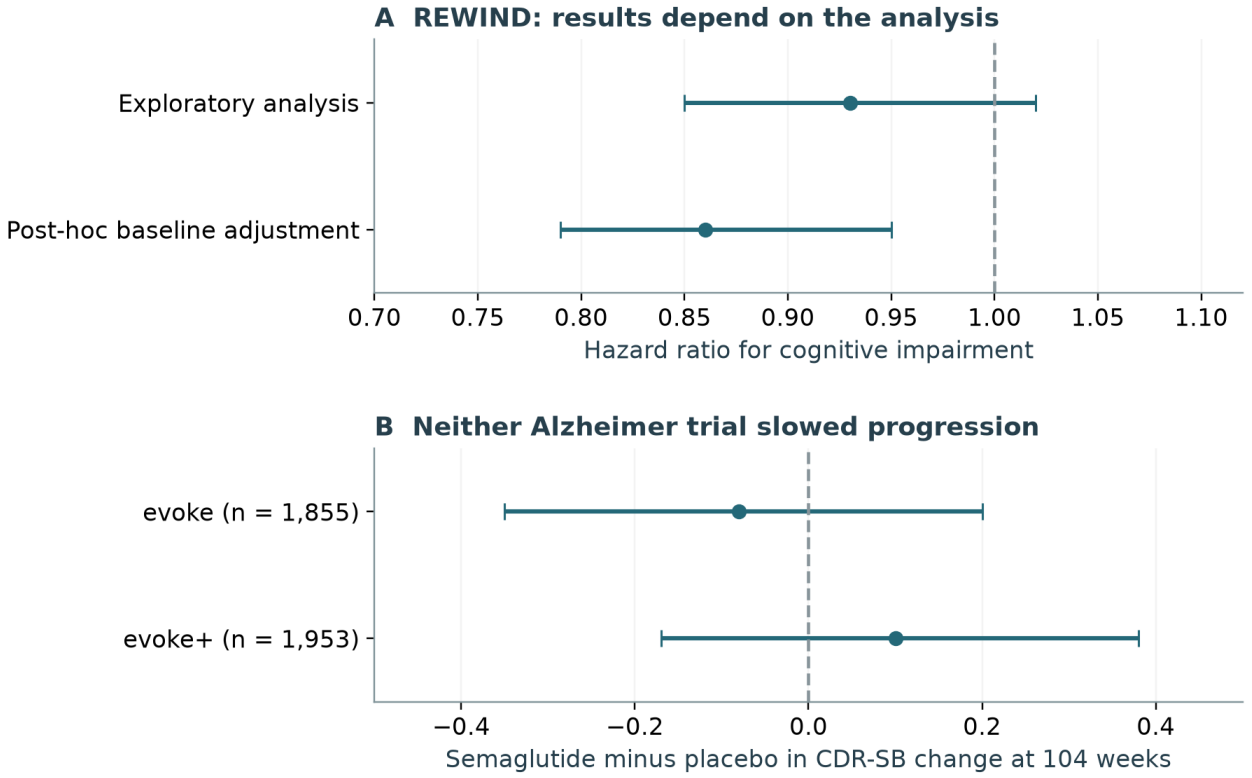


Figure 18. Cognitive outcomes do not follow automatically from a plausible mechanism. Panel A: two analyses of the same exploratory REWIND outcome, not independent trials. Panel B: two separate Alzheimer trials. Bars are 95% confidence intervals; lower values favor active treatment. The panels use different populations, drugs, endpoints and scales and must not be pooled. CDR-SB measures dementia severity through cognition and function. [56, 72]

The ELAD trial gives a mixed liraglutide result. Among 204 randomized participants with mild-to-moderate Alzheimer disease and no diabetes, the 52-week primary brain glucose-metabolism endpoint did not differ significantly from placebo: −0.17 in PET standardized uptake value (95% CI −0.39 to 0.06). A secondary cognitive composite favored liraglutide by 0.15 z-score units (0.03–0.28), but the secondary analyses were hypothesis-generating, without confirmatory multiplicity control. Daily-living and global clinical ratings did not show significant differences. Only 154 participants had the primary follow-up scans. The cognitive signal deserves further testing; it is not a 15% improvement, an established disease-modifying effect or ADHD evidence. [188]

9.12 Integrated health interventions need a measured cognitive endpoint

US POINTER provides a direct test of a broad lifestyle package in a different population: 2,111 adults aged 60–79 at risk of cognitive decline. Both groups received lifestyle guidance; the structured programme added support and intensity across exercise, diet, cognitive and social activity, and cardiovascular monitoring. Global cognition improved by an additional 0.029 SD per year (95% CI 0.008–0.050) over two years. Figure 19 shows that the domain estimates were not uniform. The gain is modest, and the package cannot identify which component caused it. It does not establish ADHD benefit or dementia prevention. [97]

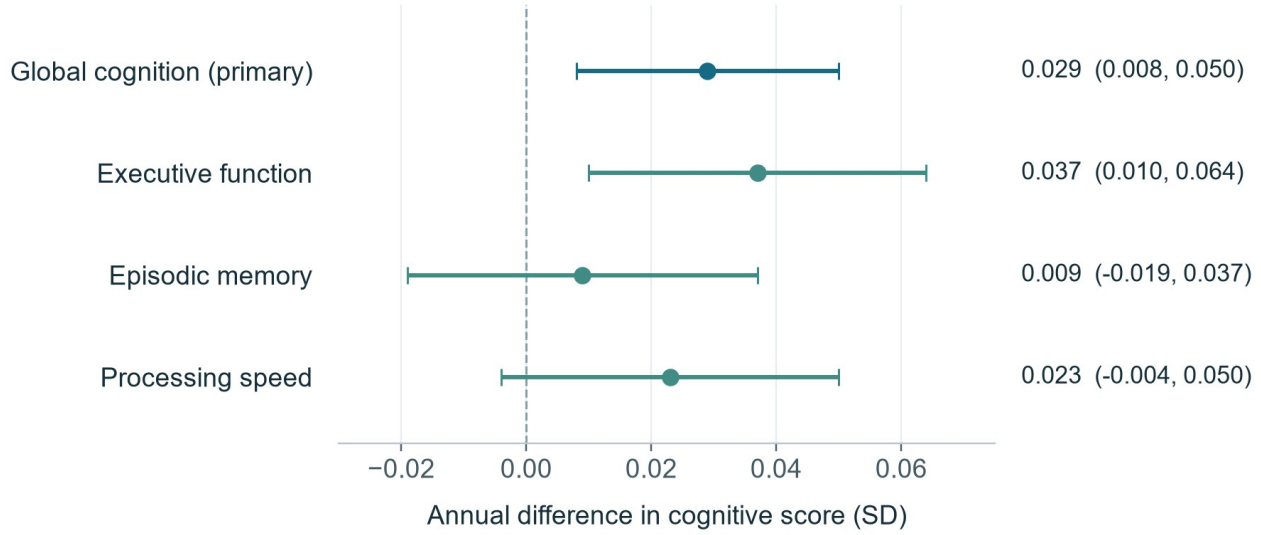


Figure 19. A small average cognitive gain from a structured lifestyle programme. US POINTER Table 2: annual differences versus self-guided intervention, with 95% CIs. Global cognition was primary; the domains are related secondary outcomes from the same participants. Higher values favour the structured programme. Memory and processing-speed intervals include zero. Everyday clinical meaning and durability remain separate questions. [97]

Vascular treatment supplies another boundary. SPRINT randomized 9,361 adults aged 50 or older with hypertension and elevated cardiovascular risk, excluding diabetes and prior stroke. Assigned blood-pressure treatment lasted 3.3 years. After a median seven years of cognitive follow-up, intensive versus standard treatment gave HR 0.89 (95% CI 0.79–0.99) for mild cognitive impairment or probable dementia, but 0.86 (0.72–1.02) for probable dementia alone. Extended ascertainment reached 4,232 of 7,221 eligible participants. This supports a specified vascular intervention and outcome, while leaving attrition and dementia-specific uncertainty visible. It cannot be carried over to normal blood pressure, orthostatic intolerance or an ADHD treatment target. [98]

9.13 A corrected health measure may not produce a cognitive benefit

Vitamin B12 provides a useful example. A trial's eligible cohort included 201 adults aged 75 or older with moderate biochemical deficiency, but without anemia or neurological or cognitive symptoms. Daily B12 for 12 months improved vitamin status. It did not improve the main cognitive measure: the adjusted difference on the first three learning trials of the California Verbal Learning Test was −1.4 words (95% CI −2.9 to 0.1), analyzed in 91 treated participants and 93 controls. Cognitive outcomes were secondary to the trial's neurological outcome. This does not argue against treating severe or symptomatic deficiency. It shows why a corrected blood result cannot stand in for a demonstrated cognitive benefit. [156]

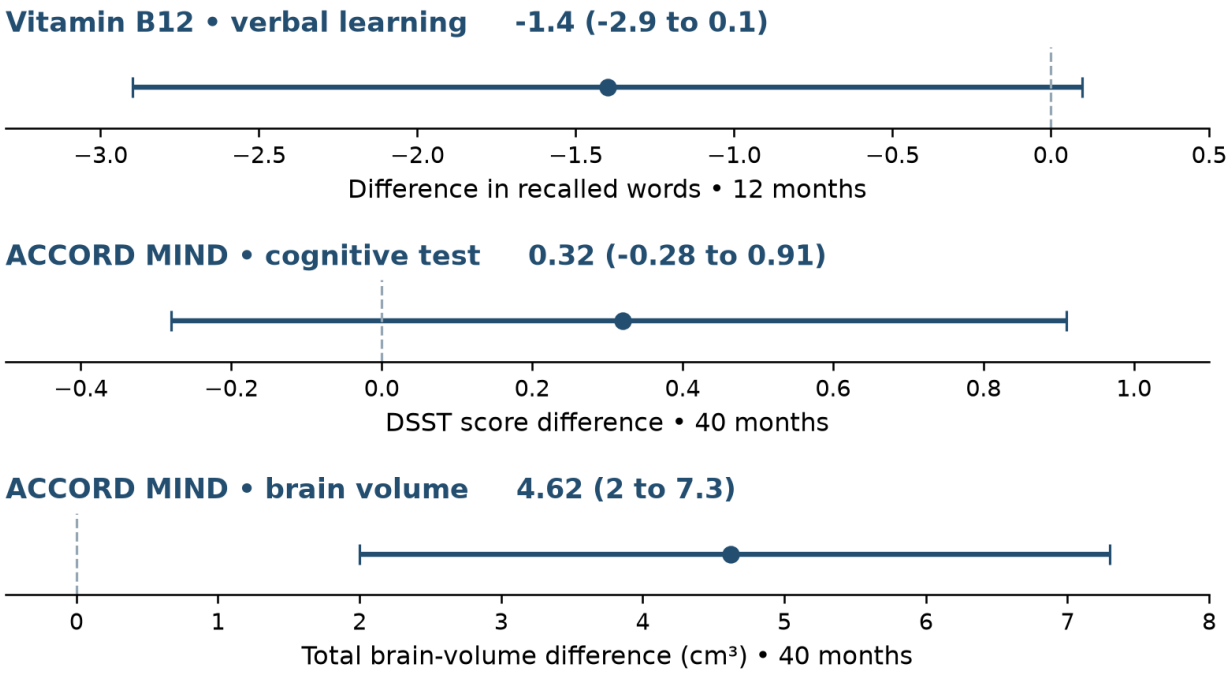


Figure 20. Published between-group estimates with 95% CIs, on separate axes and in different units. B12 verbal learning: 184 analyzed. ACCORD cognitive follow-up: 2,794; MRI subset: 503. The two cognitive intervals cross zero; the brain-volume interval does not. Do not pool these outcomes. The intensive glucose strategy was stopped for increased mortality in the parent trial. [156, 157]

ACCORD MIND makes a related point about glucose and brain imaging. It enrolled 2,977 adults with type 2 diabetes and high cardiovascular risk. Targeting HbA1c below 6%, versus 7.0–7.9%, did not improve the principal cognitive outcome, the Digit Symbol Substitution Test, at 40 months: the adjusted difference was 0.32 points (95% CI −0.28 to 0.91). The cognitive analysis included 2,794 people with follow-up. An MRI subset of 503 nevertheless showed 4.62 cm³ more total brain volume with intensive treatment (95% CI 2.00–7.30). The intensive strategy was stopped because of increased mortality in the main trial. These results concern that treatment strategy and population; they do not test modern GLP-1 therapy or ADHD. [157]

For this paper, the lesson is practical. If a treatment changes a laboratory value or an image, we still need to ask whether thinking, symptoms and everyday function changed. A null cognitive result should remain in the account, even when another outcome looks encouraging.

9.14 Hearing and access to the task

The ACHIEVE trial randomized 977 adults aged 70–84 with untreated hearing loss to hearing intervention or health education. The overall three-year cognitive result was null. A prespecified analysis found less decline in the 238 participants recruited from the Atherosclerosis Risk in Communities (ARIC) cohort, but no demonstrated benefit in the other 739 volunteers. The groups differed in baseline risk, and the interaction was statistically significant. Figure 36 shows the estimates together. The often-quoted 48% slowing refers to

cognitive-score decline in the ARIC subgroup, not a reduction in dementia risk across the trial. [155]

I would also record whether participants can comfortably hear and see the task. Difficulty receiving an instruction can affect a score before we learn anything about attention or immune signalling. This is a measurement safeguard in the proposed protocol, not a claim that hearing treatment treats ADHD.

9.15 What the treatment results mean

The trials report benefits on specific sleep, cardiac, renal, liver, pain and skin outcomes in the populations studied. Some behavioural outcomes are promising too. The route from those changes to attention remains largely unmeasured in these studies. A person can benefit from treating an overlapping illness even if it does not explain their ADHD.

A useful test would follow the sequence. Among people starting clinically indicated treatment, measure the targeted illness, medication exposure, sleep, food intake, hydration, pain and objective cognition repeatedly, beginning before treatment. Keep subjective fog and daily functioning as separate outcomes. Where feasible, randomization or an active comparator producing similar weight loss would help distinguish a drug-specific effect from weight-related and contextual changes. Statistical adjustment alone cannot settle mediation.

If the bodily endpoint improves but cognition does not, that weakens the proposed cognitive route for that population, outcome and time window. If improvement follows pain or sleep but not the proposed immune marker, the simpler route deserves priority. Benefits and adverse effects may occur together. These findings support focused studies; they do not supply a peptide treatment protocol for ADHD.

10. Vagal signalling: what human interventions establish

Metabolic drugs are one way to investigate communication between body systems. Vagal stimulation tests a different route: changing neural input. It offers unusually direct human intervention evidence for an immune disease, but the step from that result to ADHD cognition still has to be tested.

10.1 Neural control can change immune disease in humans

In RESET-RA, 242 people with inadequately controlled rheumatoid arthritis received an implanted cervical vagus-targeted system and were randomized to active or sham stimulation for three months. ACR20 arthritis response occurred in 35.2% versus 24.2%: a raw difference of 11.0 percentage points. The stratification-adjusted difference was 11.8 points (95% CI 0.6–23.1), with a prespecified one-sided primary test. Device-related serious events occurred in 1.6% and were perioperative. Later follow-up was open-label. This is human immune-disease intervention evidence, not cognitive benefit. [57]

The arthritis result does not demonstrate cognitive benefit or ADHD treatment. It also cannot establish equivalence with ear stimulation or the safety of a repurposed TENS unit.

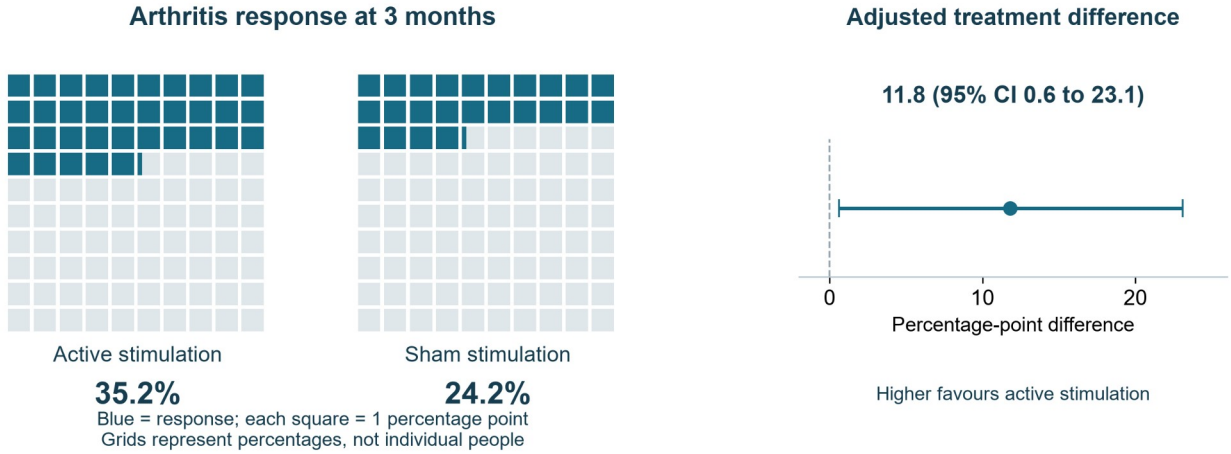


Figure 21. An implanted vagal intervention changed an arthritis outcome. RESET-RA, 242 participants: ACR20 response at three months. The observed proportions differ by 11.0 percentage points; the published stratification-adjusted contrast is 11.8 points (95% CI 0.6–23.1). The primary test was one-sided. This is immune-disease intervention evidence, not cognitive benefit or equivalence with ear stimulation. [57]

10.2 The cognition and HRV findings answer different questions

A 2026 review included 53 auricular vagus-stimulation studies, 30 in quantitative synthesis. The pooled cognitive effect was Hedges g = 0.41 (95% CI 0.30–0.53); working-memory/attention studies gave a smaller estimate, 0.185 (0.043–0.327). This follows the abstract and main results; the figure description gives a lower limit of 0.29. The populations, protocols and outcomes varied. These are standardized differences, not percentages improved, and neither estimate is ADHD-specific efficacy or proof of an anti-inflammatory route. [58]

A separate Bayesian synthesis of 16 sham-controlled studies in healthy participants found essentially no acute effect on cardiac vagally mediated HRV (g = 0.014; shortest credible interval −0.103 to 0.132). This is a Bayesian credible interval, not a conventional confidence interval. The result limits HRV as a check that auricular stimulation engaged its intended target; it does not show that every neural effect is absent. [59]

A vagal extension should therefore measure the clinical or cognitive target directly, use an appropriate sham and assess masking, adverse effects and target engagement. A rise in HRV is neither a substitute for improved cognition nor proof that inflammation has fallen. Absence of an HRV rise should not automatically erase an independently measured effect either.

11. Medications are part of the physiological state

After the wider intervention evidence, we return to the treatment benchmark closest to the paper's question: ADHD medication. These studies show why symptom relief, cognitive performance, learning and daily health outcomes must be kept separate. They also make treatment timing part of the state we are trying to understand.

Medication also affects how we interpret a difficult day. Record treatment, timing and physiological effects alongside symptoms so that a medication change is not mistaken for another process. [76, 77]

11.1 Benefit depends on what we measure

A recent crossover experiment makes the outcome distinction concrete. The main analysis included 57 adults with ADHD and 60 controls. In participants with ADHD, methylphenidate reduced response-time variability on a cost–benefit task by 0.03 seconds (95% credible interval −0.05 to −0.01 for treatment minus placebo). The estimated accuracy gain was 1 percentage point, with an interval from −1 to 3. These are Bayesian credible intervals. Participants with ADHD were already prescribed methylphenidate; this acute task experiment does not establish durable improvement in learning or everyday function. [196]

A 2018 network meta-analysis included 133 double-blind trials. At roughly 12 weeks, clinician-rated symptoms improved with amphetamines, methylphenidate and atomoxetine in both age groups, with different average estimates in children and adults. A newer adult synthesis, spanning 113 trials across several treatment types, also supported short-term symptom benefit for stimulants and atomoxetine. Table C5 gives the estimates, populations and uncertainty. These reviews share some underlying trials; they are not **independent replication**s. [76, 77]

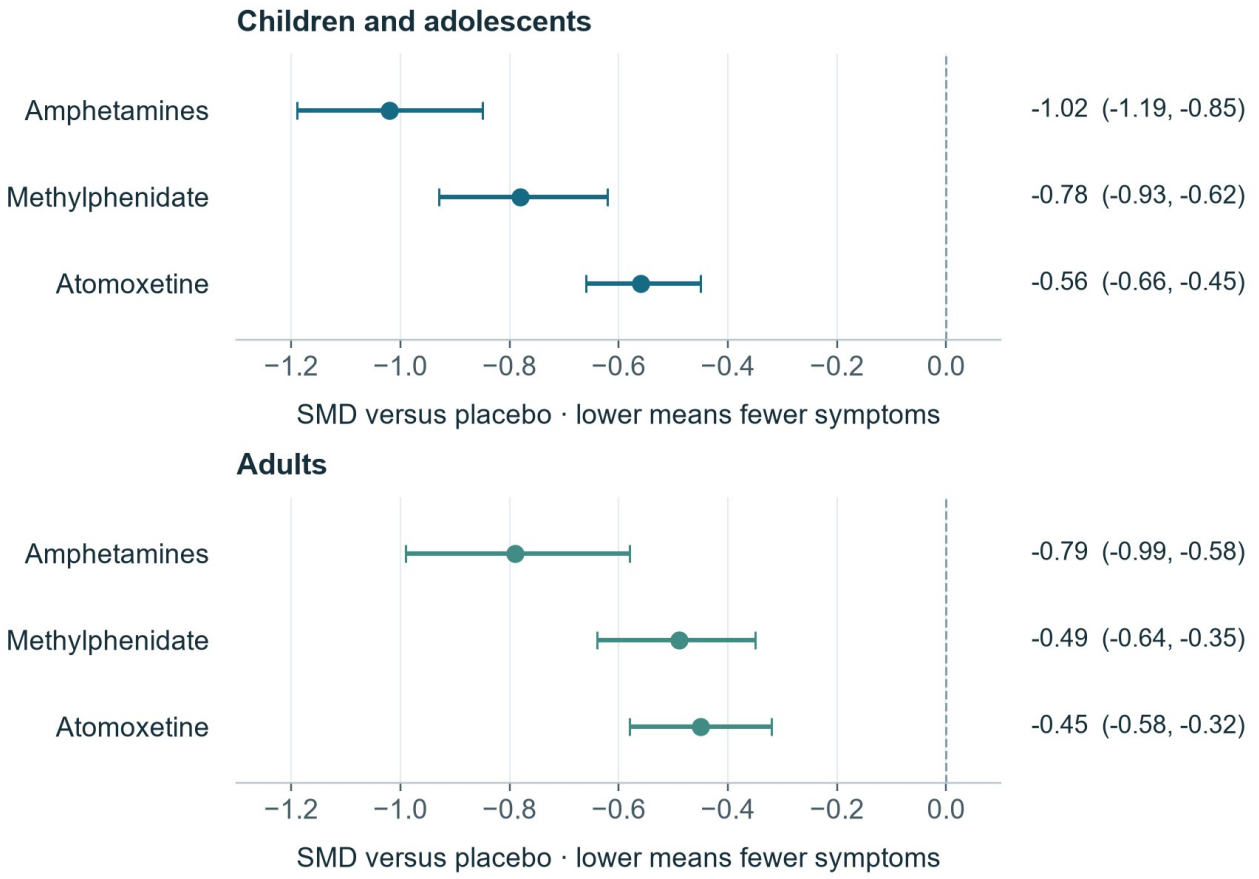


Figure 22. Direct ADHD symptom evidence provides a treatment benchmark. Selected clinician-rated placebo comparisons from Cortese et al., closest to 12 weeks; dots are standardized mean differences and bars are 95% CIs. Network estimates include indirect comparisons. Separate age panels are not a randomized test of age differences. These estimates neither measure learning nor rank treatment for an individual. [76]

Symptom relief, better task performance and learning are related, but separate. A meta-analysis of 31 placebo-controlled methylphenidate studies in 804 children found cognitive effects ranging from d = 0.20 to 0.73 across domains. Higher doses did not consistently produce larger effects on executive memory, inhibition or flexibility. A 2026 dose-response synthesis likewise found benefit plateaus for some drugs and outcomes, alongside increasing adverse-event withdrawals in some comparisons. Neither supplies a universal optimal dose for an individual. [78, 79]

Worked example 4. More work completed is not the same measurement as more learned

In a triple-masked crossover study of 173 children, extended-release (OROS) methylphenidate produced 37% more arithmetic work per minute and 53% fewer classroom rule violations than placebo. Yet it did not improve the amount of new science, social-studies or vocabulary material learned during the instructional periods. This does not erase the classroom benefit. It tells us that a child may need both effective symptom

treatment and teaching that supports learning. The result comes from a structured summer programme, not every classroom or age group. [80]

Longer-term studies ask what happens outside the symptom scale. A Swedish target-trial emulation followed 148,581 newly diagnosed people under modeled treatment-initiation strategies for two years. Medication initiation was associated with lower rates of several first events, including suicidal behaviour and transport accidents; the first accidental-injury estimate was compatible with no difference. This was an observational comparison, not random treatment assignment. Dispensing does not prove adherence, and residual confounding remains possible. The study does not identify a cognitive or immune mediator. [81]

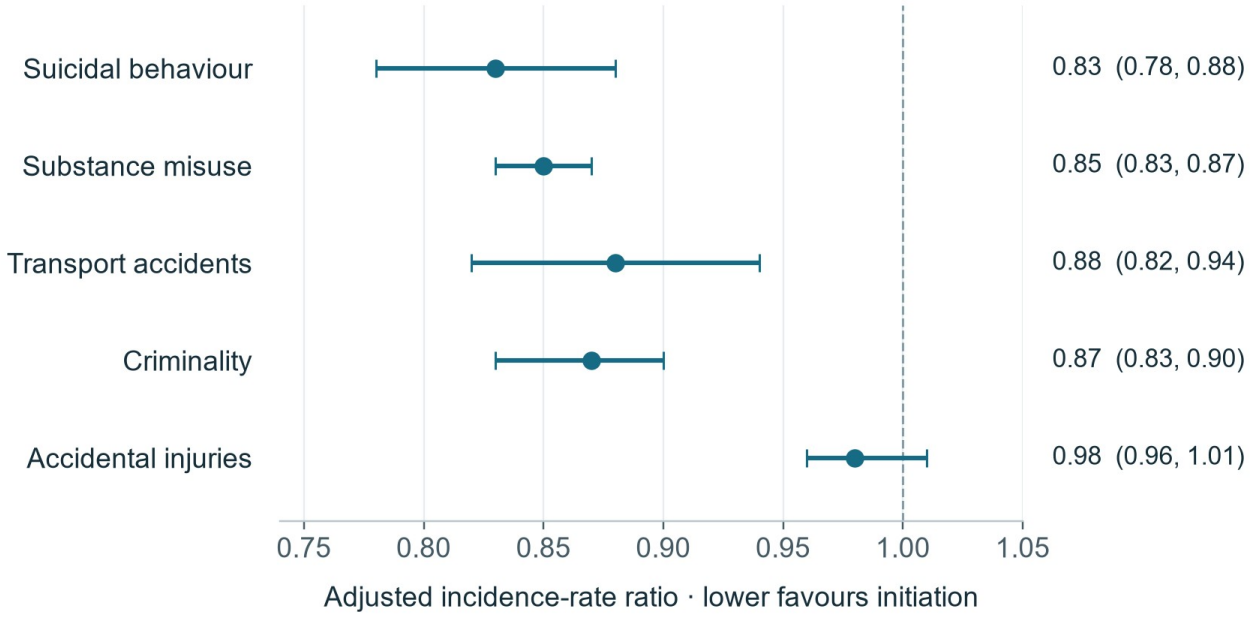


Figure 23. Longer-term events answer a different medication question. Adjusted first-event incidence-rate ratios with 95% CIs from a Swedish target-trial emulation in 148,581 newly diagnosed people. Follow-up was two years. This is observational evidence; residual confounding remains possible. Accidental-injury results are compatible with no difference. Ratios are not individual probabilities, and they do not identify a cognitive mediator. [81]

11.2 Treatment can change more than attention

Sleep is one practical example. Across nine randomized studies with objective sleep measures in 246 young people, stimulants were associated with longer sleep latency, lower sleep efficiency and shorter total sleep time. Average results do not describe every person: better daytime regulation may help an individual routine even while a particular dose or timing disrupts sleep. Record the observed pattern rather than assume that either benefit or disruption must occur. [82]

Cardiovascular effects depend on the drug. A 2025 synthesis of 102 randomized trials found small average increases in several blood-pressure or pulse measures with stimulant and some nonstimulant treatments, whereas guanfacine generally lowered them. Most trials were

short: the median was seven weeks, and none supplied long-term randomized evidence. “Nonstimulant” therefore does not mean physiologically neutral. Heart rate, blood pressure, symptoms and clinical events are distinct outcomes. [83]

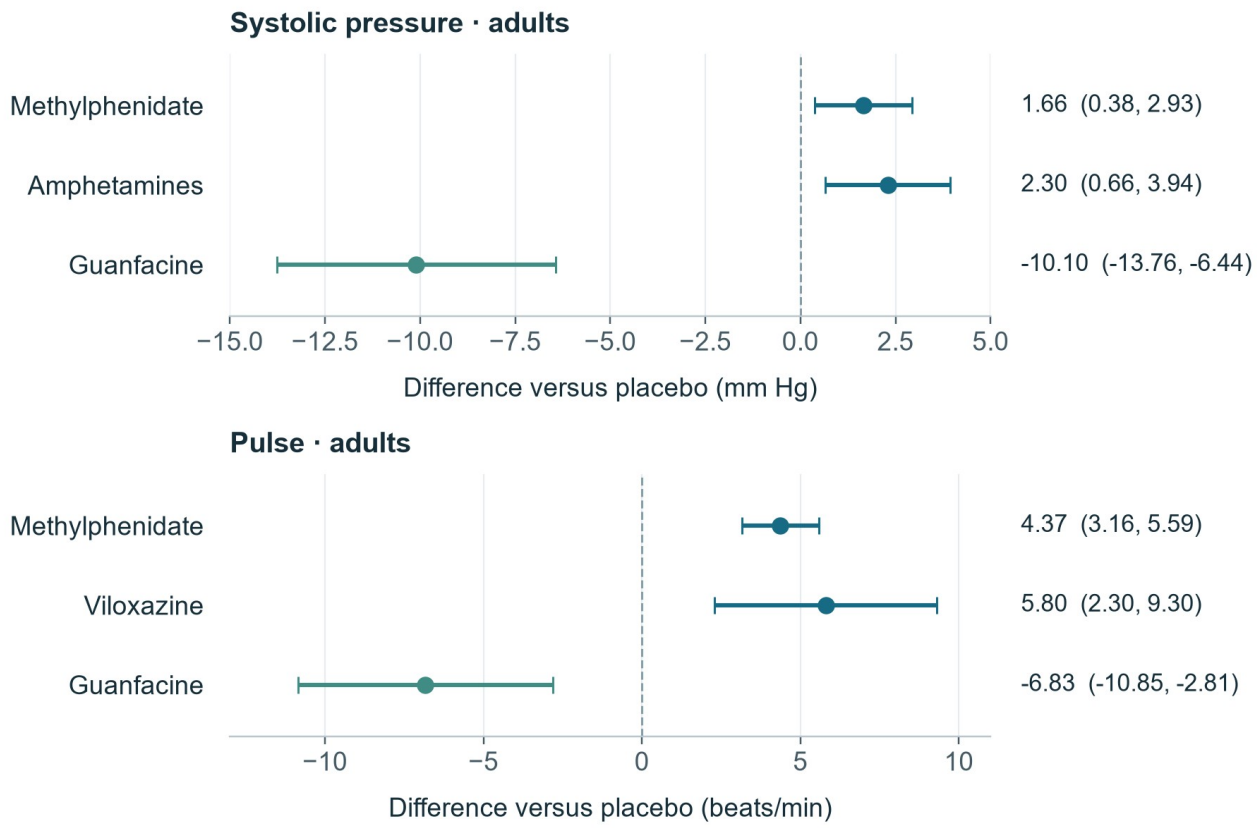


Figure 24. Medication can move physiological measures in different directions. Selected adult placebo comparisons from Farhat et al., with 95% CIs; all plotted estimates had very low evidence confidence. Median trial duration was seven weeks. Different drug selections illustrate the measures, not a complete ranking. Blood pressure, pulse and clinical cardiovascular events are separate outcomes; a lower value is not always better for a particular person. [83]

The person’s underlying condition can change the response. In 27 people with POTS, an acute randomized crossover experiment found standing heart rate of 121 versus 105 beats/minute two hours after atomoxetine or placebo, respectively; symptoms also worsened on atomoxetine. This was a POTS experiment, not an ADHD efficacy trial. It supports checking autonomic context rather than assuming that the same noradrenergic effect is helpful in every physiological state. [84]

Longer follow-up adds a different caution. A Swedish nested case-control study associated more than five years of cumulative ADHD-medication use with higher odds of cardiovascular disease than nonuse: adjusted OR 1.23 (95% CI 1.12–1.36), with prominent hypertension and arterial-disease associations. This does not establish causation or a 23-percentage-point risk increase. Surveillance, severity and other confounding can affect the estimate. Benefits on one outcome and risks on another can coexist. [85]

11.3 Medicines for overlapping conditions matter too

An allergy medicine can change attention through sedation as well as through symptom relief. In a randomized crossover study of 40 adults with seasonal allergic rhinitis, diphenhydramine impaired simulated driving, whereas fexofenadine performed similarly to placebo. Subjective drowsiness was a poor guide to impairment. This is a concrete reason to distinguish centrally sedating H1 drugs from other antihistamines. It does not show that allergy-related cognitive symptoms reveal a histamine cause of ADHD. [89]

Migraine treatment is another example. Topiramate's prescribing information reports cognitive adverse reactions in 22% of participants receiving 100 mg/day in six-month migraine-prevention trials, versus 10% on placebo. That is a trial exposure, not a dose recommendation. The endpoint was reported adverse reactions, not a dedicated ADHD test. A change in concentration during migraine prevention could reflect attacks, sleep, pain relief, treatment effects or several of these together. [88]

GLP-1 medicines can change appetite, gastrointestinal symptoms and gastric emptying. Semaglutide labeling flags possible effects on absorption of oral medicines, but does not establish a specific interaction with ADHD stimulants. A reported change in stimulant response after starting a GLP-1 drug deserves a medication and timing review; it cannot by itself demonstrate altered absorption or an immunometabolic mechanism. [90]

Tirzepatide has a specific hormonal-medication interaction worth distinguishing from a general absorption concern. Its US label advises switching to a non-oral contraceptive or adding a barrier method for four weeks after starting and after each dose escalation because oral hormonal contraception may be less effective. This is a prescribing boundary relevant to the endometriosis and hormonal-state discussion, not evidence of an established stimulant interaction. [143]

Mental-health safety and mental-health efficacy also answer different questions. In January 2026, the FDA requested removal of the suicidal-behaviour/ideation warning after its review found no increased risk with GLP-1 medicines. That finding does not establish an antidepressant effect or replace assessment of an individual's symptoms. [142]

11.4 A prescribed dose is not the whole exposure

Some interactions are well defined. In normal CYP2D6 metabolizers, fluoxetine or paroxetine can increase atomoxetine exposure about six- to eightfold. That is an interaction with specified enzyme inhibitors, not an effect of all antidepressants. Guanfacine exposure changes with CYP3A4 inhibitors or inducers; sedation can add to other sedating medicines. Abrupt guanfacine withdrawal can cause rebound hypertension. These label findings explain why co-treatment and medication changes belong in the record. [86, 87]

Amphetamine-product labeling also identifies serotonin-syndrome risk with serotonergic co-medications, including some antidepressants and triptans, and contraindicates concurrent or recent MAOI use. This is a specific prescribing interaction, not evidence that all migraine or mood treatments are incompatible with ADHD medication. The examples here are not an

exhaustive interaction check or a self-adjustment guide. Prescribing and changes remain an individual clinical decision. [91]

11.5 How medication fits the proposed model

Use a treated baseline when that is the person's ordinary life. Record formulation, prescribed dose, actual dose time, missed doses and co-medications, alongside sleep, food intake, symptoms and the task. A late-day decline might involve waning drug effect, accumulated demand, poor sleep or pain. Calling it "inflammation" before comparing these explanations would make the theory less useful.

Separate three questions: did the medication improve ADHD symptoms; did it change a measured bodily state; and did that bodily change account for any benefit or difficulty? The first can be answered without proving the third. A medication response does not diagnose a single neurotransmitter deficiency, and a side effect does not validate this paper's wider mechanism.

For research, medication is both an exposure and a possible source of confounding. Someone may change a dose because symptoms worsened; a same-day association can then run in the opposite direction from the drug's effect. Stable regimens, recorded timing, prespecified treatment-change analyses and clinically supervised designs are more informative than treating "medicated: yes/no" as a sufficient adjustment. No unsupervised withdrawal is needed to make the model testable.

12. Mental health, trauma and the wider environment

Cognitive complaints also occur with mental health conditions, illness, treatment and environmental demands. Which comparisons help us understand ADHD, and which could mislead us? This chapter follows those possibilities, then returns to what they change about the proposed study.

Someone can have ADHD, depression, painful bowel disease and a difficult history at the same time. The task is to find out what each contributes. Counting overlaps does not tell us which condition came first, which route matters today, or whether treating one will change the others.

12.1 Mental health belongs inside the model

A meta-analysis of PTSD included 60 studies and 4,108 participants. The largest average differences included verbal learning (d = −0.62), processing speed (−0.59) and attention/working memory (−0.50). These are differences between groups, not percentages of ability lost. Comparison groups included people with and without trauma exposure; study selection, coexisting ADHD and other clinical differences affected estimates. The findings

establish cognitive relevance, not whether the difference began before the trauma or resulted from a particular immune pathway. [99]

Depression also needs more than a mood score on test day. A synthesis of 252 studies compared 11,882 people in remission after depression with 8,533 controls. Differences remained on 55 of 75 cognitive variables. That is 73% of the variables examined, not 73% of people, and it is not proof of permanent damage. Recovery of mood and recovery of cognition should be measured separately. [100]

For this hypothesis, depression, anxiety, intrusive memories and dissociation are candidate contributors, outcomes or confounders depending on the question. Their role cannot be decided after seeing which adjustment gives the preferred result. Bipolar disorder, psychotic disorders and substance-related conditions also require their own clinical histories; a shared complaint of poor concentration does not make their mechanisms interchangeable. This edition does not attempt a separate evidence review of each.

12.2 Trauma is relevant; ADHD is not a synonym for trauma

The temporal evidence is mixed. In the E-Risk cohort of 2,232 British twins, childhood abuse/neglect did not predict young-adult ADHD after accounting for childhood ADHD. Childhood ADHD predicted later maltreatment particularly with coexisting conduct disorder. This identifies a need for protection, never responsibility on the child's part. [101] The Boricua Youth Study, beginning with 2,480 children and adolescents, found some prospective evidence involving physical abuse and ADHD, as well as concurrent associations. Different exposure windows, informants and adjustments can change the answer. Neither cohort settles every form of adversity or trauma-related inattention. [102]

The distinction we need is between developmental history and present-day amplification. Trauma may be relevant to current sleep, arousal, avoidance or the demands of a setting without explaining the origin of someone's ADHD. A study should ask about timing and present symptoms with consent, allow people to decline detailed disclosure, and avoid turning an adversity score into a biological diagnosis.

12.3 Stress, autoimmunity and infection: association has a direction

In Swedish registers, 106,464 people diagnosed with stress-related disorders were compared with 1,064,640 matched people without that diagnosis. Subsequent autoimmune disease rates were 9.1 versus 6.0 per 1,000 person-years; the adjusted hazard ratio was 1.36 (95% CI 1.33–1.40). These were diagnosed stress-related disorders, not an experiment in everyday stress. The study also used siblings, but residual differences in health, exposure and care-seeking remain possible. It measured autoimmune diagnoses, not cognition. [103]

Other studies start with immune disease or infection and follow later mental health diagnoses. A Danish cohort of 3.56 million people recorded 91,637 first mood-disorder diagnoses through hospital, outpatient or emergency contacts. Figure 25 shows higher subsequent diagnosis rates after autoimmune disease, hospitalized infection, or both. These histories

may mark several pathways, including illness burden and treatment. They do not show that infection persists or that mood symptoms are caused by brain inflammation. [104]

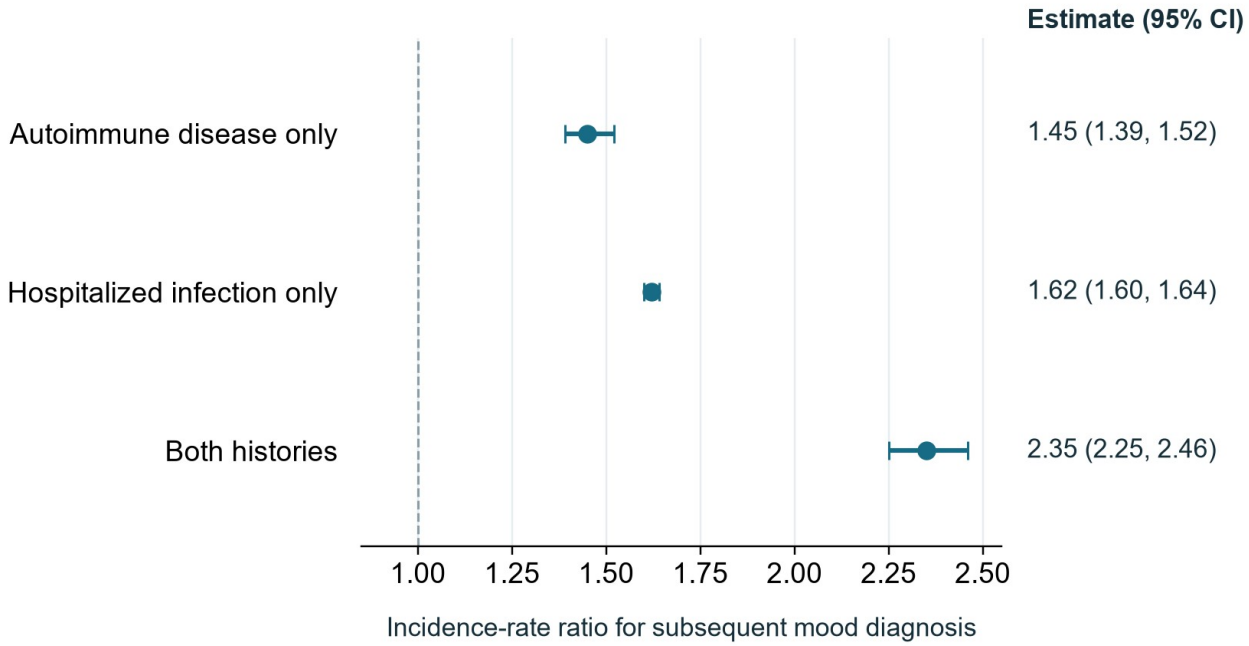


Figure 25. Immune and infection histories predict later mood diagnoses in a register study. Table 2 estimates, with 95% CIs, adjusted for age, sex and calendar year; the reference is neither history. “Only” excludes the other exposure category. A ratio of 2.35 does not mean a 235% probability. Statistical interaction depends on the scale; the joint estimate is not proof of biological synergy. [104]

An immune-mediated condition can have cognitive relevance without establishing a microbiome cause. In an inflammatory bowel disease synthesis, all 12 included studies were cross-sectional (687 participants; 11 studies contributed to meta-analyses), generally involving younger adults with inactive disease. Attention and executive measures differed from controls, while learning/recall and broad screening estimates were inconclusive. Pain, fatigue, nutrition, mood and medication remain competing routes. This is not evidence that cognition tracks a flare within one person. [105]

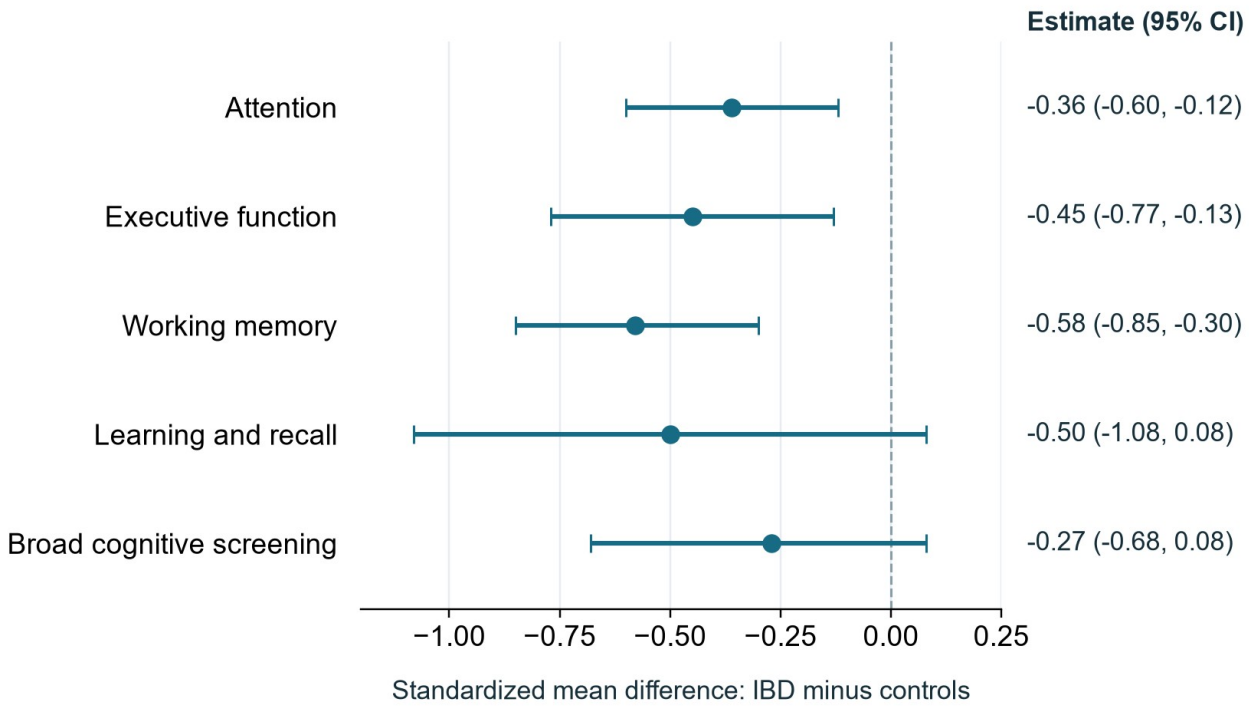


Figure 26. Cognitive domains do not all give the same answer in inflammatory bowel disease. Published standardized differences with 95% CIs; negative values indicate poorer average performance. Working memory overlaps with executive function, so these are not independent replications. The broad-screening interval is reproduced as published. The studies cannot identify a cause or measure decline over time. [105]

An autoimmune label alone tells us too little. We need the diagnosis, treatment, timing and organs involved before proposing a route to cognitive change.

12.4 Infections can be causal in a specific disease without explaining ADHD

Epstein–Barr virus and multiple sclerosis provide a stronger example of temporal evidence. The underlying military cohort included more than 10 million people and 955 MS diagnoses; the analysis with available serum samples included 801 cases and 1,566 matched controls. Among those initially EBV-negative, becoming antibody-positive rather than remaining negative was associated with an MS hazard ratio of 32.4 (95% CI 4.3–245.3). The large estimate has wide uncertainty. It concerns EBV seroconversion, not a mono diagnosis or a 32% absolute risk. The absence of a similar cytomegalovirus association strengthens the specific etiological case. It does not establish ongoing infection as the cause of ordinary fog or an antiviral treatment for ADHD. [106]

Long COVID already supplies a distinct postinfectious setting in Section 8. Acute infection, postinfectious symptoms, latent infection and active infection are different exposures. Combining them into “infection load” would lose the timing and biological distinctions the hypothesis needs. Prospective work should record documented episodes, severity, recovery, sleep and treatment, rather than infer an infectious cause from a symptom checklist.

Autoimmune encephalitis is a separate clinical boundary. The diagnostic approach concerns a subacute syndrome—typically progression within three months—with memory, mental-status or psychiatric changes and supporting neurological, laboratory or imaging features after alternatives are considered. Chronic concentration difficulty alone is insufficient; normal routine tests do not by themselves exclude every case. Rapid new confusion, seizures or neurological decline warrant prompt assessment rather than interpretation through this framework. [107]

There is also a specific molecular lead in MS. Lanz and colleagues identified antibodies that cross-reacted with an EBV protein, EBNA1, and a nervous-system protein, GlialCAM. This is an example of molecular mimicry: an immune response can recognize both a viral target and part of the body. The finding concerns a subset of MS biology. It is not evidence that the same mechanism explains ADHD or ordinary post-mono fog. [169]

Lupus offers another, still limited, connection. A 2025 study examined B cells from 11 people with systemic lupus erythematosus and 10 controls. Mean EBV-infected cell frequencies were about 25 versus 1 per 10,000 sequenced B cells. The researchers also found programmes that could help infected, self-reactive B cells present targets to other immune cells. That is a cellular frequency difference in established disease, not a 25-fold lupus risk after mono. It supports mechanistic work, not a treatment conclusion or a common explanation for cognitive symptoms. [170]

12.5 Gut bacteria: a plausible route with several missing steps

A microbiome study analyzed 1,054 Rotterdam participants and replicated associations in 1,539 Dutch HELIUS participants after exclusions. Thirteen microbial taxa were associated with depressive symptoms. The association replicated. Whether those organisms can guide a diagnostic test or improve mood when altered remains unproven. The genetic analyses were exploratory and did not remove uncertainty about direction. [108]

The microbiome argument requires several steps. Bacteria must differ in a relevant way, their activity must change a signal, and that signal must affect a meaningful outcome. Finding a microbial difference does not complete that sequence. A bacterium’s ability to produce a chemical says little about how much reaches the human brain. Diet, bowel transit, antibiotics, other medication and sample handling need to be recorded before assigning a causal role.

The intervention evidence deserves the same restraint. A depression pilot included 49 treated participants in its analysis, testing a multistrain probiotic added to ongoing antidepressants for eight weeks. Its purpose included feasibility and estimating effects for a later trial. Figure 27 shows all four week-eight symptom measures: two intervals cross zero. These correlated outcomes are not four replications, and none was an objective cognitive endpoint. [109]

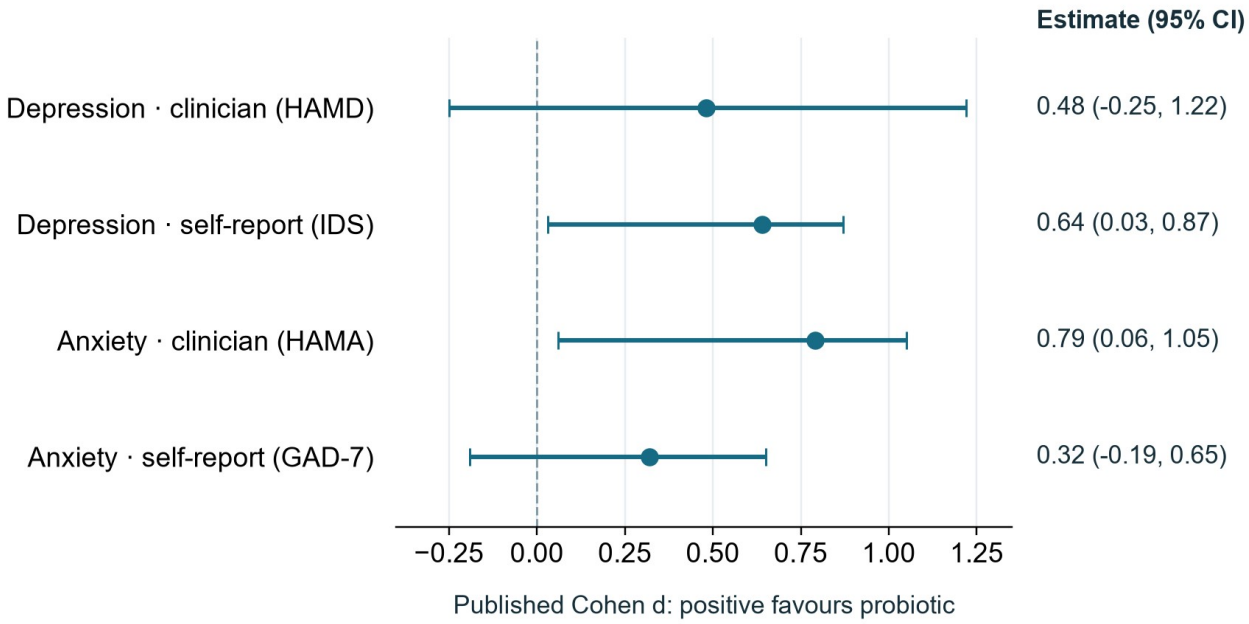


Figure 27. A promising pilot does not give one uniform answer. Published week-eight Cohen d estimates and 95% CIs from Table 2, before the small-sample correction; asymmetric intervals are retained as printed. HAMD/IDS measure depression; HAMA/GAD-7 measure anxiety. Positive values favour the probiotic. Small sample, multiple outcomes and adjunctive treatment limit conclusions; these are not ADHD treatment estimates. [109]

A larger gut-directed trial adds a provisional treatment signal. RECOVERY randomized 463 adults with post-COVID symptoms to the synbiotic SIM01 or placebo for six months. Reported alleviation favoured SIM01 for concentration difficulty (OR 2.64, 95% CI 1.69–4.14) and memory loss (1.97, 1.27–3.04). These were interview-based symptoms, **not objective cognitive gains**. Quality of life, physical activity and the cytokine analysis did not show significant between-group results. The single-centre design, symptom measurement, missing-data assumptions and investigators' commercial interests make independent replication valuable. The trial does not establish an inflammatory mechanism or ADHD efficacy; its historical outcome-registration changes were not independently audited here. [197]

A separate 12-week pilot involved 38 children assigned to an ADHD diagnostic group and 42 to an autism group, with overlapping diagnoses placed in the latter. Its adjusted overall comparisons did not show clear probiotic–placebo differences on the main behavioral or neuropsychological measures. Within-group improvements and exploratory age findings do not overturn that comparison. The formulation also included vitamin D in the active product, further limiting attribution to bacteria alone. [110]

The next trial should specify the strain or formulation, population, primary outcome and time point before recruitment. Symptom relief in the gut, a changed stool profile and improved attention would be separate outcomes. These studies do not justify antibiotics, restrictive diets, stool testing or microbiota transplantation as treatments for ADHD.

What newer gut studies add

A 2026 ADHD analysis examined stool from 147 participants at baseline and 106 at follow-up from an earlier synbiotic trial. Treatment changed microbial composition and some circulating short-chain fatty acids without changing overall within-sample diversity. Medication-associated differences in bacterial vitamin B12 synthesis genes were also reported. A microbial gene pattern is not a diagnosis of vitamin deficiency, and medication use was not randomized. The parent nine-week trial had 182 completers and found similar improvement in core ADHD symptoms with synbiotic and placebo, with **no significant daily-function benefit**. The newer biological measurements do not reverse that clinical result. [213, 214]

The PROMOTe trial gives a narrower positive cognitive example. Thirty-six twin pairs aged 60 or older were studied for 12 weeks, with one twin assigned prebiotic and the other placebo. Both received an amino-acid supplement and exercise advice. The **primary chair-rise outcome** was negative; a secondary cognitive factor favoured the prebiotic. More cognitive data were missing than physical data, and most participants were women. This is a small older-adult result, not established ADHD treatment or dementia prevention. A 2025 correction revised microbial feature counts, supplementary figures and ingredient units; it did not announce a changed cognitive treatment estimate. [215, 216]

Diet can change immune measurements without demonstrating a brain benefit. In a 2021 study with 36 analyzed participants, a fermented-food diet increased microbial diversity and reduced several circulating inflammatory proteins. The primary cytokine-response score was unchanged. The study compared two dietary programmes, lacked a usual-diet control, and two participants chose their allocation. It measured no cognitive outcome. “Gut health” needs a defined meaning: bowel symptoms, nutrient absorption, a microbial function or a measured metabolite. Diversity alone is not a health score, and these results do not supply one diet for everyone. [220]

Hookworms test whether immune change becomes clinical benefit

Hookworms are parasitic worms, not gut bacteria. Researchers have tested controlled infection because it can alter immune regulation. That makes them relevant to the hypothesis, but the outcome still has to improve. The trials below show why immune modulation cannot simply be called immune improvement. [217, 218, 219]

Table 4. Human hookworm trials answer different questions

Study	Primary result	Other findings and limits
Metabolic pilot 40 randomized Pierce 2023 [217]	Safety was primary; metabolic outcomes were secondary.	At 12 months, median HOMA-IR change was −1.1 with lower-dose infection versus +0.8 with placebo (P=.039); higher dose did not differ significantly. Per-protocol analysis, attrition and baseline imbalance.

Study	Primary result	Other findings and limits
Relapsing MS 71 randomized Tanasescu 2020 [218]	MRI lesion primary outcome not significant (P=.19).	Regulatory T cells increased. Biological target engagement did not establish clinical efficacy.
Celiac disease 54 participants Croese 2020 [219]	Week-42 completion: placebo 57%, lower dose 37%, higher dose 44% (P=.61).	Fewer gluten-related symptoms did not establish sustained gluten tolerance or protection of intestinal tissue. Small, unequal groups.

HOMA-IR is an indirect fasting glucose–insulin estimate, **not a brain measurement**. The metabolic pilot reported gastrointestinal adverse events in 44% of treated participants and three early deworming treatments. Its secondary signal does not establish diabetes prevention or cognitive benefit. These studies do not justify self-infection, abandoning celiac treatment or adding hookworms to an ADHD protocol. The useful question is whether a specific immune effect predicts a meaningful outcome, with harms measured alongside benefits. [217, 219]

12.6 Environment is more than a toxin list

The controlled particle-exposure experiment discussed earlier found selective effects after four hours in 26 adults; some cognitive tasks did not change. [23] A 2026 double-blind crossover study, HIPTox, exposed 15 healthy adults aged at least 50 with a family history of dementia to four pollutant mixtures and clean air. Cognitive responses differed by source and task; some processing-speed comparisons improved. With only 13 completing all five visits, this is a small, mixed experiment, not evidence that pollutants help cognition or uniformly impair every domain. It cannot determine an ADHD-specific susceptibility. [112]

The protocol should also record noise, heat, light timing, shift work, housing disruption, financial strain and access to recovery as candidate contextual exposures. Their relevance is a proposal for measurement here, not a claim that each has a proven immune mechanism. A noisy room may impair performance through distraction without changing inflammation. Poor sleep may connect a difficult environment to the next day's attention. Both are legitimate explanations to test.

Environmental causes also have different time scales. A developmental exposure that changes baseline ability is not the same as a short-term trigger of fluctuation. Record dose or a defensible exposure proxy, timing, task conditions and the opportunity to avoid exposure. Do not turn unexplained symptoms into a claim about unspecified toxins, mold or a need for detoxification.

Brake dust is a distinct exposure

Brake wear adds particles even when there is no tailpipe exhaust. A 2025 experiment compared particles from four brake-pad types with diesel particles in cultured human alveolar cells. Copper-rich formulations produced greater oxidative stress, inflammatory signalling and

metabolic disruption under the tested conditions. Removing available copper reduced several effects. These were laboratory lung-cell outcomes, not measured brain injury, cancer incidence or cognition in commuters. “More toxic than diesel” is too broad unless the particle dose, composition and endpoint are stated. [221]

A 2026 mouse study adds an important dose qualification: particle rankings changed when exposure was expressed by mass versus surface area. Particles were delivered by aspiration, which differs from normal inhalation. The result supports measuring composition and dose rather than treating all particulate matter as identical. The human pollution experiments already discussed in Sections 4 and 12.6 concern different exposure mixtures. They cannot identify brake dust as the cause of their cognitive findings. [222]

Microplastics need measurement as well as mechanism

A 2025 postmortem study reported more plastic-associated material in brain than liver or kidney, with higher concentrations in 12 dementia cases. The authors explicitly did not infer causality. Disease could alter tissue volume, barrier function or clearance, and postmortem sampling cannot establish what came first. A subsequent correction addressed images, scale bars and preparation details. Published criticism raised analytical and quality-control concerns, including possible interference from lipid-rich tissue; the authors responded. The exchange means that exact brain-burden estimates remain contested. Particle detection deserves investigation, but it does not establish that plastics caused dementia, ADHD or someone’s current symptoms. [223, 224, 225, 226]

Exposure estimates are also being revised. A 2026 analysis compiled 2,782 atmospheric measurements from 76 studies and compared them with size-aligned models. The modelled global median concentrations exceeded observations by two to four orders of magnitude. That finding concerns atmospheric estimation, not a direct remeasurement of brain tissue or proof that plastics are harmless. Size detection limits, contamination controls and the distinction between particle number and mass matter at both ends of the proposed exposure–health relationship. [227]

The vascular evidence is prospective but **still observational**. Among 257 people followed after carotid surgery, the composite of heart attack, stroke or all-cause death occurred more often when plastic-associated material was detected in plaque. Figure 28 shows crude proportions; the adjusted hazard ratio was 4.53 (95% CI 2.00–10.27). The selected surgical population, residual confounding and contamination concerns limit causal interpretation. Cognition was not measured, and removing an exposure was not tested. [228]

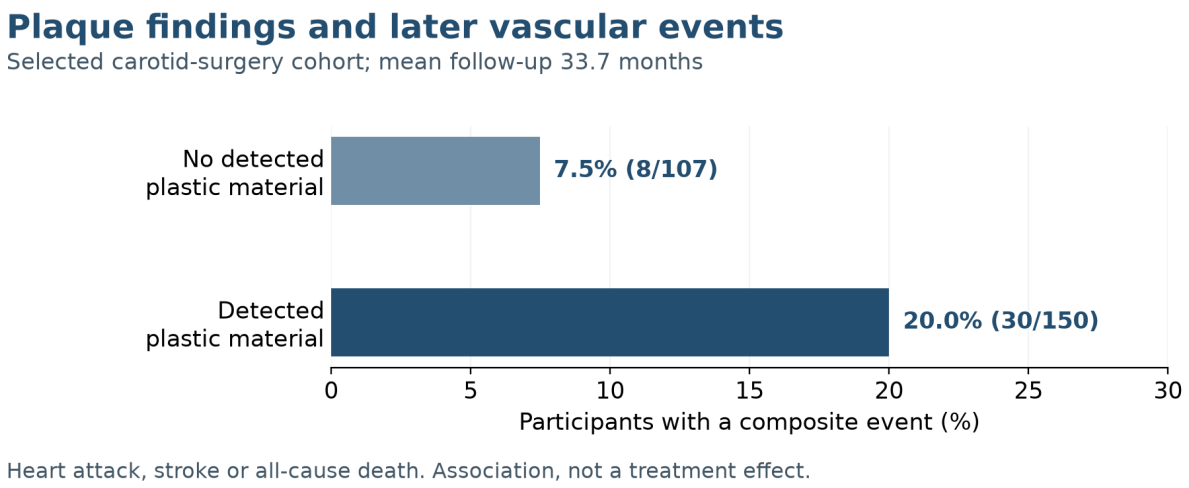


Figure 28. Calculated proportions from Marfella et al. Counts are participants with an event. Bars are unadjusted and do not depict the adjusted hazard ratio or a cognitive outcome. No confidence intervals are inferred from these bars. [228]

12.7 Physical health across the life course

The 1970 British Cohort Study connects childhood traits with later physical health. In 10,930 participants, high ADHD traits at age ten were associated with an estimated 42.1% probability of physical multimorbidity by age 46, versus 37.5% without high traits. These are model estimates based on reported conditions, not percentages diagnosed with ADHD. Smoking, psychological distress and BMI contributed indirect associations, without proving causal mediation. Missing-data selection and limited ethnic diversity matter. Figure 41 gives the uncertainty. The result supports attention to cumulative health and access to care; it does not identify one shared disease process. [184]

12.8 Cancer shows why the route and outcome matter

Cancer-related cognition is a boundary comparison for this framework. It lets us examine symptoms, objective performance, treatment and recovery in a setting where the illness and its treatment may both contribute. The purpose is to improve the questions we ask about change, not to imply that ADHD and cancer share a cause.

A large prospective study assessed 581 patients with breast cancer and 364 noncancer controls. Clinically important worsening on a self-report cognitive questionnaire was more frequent after chemotherapy and remained more frequent six months later. Figure 29 shows the published proportions. These are reported changes, not diagnoses of objective impairment. Without an untreated cancer comparison, the study cannot isolate chemotherapy from illness and other accompanying changes. [229]

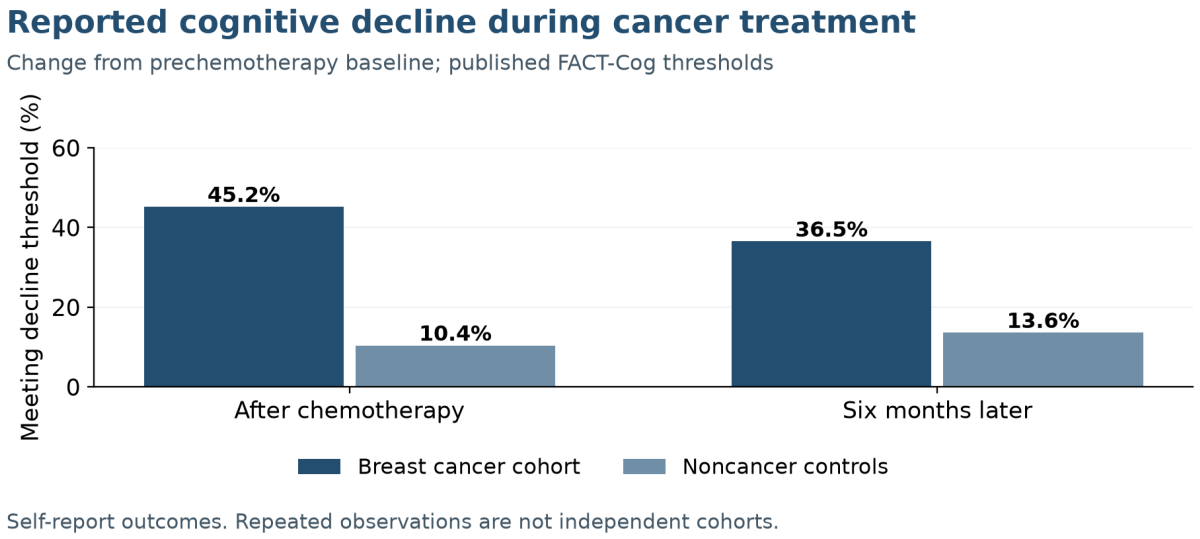


Figure 29. Janelsins et al. reported these percentages of clinically important self-reported decline. Cohort totals are 581 patients and 364 controls; no counts are reconstructed from rounded percentages. Observational comparison, not a randomized chemotherapy effect. [229]

A companion report from the same research cohort assessed objective performance in 580 patients and 363 controls. Visual-memory trajectories differed between groups, with a significant group-by-time effect. It is a complementary measurement, **not an independent replication**. Together, the reports support measuring both cognitive experience and performance, while leaving the responsible pathway unresolved. [230]

A 2024 study measured 77 patients before, during and after chemotherapy. A reliable-change classification based on four objective tests identified decline in 35% of participants. Lower microbial diversity tracked poorer objective cognition independently of the measured circulating cytokines. Subjective cognitive changes did not track the same microbial or inflammatory measures. Treatment can change several systems at once, so this does not prove microbial mediation or that increasing diversity restores cognition. A negative cytokine association also cannot exclude every inflammatory route. [231]

Microbiome intervention during immunotherapy

The 2026 TACITO trial directly tested a gut intervention alongside cancer therapy. Fifty people with metastatic kidney cancer were randomized; 45 received at least the first assigned treatment. Donor fecal microbiota transplantation (FMT) was compared with placebo FMT alongside pembrolizumab and axitinib. The primary 12-month progression-free outcome was 70% versus 41%, but did not meet the prespecified significance threshold (P=.053). A secondary progression-free survival result favoured donor FMT in the treated analysis; the all-randomized sensitivity analysis was inconclusive (HR 0.66, 95% CI 0.33–1.35). Much donor material came from one person. This is a promising, limited cancer-treatment experiment. It **did not measure cognition** and cannot validate FMT for ADHD. [232]

Removing an infection can help when the target is specific

A Korean trial randomized 1,838 people with Helicobacter pylori infection and a first-degree relative with gastric cancer. In the modified intention-to-treat population, gastric cancer developed in 10 of 832 treated participants versus 23 of 844 assigned placebo over a median 9.2 years: HR 0.45 (95% CI 0.21–0.94). This is direct evidence for a specific infection-targeted prevention strategy in a selected risk group. It measured no cognitive outcome. It does not justify antibiotics for unexplained fog, and it should not be combined with the hookworm trials into a claim that more or fewer microbes are always better. [233]

What these connections change

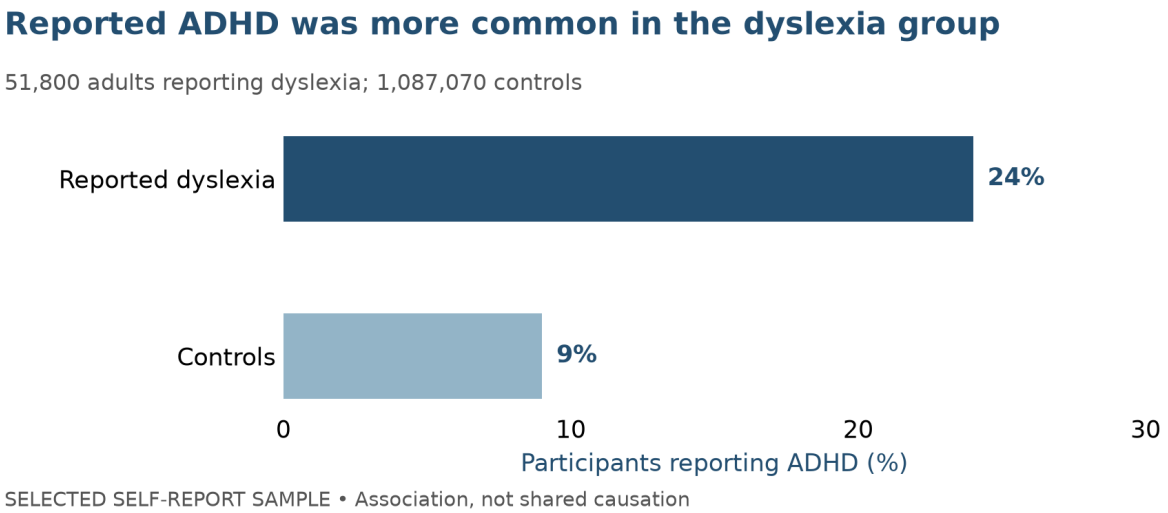
I want to keep the whole person in view without losing track of the specific explanation being tested. A lung exposure could affect function through respiratory symptoms or sleep; a gut intervention could affect comfort without changing attention; cancer treatment could affect cognition through several routes at once. These are candidate explanations. A shared inflammatory marker cannot decide between them. We need to predict which change comes first, which outcome follows and where the relationship should fail.

12.9 Autism dyslexia and other neurodevelopmental profiles

Autism and dyslexia belong in this discussion. The question is how a person's developmental profile interacts with changing bodily state and everyday demands. We should not turn a model of fluctuating attention into a claim that these conditions are temporary states, or that one immune or metabolic explanation accounts for all of them.

Dyslexia and learning differences

A large genetic study included 51,800 adults reporting a dyslexia diagnosis and 1,087,070 controls. ADHD was reported by 24% of the dyslexia group and 9% of controls. The study found genetic overlap with ADHD, but no statistically significant genetic correlation with autism in that analysis. These are selected, self-report data, not universal diagnostic prevalence estimates. **Shared genetic liability** does not identify histamine, gut inflammation or another daily physiological mechanism. [234]



Visual guide B. ADHD was reported by 24% of adults reporting dyslexia and 9% of controls in the cited genetic study. Group sizes describe the source groups; the rounded percentages are not reconstructed exact case counts. Self-report and selection limit generalization. This comparison does not identify a daily immune or metabolic mechanism. [234]

A 2025 Netherlands Twin Register study extended the comparison to dyscalculia, involving up to 19,125 twin children and 2,150 siblings assessed at ages 7 and 10. Across the conditions, having one was associated with a 2.1–3.1-fold likelihood of a second. Yet 77.3% of children with at least one had only one. The models generally favoured correlated genetic influences over one trait causing another, apart from a reading-to-spelling pathway. That interpretation depends on the modelling assumptions; it is not an experimental demonstration that environment or support cannot matter. [235]

For our framework, reading performance needs its own baseline. A difficult reading task can mix decoding demands with attention, effort and fatigue. A lower score after poor sleep would not show that sleep caused dyslexia. We should compare change within the same person and include a task with limited reading demands before calling the result a general loss of executive function.

Autism and ADHD overlap without being interchangeable

In a Swedish register cohort of 1,899,654 people, 13,793 had both recorded autism and ADHD: approximately 48% of the 28,468 autism cases and 17% of the 82,398 ADHD cases. **The denominator changes the percentage.** Family associations were consistent with shared genetic liability, while shared environmental contributions remained possible. Historical diagnostic rules, medication-based ADHD ascertainment and selection of people reaching services limit generalization. This is evidence of overlap, not evidence that autism is a form of ADHD. [236]

Sleep provides a more direct intervention example. A 13-week placebo-controlled trial randomized 125 children and adolescents with persistent insomnia after behavioural intervention; 96.8% were autistic and the remainder had Smith–Magenis syndrome.

Prolonged-release melatonin improved the primary caregiver-reported total-sleep-time outcome. The reported sleep-response rate was 68.9% versus 39.3%, using the trial's combined sleep-duration and/or sleep-latency criterion. This supports treating a defined sleep problem in that population. It does not establish a change in autism, objective cognition or the proposed resilience index. [237]

The gut connection also needs a direction check. A stool-metagenomics study of 247 participants found little direct association between autism diagnosis and the microbiome. Its results supported a pathway involving dietary preferences, dietary diversity and microbial diversity. The observational design does not settle every gut–brain question, but it gives us a concrete alternative to assuming that a microbial difference caused the developmental condition. Gastrointestinal discomfort and nutrition can still deserve attention in their own right. [238]

What this changes in the proposed study

Record autism, dyslexia and dyscalculia separately, including how they were identified. Developmental language disorder, developmental coordination disorder, tic disorders and intellectual disability are also relevant candidates for the intake profile; this section does not claim the same physiological pathway has been demonstrated in each. Describe communication, learning, sensory and motor requirements alongside diagnoses. Avoid a single combined category that hides those differences.

A practical example is an autistic participant who finds the testing room overwhelming, or a participant with dyslexia who must read long instructions before a timed task. The design could mistake an access barrier for poor resilience. Offer accessible instructions and a suitable environment, record accommodations, and keep them consistent across visits. Measure sleep, pain and illness alongside performance rather than inferring bodily dysfunction from a diagnosis.

The falsifiable question is whether a measured physiological change predicts later cognitive change within each profile, and whether those relationships differ across profiles. Prespecify a small number of comparisons and recruit enough participants to estimate them. A similar relationship across groups would support a broader state–cognition account while weakening an ADHD-specific claim. Helping someone sleep, communicate or manage pain is a worthwhile outcome without claiming to reverse their neurodevelopmental identity.

12.10 What the wider evidence changes

The wider evidence adds competing explanations that the model has to address. In a 60-person randomized trial, infliximab did not improve the overall depression outcome versus placebo. A baseline-inflammation interaction and exploratory subgroup were interesting, but did not establish general efficacy or a clinical CRP threshold for treatment. We cannot rescue the hypothesis by inventing a new subgroup after every null finding. [111]

Worked example: three routes to the same difficult morning

Imagine someone with ADHD and bowel disease who reports poor concentration after a painful, sleepless night. One possibility is a direct inflammatory effect; another is that pain disrupted sleep; a third is that medication or worry changed arousal. This is hypothetical. Measure the proposed sequence rather than naming the route in advance. If sleep and pain explain the later performance change and the immune measure adds no predictive value, the direct immune claim weakens even though the person's illness and difficulty are real.

Table 5. How the wider evidence changes the next study

Area	What to distinguish	Test that could weaken the claim
Mental health and trauma	Developmental history; current mood, trauma symptoms, sleep and cognition	Physiological measures add no prediction beyond these simpler measures.
Autoimmune and infection history	Specific diagnosis, severity and timing; symptoms versus cognitive tests	No within-person relationship despite reliable measurement and adequate change.
Gut microbiome	Composition, function, treatment exposure and clinical outcome	A microbial shift occurs without the prespecified cognitive benefit.
Environment	Measured exposure, timing, direct distraction and sleep disruption	Association disappears with exposure timing or task conditions properly controlled.

These are proposed tests, not findings. A null result is informative only if measurement, sample size and exposure variation were adequate. Failure of a specific route should narrow the theory rather than invalidate the person's experience.

13. What the integrated hypothesis adds

The branches above do not converge on one demonstrated cause. They do give us a more precise choice of explanations. The next step is to bring them back to the same person, the same task and a measured change over time.

The proposal is to test how particular bodily changes affect cognition. Different routes may produce different outcomes; an inflammation label does not explain those differences.

One possibility is an additive practical effect: ordinary sleep disruption matters more when a task is already difficult. Another is a specific interaction: an endocrine or orthostatic change is associated with a larger cognitive response in a defined group. Perception and performance may also differ: thinking feels much harder while accuracy is preserved. Some effects may arrive too late for a same-day measure to capture. Each possibility requires its own outcome and time window.

The network can also contain feedback. Pain may disturb sleep; altered sleep may accompany worse vigilance; symptoms may change behaviour and treatment, which change the next measurement. Neural regulation may modify immunity rather than merely receive its signals. These are proposed structures to compare, not permission to infer a complete feedback loop from two correlations.

A proposed test is cross-state transfer: freeze a relationship learned during one prespecified state and test it during another state observed independently. Success would favour some shared predictive structure, but would not prove a common biological cause. Failure could reflect state specificity, weak measurement or exposure differences. Those alternatives must be planned rather than invented after the result.

The remaining possibility is that different biological routes may sometimes converge on vigilance, effort allocation or the practical ability to meet demands. We do not yet know how often that happens, how ADHD changes it, or whether the added measurements will outperform careful clinical and behavioural assessment.

14. A research programme broad enough to test the theory

A broad account becomes useful only when the study is small enough to interpret. I would begin with a shared daily measurement core, add a limited mechanism module, and test it against simpler explanations. The sections below move from that core to targeted experiments, the provisional score, the possible role of AI and the first priorities.

14.1 A common measurement core with targeted modules

The initial feasibility study remains focused: a well-characterized adult ADHD group and a comparison group, followed for a proposed four to six weeks with two brief scheduled assessments daily. Duration and frequency are design proposals, not proven optima. The first goals are completion, measurement quality, exposure variation and uncertainty. A confirmatory sample should follow simulation-based power and precision analysis; many trials within a few people do not replace enough participants.

Each window would pair a sustained-attention task with sleepiness, fog, fatigue, pain/headache and current context. Sleep diary and validated actigraphy, when feasible, would record duration and timing. Clock time, time since waking, medication timing, caffeine and relevant treatment changes would be captured. An independent daily-function measure would record what could or could not be done. People with lived experience should help select meaningful, tolerable outcomes rather than having the functional threshold invented solely from a laboratory score. Medication recording would include formulation, actual administration time, missed doses and co-medications, with treatment changes handled explicitly rather than reduced to a yes/no covariate. Baseline history would also record

relevant sensory limitations and known medical or nutrient problems, with clinical follow-up when indicated; the research protocol would not substitute for a diagnostic assessment.

The candidate primary confirmatory test remains the ADHD-by-**within-person** prior-night sleep-duration association with target-omission probability. It should be finalized only after feasibility work establishes reliable change, adequate exposure variation and a meaningful effect scale. Commission errors, response variability, fog and function remain separate outcomes. If omissions are too rare to measure reliably, the task must be redesigned before confirmation—not replaced after seeing the preferred result.

Sleep is a benchmark, not the whole hypothesis. The broader modules below can use the same outcome definitions while selecting a small set of route-specific measurements. This makes findings comparable without pretending that every mechanism needs to be measured in every participant.

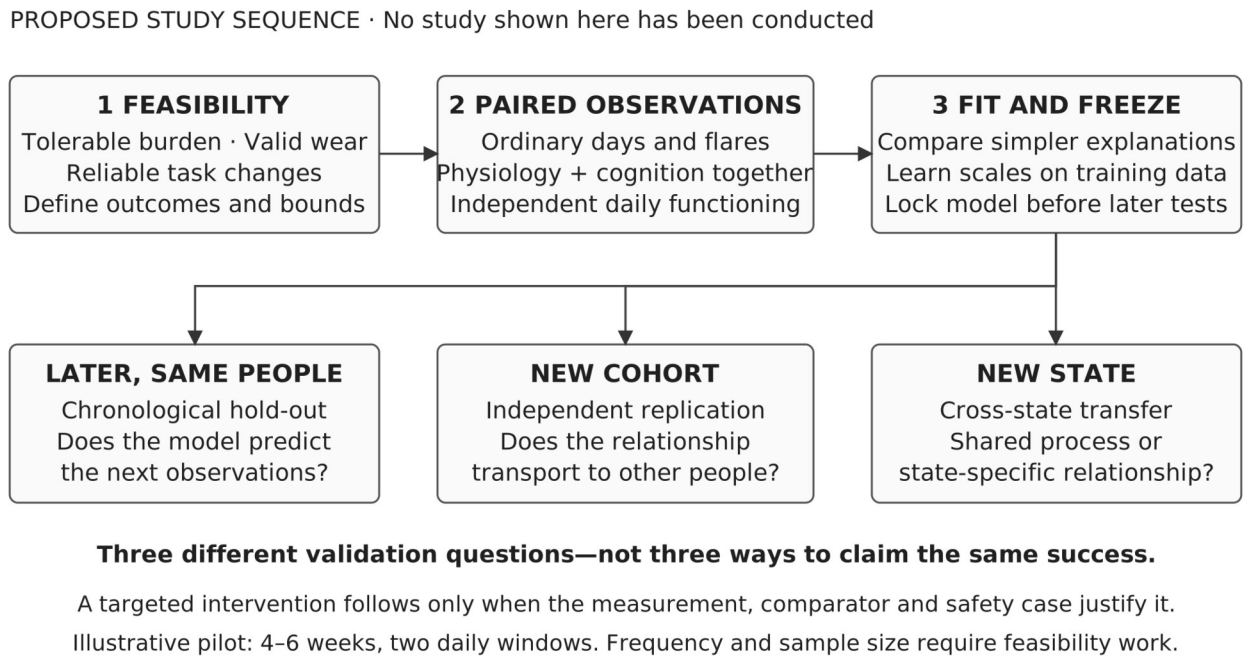


Figure 30. A shared measurement core, followed by different validation questions. Proposed workflow, not a completed study. Later observations in the same individuals, new participants and a new physiological state test different kinds of generalization. All preprocessing and subgroup decisions must be fixed without access to the observations used for testing. The illustrative pilot duration is not a power calculation or a registered protocol.

14.2 Modules and the missing evidence each would address

Table 6. Proposed mechanism modules: a common core with separate questions.

Module	Paired measurements	The question it can answer
Sleep, timing and vigilance	Prior sleep, chronotype, repeated attention, sleepiness; optional waking EEG	Does vigilance instability add predictive information beyond sleep and medication timing?
Migraine and orthostatic state	Attack/recovery timing, pain, treatment, HR/BP; supervised carbon-dioxide and Doppler measures where justified	Are bodily response, cognitive response and practical consequence separable?
ADHD × endometriosis	Repeated endocrine measures or validated phase, pain, sleep, migraine and attention	Is the association additive, interactive, or carried through specified pain/sleep routes?
Allergy and histamine	Documented allergic state, symptom burden, sleep and preselected mediator measures when justified	Does an identified immune-linked change precede or explain cognition beyond allergy discomfort and sleep?
Post-infectious illness	Infection history, pre-existing ADHD ascertainment, low-burden repeated measures and delayed symptom sampling	Does pre-existing ADHD modify cognitive change or recovery, without conflating acquired symptoms with developmental ADHD?
Metabolic or vagal intervention	Clinically justified target, credible control, co-treatment and target measures, cognition and functioning	Does the intervention affect the specified outcome, and what evidence distinguishes the possible routes?

These are separate studies or nested modules, not a recommendation to perform every assay or challenge in one cohort. Each needs prespecified primary outcomes, an appropriate comparator and its own ethics and safety justification.

The wider modules need a small prespecified core: current depression/anxiety and relevant trauma symptoms; recent infection and diagnosed disease activity; gastrointestinal symptoms, diet and medication changes; and a limited set of timed environmental exposures. Baseline history and repeated state measures serve different purposes. Choose a primary comparison and outcome, address missing data on the worst days, and test incremental prediction in held-out observations. Exploratory microbial or immune subgroups need **independent replication** before they become explanations.

14.3 Design details that prevent a misleading answer

The timing/EEG substudy should counterbalance visits across days where possible and distinguish clock time from biological circadian phase. The orthostatic study needs repeated-

task control sessions, planned recovery observations and separate models of physiology and cognition. It must not attribute task-induced cardiovascular change to posture. [42, 43]

The endometriosis module should observe more than one cycle where feasible and document diagnosis ascertainment, hormone treatment, irregular cycles, amenorrhea and reproductive stage. Calendar days alone should not stand in for measured endocrine state. The combined ADHD–endometriosis group having the worst average score is not sufficient evidence of interaction. Record bleeding history, clinically indicated iron assessment, fatigue and relevant treatment as possible contributors. Avoid a universal supplement or biomarker panel. Separate menopausal stage from calendar phase.

The allergy and post-infectious modules need ordinary-day sampling as well as flares. A preselected mediator panel should specify assay limitations, timing and handling. Biomarkers are not interchangeable with diagnoses, and a negative result should not be dismissed by invoking an unspecified hidden compartment. The post-infectious module must protect participants with post-exertional worsening and accommodate missed or shortened assessments. [46]

A later intervention trial may enrich for a relationship that already predicted independent observations. Randomization can estimate an intervention's effect; it does not automatically prove the pathway responsible. Blocking a receptor need not lower the amount of its mediator. Conversely, a blood-marker change may not reflect the target relevant to cognition. No personal response to several simultaneous treatments is used as evidence here.

A clinically indicated GLP-1 or other peptide treatment can be studied within this design, using the sequence specified in Section 9.15. Record discontinuation and tolerability, because analyses restricted to people who remain on treatment may select a different group. A mediator analysis should state its assumptions and compare pain, sleep and weight routes before attributing cognitive change to immune or neuronal action.

14.4 Testing the endocrine immune and metabolic connections

These findings make the central question more specific. We need to know whether a changing physiological state predicts a changing outcome, and whether that relationship differs between people. A positive finding for PMDD does not validate a mast-cell pathway; a GLP-1 treatment response does not establish insulin resistance as the cause of attention problems.

For the endocrine module, compare prospectively characterized ADHD, PMDD, both and neither. Record daily symptoms and a brief cognitive task across repeated cycles, with sleep, pain, migraine and medication timing. Hormonal measurements should test a prespecified timing or change hypothesis. If endocrine data add no reliable prediction beyond those simpler observations, that weakens the endocrine contribution to this version of the model.

For the metabolic module, distinguish diabetes from other populations and record treatment, meals and any clinically indicated glucose data. Test whether glucose state adds useful **within-person prediction**. We should not deliberately induce hypoglycemia in an ADHD pilot

or infer brain insulin resistance from fatigue. For the immune module, define the disease and mediator measurement before testing its association with cognition.

A combined hormone–mast-cell–barrier hypothesis needs temporally aligned evidence for each step. If an immune marker changes without the predicted cognitive change, or the association is accounted for by pain and sleep, the proposed direct route loses support. Interaction terms should be limited and prespecified; dividing a small sample into ever narrower subgroups would make almost any story look plausible.

14.5 Testing gut and environmental routes without broadening every visit

The gut module should pair a limited set of stool and circulating measures with diet, bowel transit, gastrointestinal symptoms, medication and the existing cognitive core. A microbial change should precede the proposed metabolite and cognitive changes if that sequence is part of the claim. Replicate it before assigning a biomarker-defined subgroup or testing a formulation. Stool diversity should not be added to the Neurophysiological Resilience Index as an assumed benefit.

The environmental module should observe ordinary variation or a separately justified exposure-reduction intervention. Record particle size, source information, timing, noise, heat and sleep; a generic PM reading cannot uniquely identify brake wear. Microplastic assays require validated blanks, recovery testing and independent laboratory agreement before they become explanatory variables. Deliberate pollution exposure, parasitic infection and cancer treatment are not components of the proposed ADHD study.

Clinically indicated treatment cohorts can supply separate tests of recovery. Track fatigue, pain, sleep, mood, anemia and co-medication alongside cognition before and after treatment. If these simpler measures explain the change and a proposed microbial or immune measure adds no reliable prediction, that route loses support. Cancer or infection recovery may inform the model without validating it in ADHD.

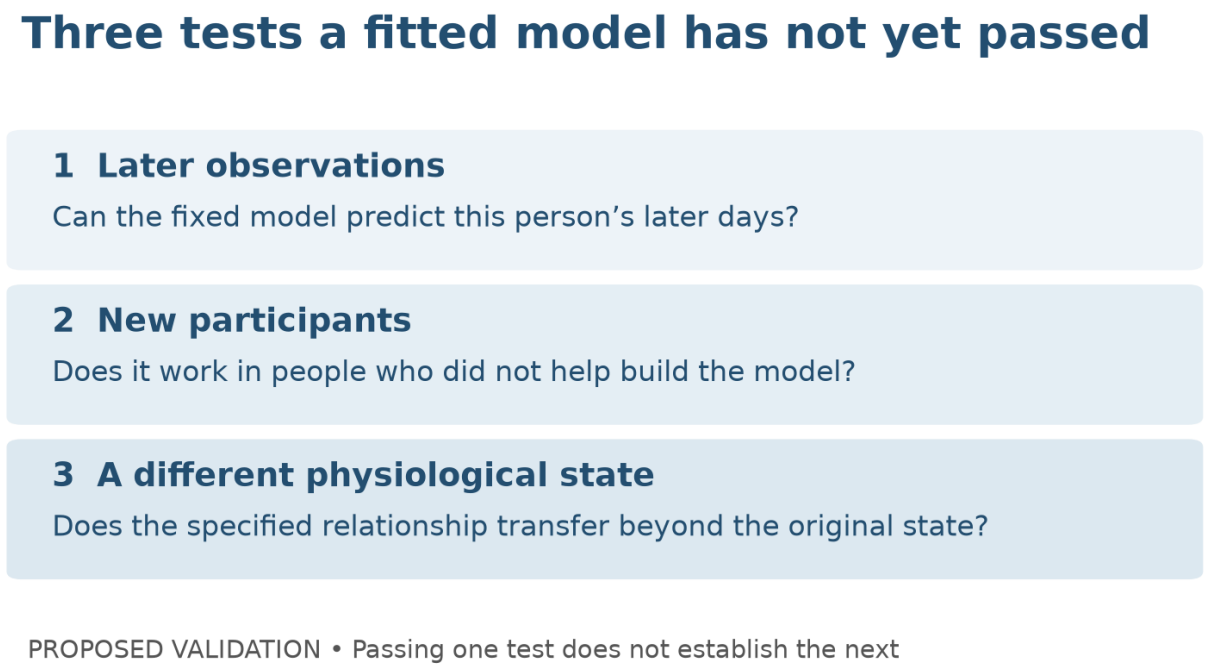
14.6 Can we put a number on resilience?

We can propose one, provided we are clear about what it measures. The Neurophysiological Resilience Index (NRI) in Appendix J summarizes how far a specified cognitive outcome falls below its expected reference trajectory, and for how long, within a fixed observation window. It is a proposed summary of a response, **not a clinical scale** or a measure of how healthy someone is.

Existing resilience research shows why the definition matters. A study of 170 older adults after knee replacement compared recovery trajectories with outcomes relative to model predictions. These approaches sometimes described the same person differently. A study in 248 adults combined cortisol, blood pressure and heart-rate responses to a laboratory stressor; that index is a precedent for measurement, not a validated ADHD score. Neither study validates the formula proposed here. [178, 180]

I would keep the curve and the four-part profile beside the number: functional margin, physiological reactivity, cognitive sensitivity and recovery. They answer different questions and should not be averaged into a single biological score. A strong bodily response can be appropriate. Maintaining task performance can also require considerable effort or leave less capacity for the rest of the day.

The first test is whether the number is reproducible and predicts independently measured later function better than simpler measures such as baseline performance, the largest decline or sleep and pain alone. If it cannot do that, a more elaborate name will not make it useful. The word neurophysiological describes the hypothesis being tested; the task-derived score alone **does not establish a physiological mechanism**.



Visual guide C. Later observations, new participants and a new physiological state test different claims. The model and preprocessing rules must be fixed using development data before each evaluation. These are proposed validation steps, not completed results.

14.7 AI assisted research and the next tests

AI is becoming useful at finding candidates that researchers could not realistically test one by one. For this hypothesis, genes, immune signals, microbial products and brain circuits generate far more possible relationships than any single study can examine. A pattern still needs testing. We still need to know whether it survives a new dataset, an experiment and, eventually, a meaningful test in people. The examples below were published in 2024–2026 and span very different stages of that process. None establishes an AI-discovered treatment for ADHD.

Predicting molecular interactions. AlphaFold 3 extended structure prediction to complexes containing proteins, nucleic acids and small molecules. This gives researchers a way to

propose how a peptide or drug might fit a biological target. For our framework, that could help prioritize experiments on receptors involved in immune or metabolic signalling. A predicted fit does not establish receptor activation, safe exposure, access to the brain or a useful clinical effect. Those are **separate measurements**. [144]

Reading the regulatory genome. AlphaGenome, published in January 2026, accepts about one million DNA bases at a time and predicts several kinds of gene regulation. It matched or exceeded the strongest external models in 25 of 26 variant-effect evaluations. This is a result across benchmark tasks, not a 96% diagnostic accuracy rate. It could help choose which non-coding variants to investigate when looking at overlapping conditions. It does not show that ADHD, endometriosis and autoimmunity share one causal mechanism. That would require disease-specific genetic evidence, appropriate cell types and experimental validation. [145]

Designing antimicrobial peptides. A 2025 ProteoGPT study combined sequence generation with laboratory testing. Of 42 generated peptides synthesized for initial testing, 26 showed potential bacterial inhibition in a disk-diffusion assay. That is a selected laboratory sample, not a patient response rate or the success rate of every AI proposal. Further-selected candidates were tested in mouse infection models. These experiments supply no evidence that those peptides treat chronic cognitive symptoms or restore a human gut microbiome. Antimicrobial activity can be useful while its effects on a person's wider microbial community remain uncertain. [146]

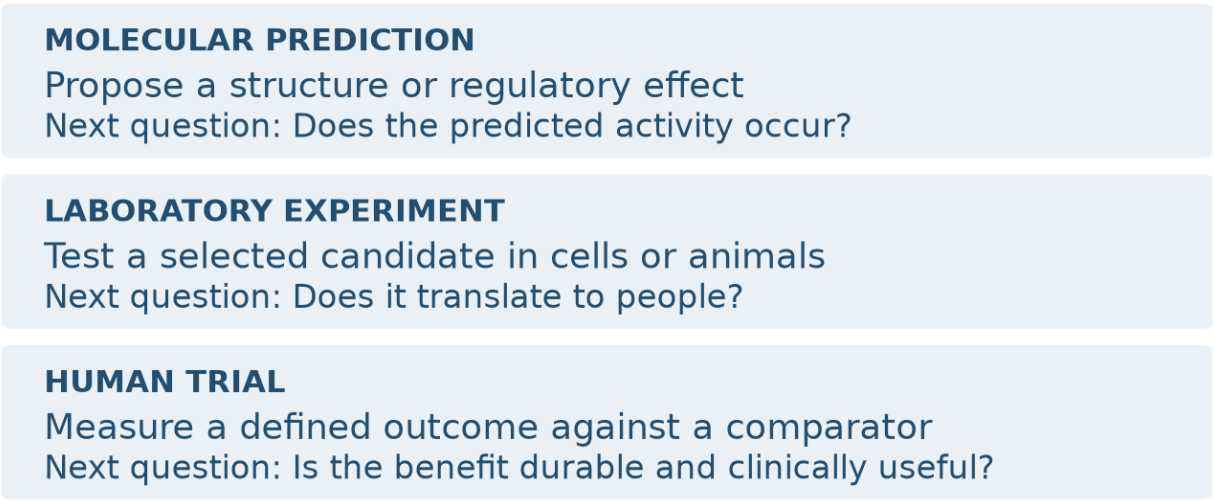


Figure 31. Conceptual guide to the stages represented by the AI examples. These are separate research programmes, not one demonstrated discovery pipeline. Success at an earlier stage does not establish ADHD treatment efficacy. [144, 145, 146, 147]

An AI-assisted candidate in a human trial. Rentosertib is a small molecule, not a peptide. Its target selection and molecular design used AI. A 2025 phase 2a trial randomized 71 people with idiopathic pulmonary fibrosis across three doses and placebo for 12 weeks.

Safety was the primary endpoint. In the 60 mg once-daily arm, mean forced vital capacity changed by +98.4 mL (95% CI 10.9 to 185.9), versus −20.3 mL (−116.1 to 75.6) with placebo. These are within-arm changes, not the confidence interval for a treatment contrast. Sixteen participants withdrew across the trial. This is an early clinical signal that justifies further testing; it does not establish durable benefit, cognitive benefit or superiority of AI drug development over conventional discovery. [147]

Finding clinically useful subgroups. Machine learning also operates at the level of people. A 2024 brain-imaging study identified six circuit profiles in 801 people with depression and anxiety; treatment-related analyses involved 250 participants. The profiles differed in symptoms, cognitive performance and treatment response. These are candidate ways to divide a heterogeneous clinical population, not six settled diseases. We face the same problem with any proposed state-sensitive subgroup: it must work in new people, beyond the dataset used to define it. A trial must also test whether using the classification improves treatment decisions. [148]

An adjacent 2025 study gives a more direct connection to cognition and an ADHD-related medication. Seventeen participants with a prospectively defined cognitive subtype of depression completed at least six weeks of guanfacine immediate-release treatment. Circuit measures and cognitive performance improved, and 76.5% met the study's depression-response threshold. This was an open-label, single-site, per-protocol study without randomization or placebo. Expectation, repeated testing and selection of completers remain plausible explanations for some change. It is a useful treatment hypothesis, not proof of an AI-selected therapy or evidence that depression and ADHD are the same disorder. [149]

I would start by asking whether repeated sleep, pain, autonomic, medication and cognitive measurements predict a person’s next difficult day. Compare that model with their usual performance and recent sleep. For later-day prediction, preserve time order; for new-person prediction, keep participants separate between training and evaluation. Fit preprocessing rules on training data only, so evaluation data cannot influence the model. Report uncertainty and test performance across sites, devices and demographic groups. If the added measurements contribute little, use the simpler model.

The stronger test comes after prediction. If a model identifies an actionable state, does acting on that information improve everyday function in a controlled study? A plausible explanation generated after seeing the result is not enough. The predicted sequence, comparison and failure criterion need to be specified first. AI may help choose what to test. The result still has to stand up to the same checks.

14.8 The next questions worth prioritizing

I would prioritize the gaps that could change the core explanation before expanding the list of associated conditions. These are proposed tests, not findings.

First, establish the daily signal. Can measured sleep and physiological change predict later attention and independently recorded daily function, beyond medication timing, mood, pain,

workload and prior performance? Predefine what improvement in prediction would be useful. If a simpler model performs just as well, the extra measurements have not earned their place.

Second, test the ADHD interaction directly. Follow people with and without established ADHD through documented infection and recovery, ideally with pre-infection measurements. Distinguish a lower starting level, a larger decline and slower recovery. Mono is a useful candidate because the illness has a recognizable onset; the studies here do not yet supply that comparison.

Third, connect a proposed mechanism in the same people. For the immune and histamine account, a peripheral marker, a credible measure of the proposed intermediate process, and subsequent cognition need to be linked with explicit timing. For the autonomic account, circulation, symptoms and cognition need joint measurement. Findings from separate cohorts cannot be assembled into proof of either chain.

Fourth, separate treatment benefit from its route. In a trial of treatment for an overlapping condition, measure cognition as well as disease improvement, sleep, pain and medication exposure. A GLP-1, immune or vagal intervention improving its primary disease outcome does not establish an ADHD benefit. Even a cognitive benefit would leave open whether it arose through symptom relief, a direct drug effect or the proposed pathway.

Finally, make the weakest days observable. Reduce assessment burden, record why data are missing and test whether monitoring itself changes reports. Include enough variation in age, sex, treatment and baseline impairment to state who the result applies to. The gut and environmental modules remain worth investigating, but adding more exploratory markers before these design problems are addressed risks making the explanation harder to test.

15. Interpretation and clinical boundaries

The proposed studies give the framework somewhere to fail. Before concluding, we need to say what evidence would count, which gaps remain and where the clinical boundaries sit.

A stronger account must predict more than a general “bad day.” Compare the same state–cognition relationship in people with and without ADHD and with relevant comorbidities. If the effect is similar across groups, it may explain general cognitive vulnerability without explaining anything specific to ADHD. If it predicts only self-report but not task performance, revise the outcome claim rather than treating the measures as interchangeable.

The measurement itself is part of the test. In a cognitive ecological momentary assessment study, 198 adults with type 1 diabetes and 128 community participants completed brief tests three times daily. Between-person reliability was 0.95–0.99, while **within-person** reliability ranged from 0.20 to 0.80 across tests and samples. A test can therefore separate people consistently while being less dependable for tracking change within one person. These samples overlap the broader research programme cited in Section 4.6; they are not an independent clinical replication. The proposed resilience score needs its own reliability study. [194]

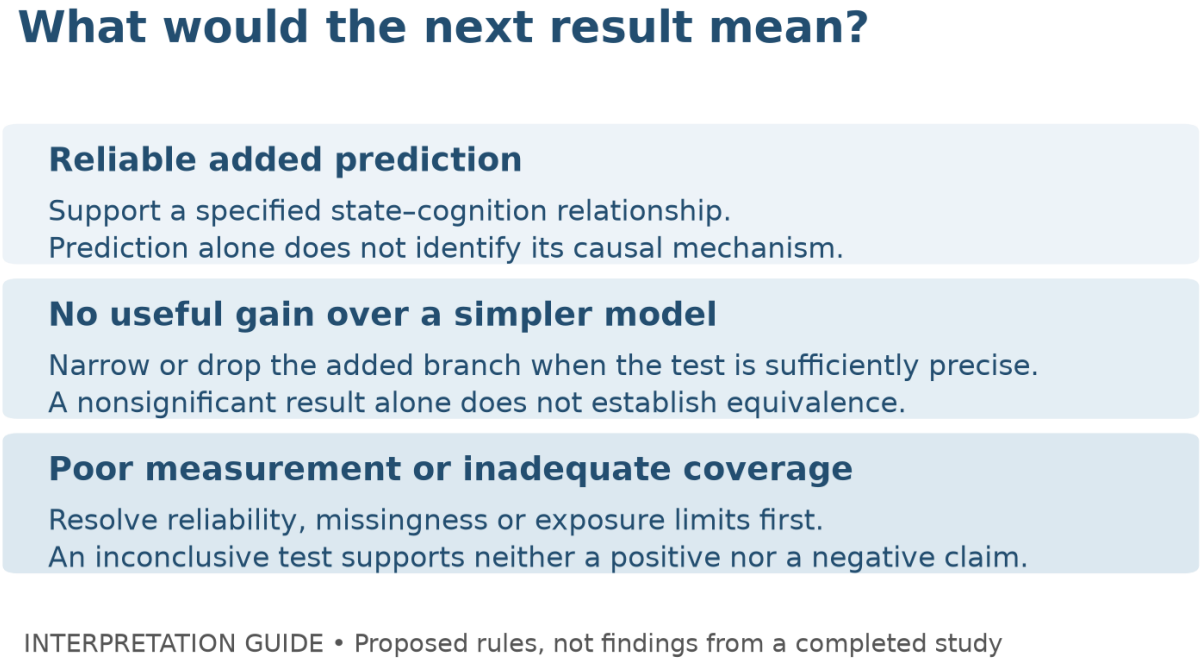
15.1 What would support—and weaken—the framework

Support would require reliable change, temporal specificity, prediction in new observations and an effect large enough to matter. The model must compete with simpler accounts that include baseline performance, context, sleep and medication. Improvement in a fitted dataset is not enough. Relationships should predict later observations without using those future observations to set means, cutoffs or subgroups.

Table 7. Results that would weaken each branch of the framework.

Claim	A result that would weaken it
Functional margin adds information	Baseline performance and context predict independent functioning equally well; the new construct adds nothing.
ADHD changes bodily reactivity	Adequately precise challenge data exclude a meaningful difference in the measured physiological response.
ADHD changes cognitive sensitivity	Estimates exclude a meaningful difference in cognitive response on the prespecified scale and exposure range.
Recovery adds an independent signal	Continuing physiology, treatment timing, practice and context account for the apparent delay.
A wider mechanism is needed	Additional measurements do not improve prediction beyond the simpler model.
Individual coupling is reproducible	Relationships disappear in later observations or another cohort, or reflect measurement error.
A pathway is shared across states	A prespecified relationship fails to transfer to the independent state; retain a narrower, state-specific account.

A nonsignificant interaction **does not establish equivalence**. Meaningful bounds and adequate precision are necessary to exclude effects. Equally, “some subgroup might respond” cannot be an unlimited escape route. A negative branch should remain negative rather than be relabelled as support for another branch. [60]



Visual guide D. A result must be interpreted alongside its precision and measurement quality. Table 7 gives branch-specific ways the framework could fail. These decision rules do not supply diagnostic criteria or treatment recommendations.

15.2 The important tests still missing

The cited evidence does not demonstrate a complete peripheral immune-to-central histamine-to-ADHD cognition chain in the same people. It does not establish that GLP-1 or vagal interventions improve ADHD through immune modulation. Nor does it establish a general ADHD-specific response across infection, migraine, hormones and autonomic challenge. More confident wording will not fill those gaps. The added fatigue, sleep-surgery and lifestyle findings narrow the predictions; they do not fill the missing ADHD-specific causal tests.

The famotidine report gives the sample flow and mean enrollment timing: 60 randomized, 50 analyzed, 45 completed, with enrollment about 29 days after symptom onset. The authors report that its registry did not explicitly distinguish primary and secondary outcomes; an independent registry comparison remains outstanding. The adult waking-EEG report remains supported here at abstract level rather than by a complete methods audit; exact regional spectroscopy effects also need fuller review. Appendix I records the scope of this audit and two correction notices whose full contents remain unchecked. The paper still needs independent scientific review, and the ADHD-specific experiments still need to be done.

15.3 Clinical boundary

This framework supplies **no diagnostic criteria** for autoimmune brain disease, MCAS, POTS, impaired cerebral perfusion or any other condition. It is not a reason to replace clinical assessment with a wearable score or symptom resemblance. It does not recommend antihistamines, GLP-1 drugs or improvised electrical stimulation for ADHD. An implanted vagal device, a research auricular protocol and a general TENS machine are not interchangeable interventions.

No personal medical records, medication response or wearable trace is analyzed as data. Prescribed treatment should not be withdrawn merely to make an experiment cleaner; any research change requires appropriate clinical and ethics oversight. The same applies to exposure challenges, particularly in participants vulnerable to delayed post-exertional worsening.

The broader review adds no indication for immune suppression, antimicrobial treatment or microbiome intervention for ADHD. It supports attention to diagnosed illness and treatable symptoms while keeping the proposed mechanisms open to disconfirmation. Mental health care, social support and ordinary environmental changes do not require a biomarker explanation to matter.

16. Conclusion

I began with a familiar problem: the same person can manage a task one day and struggle with it the next. The evidence gives us reasons to measure sleep, pain, treatment, hormones and other bodily changes alongside attention. It does not give us one hidden disease that explains them all.

Following those connections matters only if it makes the original question more precise. The same performance loss can have different practical consequences. A stronger bodily response, a larger cognitive change and slower recovery remain separate possibilities. The studies of immune signalling, metabolism, circulation and infection suggest ways to distinguish them; the wider health and developmental evidence shows why context and baseline demands cannot be left out.

What I want to know is whether these measurements help us understand and anticipate a difficult day. If sleep, pain, medication and context explain it just as well, we should say so. If another measured process adds something reliable, that is the part to pursue. The point is a better explanation of someone’s life, with a test that can prove us wrong.

Appendix A. Technical model and measurement safeguards

A1. Separate within-person change from between-person differences

Within-person change and between-person differences answer different questions. Multilevel estimation separates them; neither establishes causation nor guarantees prediction for a new individual. [61, 62]

For participant i at assessment t, a proposed association model is:

$$g(\mu_{it}) = b0 + u_i + bA*A_i + (bW + v_i)*Xwithin_{it} + bB*Xmean_i + gamma*(A_i*Xwithin_{it}) + theta*C_{it} + f(time, practice)$$

Here mu is the expected outcome; A indicates ADHD ascertainment; Xwithin is deviation from the person's mean; Xmean captures between-person differences; and C contains prespecified context and treatment variables. Random intercepts and slopes provide partial pooling. For omissions, a binomial logit model requires known target opportunities.

The ADHD-by-state term estimates differential sensitivity on the chosen scale. Report marginal contrasts with uncertainty and exposure overlap, not coefficients alone. Physiology, independent daily function and a validated functional threshold need separate models. Do not extrapolate beyond states observed in both groups.

Recovery needs repeated observations in both physiology and cognition. One follow-up cannot establish a trajectory. Prespecify plausible lags or nonlinear effects; retain an additive model as a legitimate competitor.

A2. Prediction is not mediation

Specify the causal question before adjustment. Sleep may carry an allergy-related effect or confound a different exposure. Total-effect and direct-effect questions require different assumptions. Simultaneous measurement cannot resolve direction; temporal precedence alone does not establish causation.

Record treatment changes that follow symptoms. For prediction, learn centering, scaling, cutoffs and subgroups only from training data. Later observations test within-person prediction; a separate cohort tests transport. Keep those claims distinct.

A3. Reliability, missingness and participant burden

Robust group effects do not guarantee reliable individual changes. Quantify trial-level noise, practice, speed–accuracy trade-offs and slope uncertainty. Standardize device latency and presentation, and record interruptions. Calibration is helpful but does not automatically validate daily change. [63, 64]

Missingness may depend on symptoms or workload. Non-wear is missing information, not poor sleep. Record reasons where feasible, predefine usable data, test alternative missingness assumptions and minimize burden.

Estimate slopes with partial pooling before clustering. Require later prediction and replication before naming response patterns; continuous heterogeneity may fit better than subtypes. Do not build an unrestricted physiological-load score after inspecting outcomes.

A4. Repeated measurement can change what gets reported

In 90 adolescents treated for ADHD, self-rated symptoms declined across 17 days of questionnaires sent four times daily. Reports were also lower after completed rather than missed prompts. Participants were not randomized to symptom monitoring versus an appropriate control, so this does not establish that questionnaires treated ADHD. Changing circumstances, selective completion and the act of monitoring could all contribute. It is nevertheless directly relevant to a study that relies on repeated self-report. [176]

I would record prompt frequency, task exposure and assessment day; examine an initial familiarization period; and use independent outcomes where feasible. A later feasibility comparison could randomize monitoring intensity to estimate reactivity and burden. Improvement while being measured should not automatically be attributed to recovery, a drug or a physiological mechanism.

Appendix B. A short glossary

Table B1. Terms used in the manuscript.

Term	Meaning in this paper
Allopregnanolone	A progesterone-derived neuroactive steroid that modulates GABA-A signalling. Its effects depend on context; a blood level is not a direct measure of its action in the brain.
ATP / phosphocreatine	Molecules involved in cellular energy transfer. Their measured ratio does not directly reveal energy-production rate.
Autonomic regulation	Nervous-system regulation of involuntary bodily functions, including cardiovascular responses.
Biomarker	A measured biological characteristic. An abnormal value alone does not establish a diagnosis, causal pathway or cognitive impairment.
Blood–brain barrier	A regulated interface between circulation and brain tissue; not a diagnosis inferred from cognitive symptoms.
Brain fog	A subjective report of reduced mental clarity. It can be important even when a brief cognitive test is normal.

Term	Meaning in this paper
Cognitive sensitivity	The proposed change in cognition for a comparable measured physiological change; distinct from starting margin or bodily reactivity.
Confidence interval	A range produced by a statistical procedure. A 95% procedure covers the fixed target value in 95% of repeated samples under its assumptions; it is not a 95% probability assigned to this particular interval.
Confounding	An alternative explanation arising when other causes influence both the exposure and the outcome. Adjustment addresses only what the design and assumptions can support.
Credible interval	A Bayesian range containing a specified proportion of posterior probability, conditional on the data, model and prior assumptions.
CSF	Cerebrospinal fluid, which surrounds the brain and spinal cord and moves through connected spaces. Its flow is distinct from cerebral blood flow and BBB permeability.
Cytokine	An immune-signalling molecule. A blood level does not identify its source or establish what is happening in the brain.
Ecological momentary assessment (EMA)	Brief repeated assessment during everyday life, often through a phone. Between-person reliability does not establish reliable within-person change.
Estradiol	One of the estrogen hormones. A measured level, a change over time and sensitivity to that change are different questions.
Executive function	Processes involved in maintaining goals, directing attention, holding information in mind and controlling responses; not one universal ability score.
Fecal microbiota transplantation	Transfer of screened donor stool material to a recipient. Its risks, purpose and evidence depend on the clinical indication.
Functional margin	A proposed distance from an independently meaningful task requirement; not a validated ADHD biomarker or an intelligence measure.
Genetic correlation	A population-level correlation in genetic effects across traits. It is not an individual risk or the percentage of people who have both conditions.
GLP-1	Glucagon-like peptide-1, a metabolic signalling hormone. Receptor-agonist studies must be interpreted by the actual population and outcome tested.
Hazard ratio (HR)	A comparison of instantaneous event rates among people still at risk. It is not a ratio of cumulative probabilities.
HbA1c	A blood measure reflecting glycemic exposure over preceding months. It does not show the timing of short glucose excursions.
Held-out prediction	Testing a model on observations that were not used to develop it.

Term	Meaning in this paper
Helminth	A parasitic worm. Hookworms used in experimental trials are distinct from gut bacteria.
Histamine receptor	A molecular target through which histamine signals. H1, H2 and H3 interventions are not interchangeable, and effects depend on location.
HOMA-IR	An indirect estimate of insulin resistance calculated from fasting glucose and insulin; not a direct brain measurement.
HRV	Heart-rate variability: variation in intervals between beats. A cardiac measure, not a direct assay of brain inflammation or universal vagal engagement.
Hypoglycemia	Low blood glucose. Experimentally induced, clinically low values are different from ordinary variation after meals.
Immunometabolism	The interaction between cellular metabolism and immune-cell behaviour.
Insulin sensitivity	Responsiveness to insulin. Measures of peripheral metabolism and experimental brain responses are not interchangeable.
Interoception	Detection and interpretation of signals from inside the body.
Mast cell	An immune cell that can release histamine and other mediators. Its involvement in a tissue is not automatically MCAS.
Mastocytosis	A disorder involving abnormal accumulation of mast cells. It is distinct from an inference of mast-cell activation based on nonspecific symptoms.
Mediation	A proposed pathway through an intermediate variable. Statistical mediation alone does not prove that pathway is causal.
Microglia	Resident brain immune cells whose functions include responses to injury and changes in the local tissue environment.
Microplastics and nanoplastics	Small plastic particles classified by size; boundaries and detection methods vary. Particle count and mass are different measurements.
mTOR / AMPK	Cellular signalling systems involved in nutrient/energy regulation; named candidates here, not universal treatment targets.
Neurophysiological Resilience Index (NRI)	An unvalidated proposal summarizing the depth and duration of a specified task deficit over a fixed window. It is not a diagnostic or general health score.
Odds ratio (OR)	A ratio of odds between groups. Odds are probability divided by one minus probability; an OR is not a risk ratio or a percentage-point difference.
Orthostatic	Related to becoming or remaining upright.
Physiological reactivity	The size or pattern of the measured bodily response to a specified challenge.

Term	Meaning in this paper
Physiological state	A specified, measured aspect of current bodily functioning, such as prior sleep or an orthostatic response.
PMDD	Premenstrual dysphoric disorder. Diagnosis requires a recurring impairing symptom pattern and prospective daily confirmation; a screening result remains provisional.
Post-exertional malaise	Delayed or prolonged symptom worsening after activity; not simply ordinary tiredness during a task.
Premenstrual exacerbation	Premenstrual worsening of a disorder that is also present at other times. It should be distinguished from a more restricted cyclical symptom pattern.
Progesterone	A steroid hormone with reproductive and neural effects. Its metabolites can have effects distinct from the circulating parent hormone.
Progression-free survival	Time until disease progression or death as defined by a trial. A cancer endpoint, not a cognitive outcome.
Recovery	The observed return toward a prespecified reference after a disturbance. The observation window and outcome must be defined.
Synbiotic	A preparation combining live microorganisms with substrates intended to support microbial activity. Effects are specific to the tested preparation and population.
Vigilance	The ability to remain alert and maintain attention over time.
VNS / taVNS	Vagus nerve stimulation / transcutaneous auricular VNS. Route, device and protocol matter; implanted and ear-based interventions are distinct.

Appendix C. Quantitative evidence at a glance

These studies answer different questions. Do not rank diseases using incompatible statistics. HR compares instantaneous event rates among people still at risk; OR compares odds. Neither is an individual probability. SD and Hedges g are standardized differences, not percentages. Intervals are 95% confidence intervals unless labeled otherwise; see sources for full methods.

Table C1. Direct ADHD and hormonal-state evidence

Study and sample	What was measured and found	What it can establish
Sleep restriction Becker 2019 [9] 72 enrolled; 48 analyzed	6.5 vs 9.5 h in bed; measured sleep differed by 1.6 h. Parent-rated inattention d = 0.60.	Crossover symptom effect; adolescent inattention and continuous-performance tests did not show the same result. Adherence-selected analysis.
Sleep treatment Hiscock 2015 [12] 244 randomized	Parent ADHD score differences: 3 months −2.9 (−5.5 to −0.3); 6 months −3.7 (−6.1 to −1.2).	Imputed estimates from one trial. Teacher total scores did not show the same benefit; unblinded families and stimulant co-treatment.
Melatonin Van der Heijden 2007 [65] 105 randomized	Sleep onset 26.9 min earlier vs 10.5 min later on placebo after 4 weeks.	Improved sleep timing; no significant cognitive, behavioural or quality-of-life benefit. Selected insomnia population.
Menstrual symptom diaries Roberts [32]; Zaritsky [33]	32 community participants over 35 days; later pilot: 30 stimulant-treated participants over 35 days.	Within-person symptom associations. No validated universal percentage worsening, causal hormonal pathway or dose-adjustment rule.
Objective cycle cognition Jang 2025 [66]	102 articles; 3,943 participants; 730 comparisons. No systematic robust cycle-phase differences.	General-sample counterweight; does not exclude symptomatic subgroups or effort required to maintain performance.
H3-targeting drug Herring 2012 [27] 72 randomized	MK-0249 vs placebo AISRS difference −2.1 (−6.6 to 2.3) at 4 weeks.	Primary outcome null. Methylphenidate comparator improved symptoms. Histamine involvement alone does not establish treatment efficacy.

Sample counts are not interchangeable: an enrolled cohort may be larger than the analysis for a particular endpoint. The melatonin trials and behavioural sleep intervention test different interventions. AISRS: Adult ADHD Investigator Symptom Rating Scale.

Cross-system overlap

Table C2. Co-diagnosis and shared liability

Study and sample	What was measured and found	What it can establish
Endometriosis Gao 2020 [34] 854,361 women	14,144 endometriosis diagnoses. Sibling-comparison ADHD HR 1.98 (1.34–2.93).	Later recorded ADHD diagnosis; not proven disease onset, impaired cognition or an immune cause. Within-family comparison reduces some confounding.
Endometriosis genetics Li 2026 [35]	ADHD–endometriosis genetic correlation: 0.21 using LDSC; 0.17 using HDL.	Two methods on source GWAS data, not two independent cohorts. Correlation is not individual risk or a percentage of shared causation.
Migraine cohort Hsu 2022 [67] 81,441 per group	462 vs 212 newly coded diagnoses. Adjusted HR 1.92 (1.64–2.34). Ages 18–29: 1.31 (0.58–2.98).	Overall association, imprecise young-adult estimate. No cognitive outcome; variable follow-up. Counts do not estimate lifetime prevalence.
Atopic dermatitis Strom 2016 [68]	354,416 children: adjusted ADHD OR 1.14 (1.03–1.26). 34,613 adults: OR 1.61 (1.25–2.06).	Cross-sectional surveys with healthcare-use adjustment. Itch, sleep and ascertainment remain possible routes; no mast-cell mechanism established.
Dermatitis synthesis Cheng 2024 [69]	ADHD pooled OR 1.28 (1.18–1.40). Review: 24 studies, 71.4 million subjects across ADHD AND autism analyses.	The overall denominator is not the ADHD-only denominator. Large sample size does not remove confounding or prove direction.

HR and OR are different statistics. A diagnostic association cannot be converted into an expected cognitive decline. Shared datasets within syntheses should not be counted again as independent replications. LDSC (linkage disequilibrium score regression) and HDL (high-definition likelihood) estimate genetic correlation from genome-wide association study data.

Post-COVID cognition and physiology

Table C3. Different post-COVID measurements

Study and sample	What was measured and found	What it can establish
Community cognition Hampshire 2024 [70] 112,964 completers	Unresolved persistent symptoms: −0.42 SD (−0.53 to −0.31) vs no-COVID reference group.	Objective group difference; no personal pre-infection baseline. Not proof of a specific mechanism or irreversible IQ loss.
Brain barriers Greene 2024 [44] 32 MRI participants	10 recovered; 11 Long COVID without fog; 11 with fog. Greater leakage in the fog group.	Small selected imaging comparison. Association does not establish a universal cause or a sequence leading to brain fog.
Brain energy ratios Weber-Fahr 2026 [45] 27 vs 23	Authors report lower cingulate ATP/PCr and an anterior association with cognition.	Direct brain spectroscopy, but ratio ≠ ATP-production rate. Does not establish causality; exact regional estimates remain an independent verification gap.
Orthostatic testing Jamal 2022 [71] 24 referred patients	23 tilt-intolerant; 4 POTS; 15 provoked after nitroglycerin.	Highly selected sample and provocative protocol. Not 96% population prevalence or evidence that autonomic change caused cognition problems.
Famotidine Momtazmanesh 2023 [49] 60 / 50 / 45	Randomized / analyzed / completed. Week-12 endpoint MMSE +2.44 (1.47–3.41); MoCA +2.56 (1.69–3.43).	Mean enrollment about 29 days after onset. Early convalescence; low screening scores, early exclusions and carried-forward observations.
STIMULATE-ICP 2026 [50] 778 randomized	Famotidine–loratadine: adjusted FAS fatigue difference −1.48 (−2.88 to −0.08) at 12 weeks.	Open-label comparison with no study drug; supportive care in all groups. No significant group difference at week 24. Fatigue ≠ cognition.

BBB, ATP/PCr, tilt testing and cognitive outcomes came from different samples. The two drug trials differ in illness timing and outcome; one does not replicate the other. MMSE: Mini-Mental State Examination; MoCA: Montreal Cognitive Assessment; FAS: Fatigue Assessment Scale. MMSE and MoCA are cognitive screening tests.

Interventions and clinical boundaries

Table C4. Vagal and GLP-1 findings in their actual populations

Study and sample	What was measured and found	What it can establish
RESET-RA [57] 242 randomized	ACR20: 35.2% vs 24.2%. Adjusted difference 11.8 percentage points (0.6–23.1) at 3 months.	Arthritis composite response; one-sided primary test. Implanted stimulation, not ear stimulation. Related serious events 1.6%, perioperative.
Auricular VNS cognition [58] 30 pooled studies	Overall Hedges g 0.41 (0.30–0.53); working-memory/attention g 0.185 (0.043–0.327).	Mixed populations and protocols. Laboratory cognition signal, not established ADHD efficacy, mechanism or everyday durability.
Auricular VNS HRV [59] 16 studies	g 0.014; Bayesian shortest credible interval −0.103 to 0.132.	No reliable acute cardiac HRV effect under tested conditions. HRV is not a direct brain or inflammation assay.
SELECT [73] 17,604 randomized	Cardiovascular events 6.5% vs 8.0%; HR 0.80 (0.72–0.90).	Overweight/obesity plus cardiovascular disease, no diabetes. Vascular benefit; not a cognition or dementia outcome.
REWIND [72] 8,828 tested	Exploratory cognition HR 0.93 (0.85–1.02); post-hoc baseline-adjusted HR 0.86 (0.79–0.95).	Type 2 diabetes with cardiovascular risk. Initial estimate null; favorable result depends on stated analysis. Not diagnosed dementia.
evoke / evoke+ [56] 3,808 randomized	CDR-SB change differences −0.08 (−0.35 to 0.20) and +0.10 (−0.17 to 0.38).	Early symptomatic Alzheimer disease, 104 weeks. Neither showed slowed clinical progression. No inference about an ADHD-to-dementia trajectory.

These results cannot be averaged into a “brain health” effect. The targets, follow-up, populations and endpoints differ. Mouse receptor experiments [31, 51, 52, 55] supply mechanistic questions, not human efficacy estimates. ACR20 requires at least 20% improvement in tender and swollen joint counts and in at least three of five other clinical measures; it is not a measure of cognition.

Table C5. Medication benefit: four different outcomes

Study / population	What was measured	What the result permits
Symptoms, children Cortese 2018 [76]	Clinician-rated SMD: amphetamines −1.02 (−1.19 to −0.85); methylphenidate −0.78 (−0.93 to −0.62); atomoxetine −0.56 (−0.66 to −0.45).	Closest to 12 weeks; 10,068 children/adolescents in efficacy analyses. Negative means fewer symptoms. Network estimates include indirect comparisons.
Symptoms, adults Ostinelli 2025 [77]	Clinician-rated SMD: stimulants −0.61 (−0.71 to −0.51); atomoxetine −0.51 (−0.64 to −0.37).	Short-term placebo comparisons. Review included 113 trials across treatment types, not 113 medication trials. Not a measure of IQ or percentage improved.
Cognitive tasks Vertessen 2022 [79]	31 studies, 804 children. Methylphenidate effects d = 0.20 to 0.73 across domains.	A range of domain estimates, not one pooled effect or confidence interval. Executive-function benefit did not consistently increase with dose.
Classroom learning Pelham 2022 [80]	173 children: 37% more arithmetic per minute; 53% fewer rule violations. No extra curriculum-learning gain.	Randomized crossover. Author-reported percentages; structured summer setting. Productivity, behaviour and acquisition of new knowledge differ.
First health events Zhang 2025 [81]	Suicidal behaviour: 14.5 vs 16.9 per 1,000 person-years; adjusted IRR 0.83 (0.78–0.88). First accidental injuries: IRR 0.98 (0.96–1.01).	148,581 newly diagnosed people; two-year target-trial emulation. Observational, not randomized. Injury interval includes no difference; rates are not two-year percentages.

Parentheses give 95% confidence intervals unless stated otherwise. SMD/d are standardized differences, not percentages. IRR is an incidence-rate ratio. These endpoints cannot be combined into a single treatment-benefit score.

Table C6. Medication effects on the rest of the state

Study / population	What was measured	What the result permits
Objective sleep Kidwell 2015 [82]	9 studies, 246 young people. Adjusted standardized effects: latency +0.54; efficiency −0.32; total sleep time −0.59.	Stimulant versus control conditions. No intervals reproduced here. Small evidence base; not every person experiences all three effects.
Pulse and pressure Farhat 2025 [83]	Adult methylphenidate: pulse +4.37 bpm (3.16–5.59); systolic pressure +1.66 mmHg (0.38–2.93).	102 trials; 13,315 young people and 9,387 adults overall. These adult estimates had very low confidence; median trial duration 7 weeks.
Autonomic context Green 2013 [84]	27 POTS patients, crossover. Standing heart rate 121 ±17 vs 105 ±15 bpm at 2 hours after atomoxetine vs placebo; p = .001.	± values are standard deviations. Acute experiment, not ADHD treatment efficacy, long-term risk or proof of a cognitive mechanism.
Longer-term CVD Zhang 2024 [85]	10,388 cases / 51,672 matched controls. >5 years cumulative use vs nonuse: adjusted OR 1.23 (1.12–1.36).	Nested case-control analysis, base cohort 278,027. Median follow-up 4.1 years. Residual confounding; odds are not absolute risk.
Migraine prevention Topiramate label [88]	Cognitive adverse reactions: 22% at 100 mg/day vs 10% placebo in 6-month trials.	Pooled reported adverse reactions, not an ADHD-specific cognitive scale. Trial dose is descriptive, not advice.
Drug exposure Atomoxetine label [86]	Fluoxetine/paroxetine: atomoxetine exposure about 6–8 times as high in normal CYP2D6 metabolizers.	Specified inhibitors and metabolizer context. Exposure is not a symptom effect size; does not apply equally to all antidepressants.

Intervals are 95% confidence intervals unless otherwise labeled. bpm: beats per minute; CVD: cardiovascular disease. Short-term physiological changes and long-term clinical events are different evidence. Official US labels are cited for specific interactions, not as jurisdiction-specific treatment guidance.

Appendix D. Who makes it into the data

A study can begin with one group and report results from a smaller one. These two pies show that difference in studies already discussed in the paper. They describe participation, not the proportion helped, harmed or cognitively impaired.

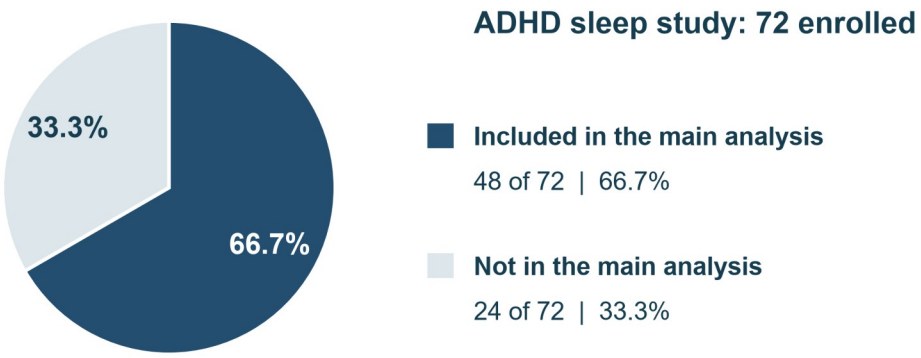


Figure 32. The sleep experiment's analysis sample. Of 72 adolescents who entered the protocol, 48 met the analysis requirements for actigraphy data and the prespecified sleep-duration contrast. The remaining 24 are calculated as 72 minus 48; this category should not be read as 24 dropouts. Percentages are calculated and rounded. [9]

This is a test in adolescents who produced the required sleep contrast. It does not tell us the average effect of assigning the protocol to every enrolled adolescent. The next study needs to account for the missed nights and incomplete measurements that will occur in ordinary life.

Completion is another selection step

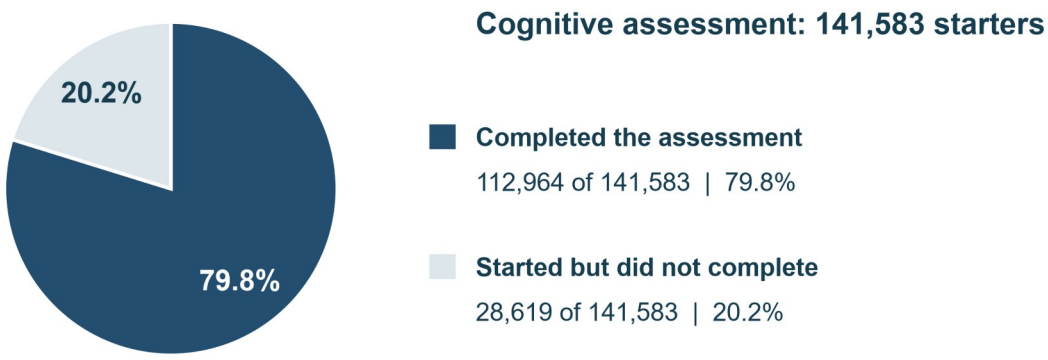


Figure 33. Completion among people who started the online cognitive assessment. Hampshire and colleagues reported 141,583 starters and 112,964 completers. The remaining 28,619 are calculated by subtraction. Percentages are conditional on starting the assessment; they are not population participation rates, infection rates or the prevalence of cognitive impairment. [70]

A large completed sample can still differ from the people who did not finish. This pie does not show why anyone stopped, whether non-completers had worse cognition, or the direction of any bias. The sample size does not resolve that uncertainty.

For our proposed study, record invitations, starts, usable assessments and reasons for missing data separately. Compare completers with non-completers where those data exist. Plan sensitivity analyses before seeing whether the missing observations help or hurt the hypothesis.

Appendix E. Two outcomes from the same trial

SELECT tested semaglutide against placebo in 17,604 adults aged 45 or older with established cardiovascular disease, BMI at least 27 and no diabetes. These waffles put two outcomes on the same visual scale.

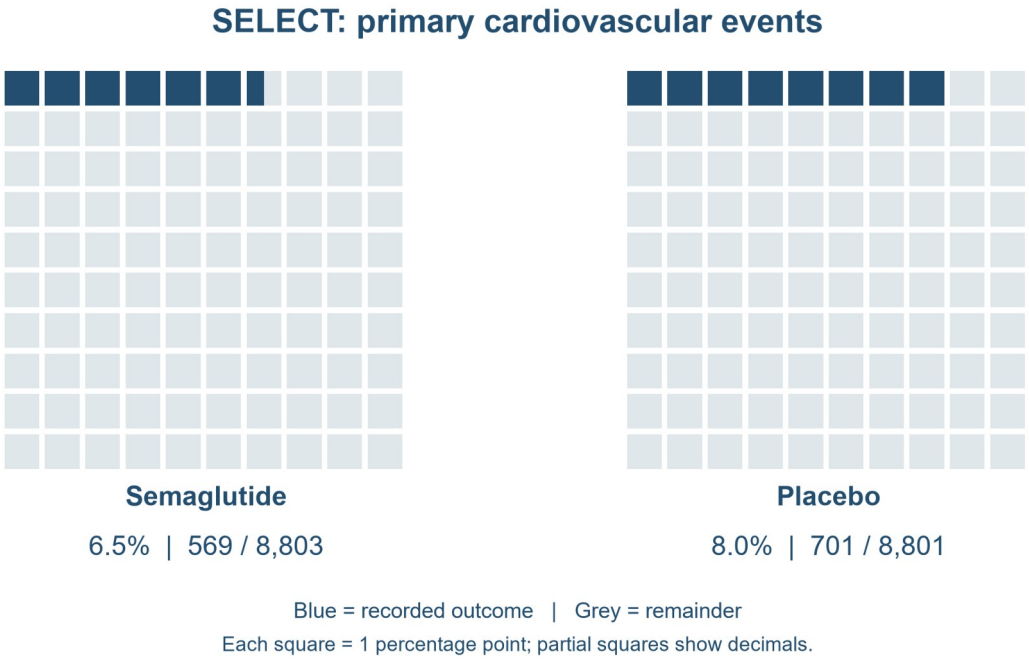


Figure 34. Cardiovascular death, nonfatal heart attack or nonfatal stroke: 6.5% versus 8.0%. Mean follow-up was 39.8 months. These are reported participant proportions, not annual risks; the time-to-event estimate was HR 0.80 (95% CI 0.72–0.90). [73]

This supports cardiovascular benefit in the studied population. It does not establish an ADHD or cognitive benefit, or identify which biological pathway produced the result.

Stopping treatment because of adverse events

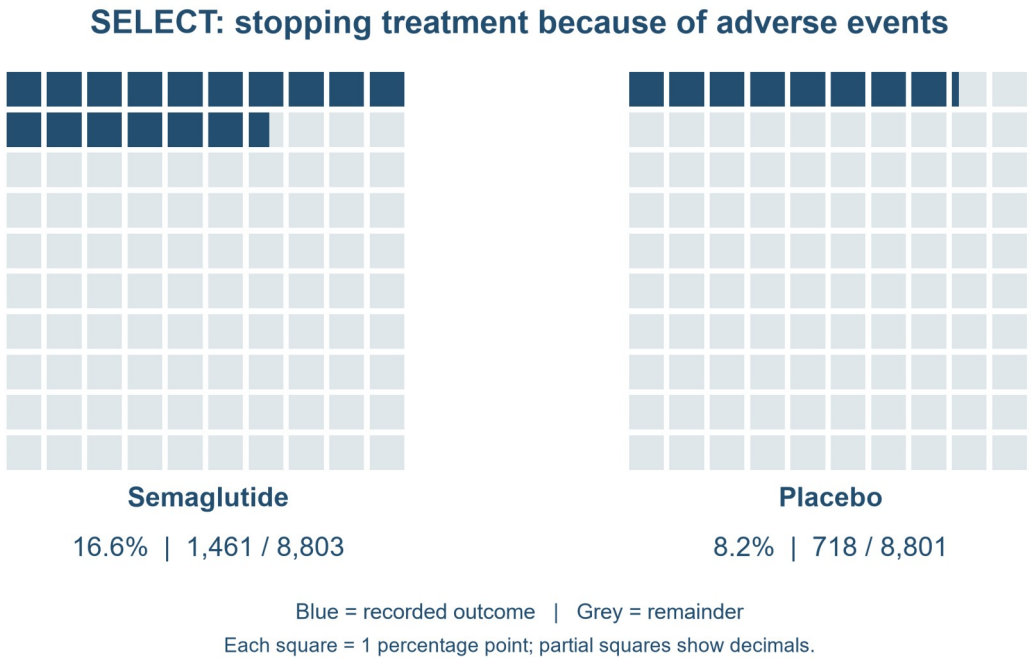


Figure 35. Adverse events leading to permanent discontinuation of trial treatment: 16.6% versus 8.2%. This is a treatment-discontinuation outcome, not withdrawal from study follow-up or the frequency of all adverse events. [73]

Grey means the specified outcome was not recorded; it does not mean symptom-free or free of other adverse events. Percentages in both charts follow the publication's rounding. The squares represent proportions, not individual participants.

The two outcomes can occur in the same person and differ in severity. Subtracting one from the other would not measure net benefit. For our hypothesis, the useful lesson is to measure function, benefit and tolerability separately.

Appendix F. Reading the overall result and the subgroups

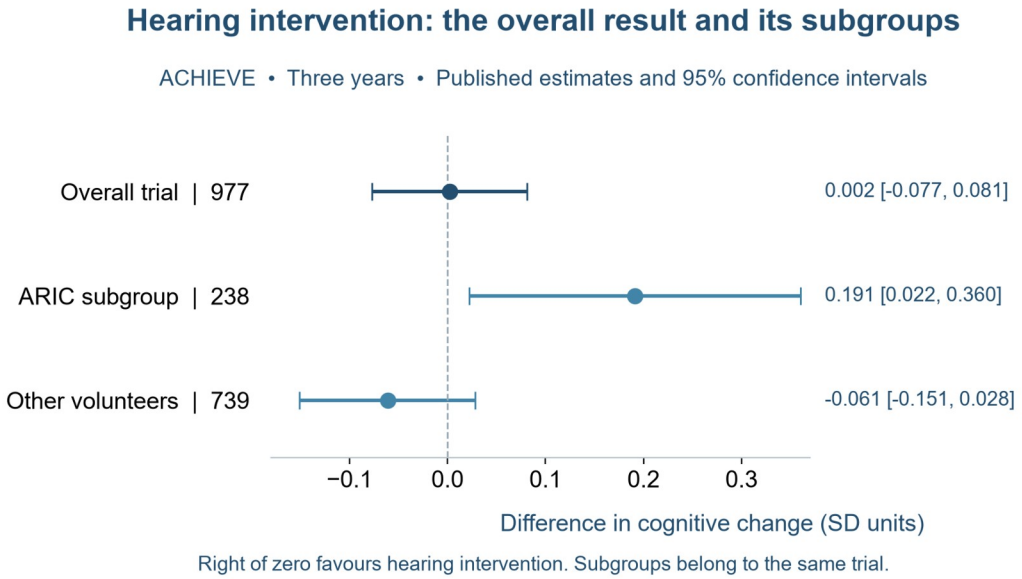


Figure 36. ACHIEVE: intervention minus control in three-year global cognitive change, in standard-deviation units, with 95% confidence intervals. Positive values favour hearing intervention. The two recruitment subgroups are components of the overall trial, not separate replications; their interaction was p = 0.010. A confidence interval crossing zero leaves the direction uncertain. [155]

The question here is whether the overall result describes every enrolled group equally well. A subgroup result can guide the next test, but it needs to remain beside the overall estimate. Pooling these three rows would count the same participants twice.

Appendix G. Recovery after mono

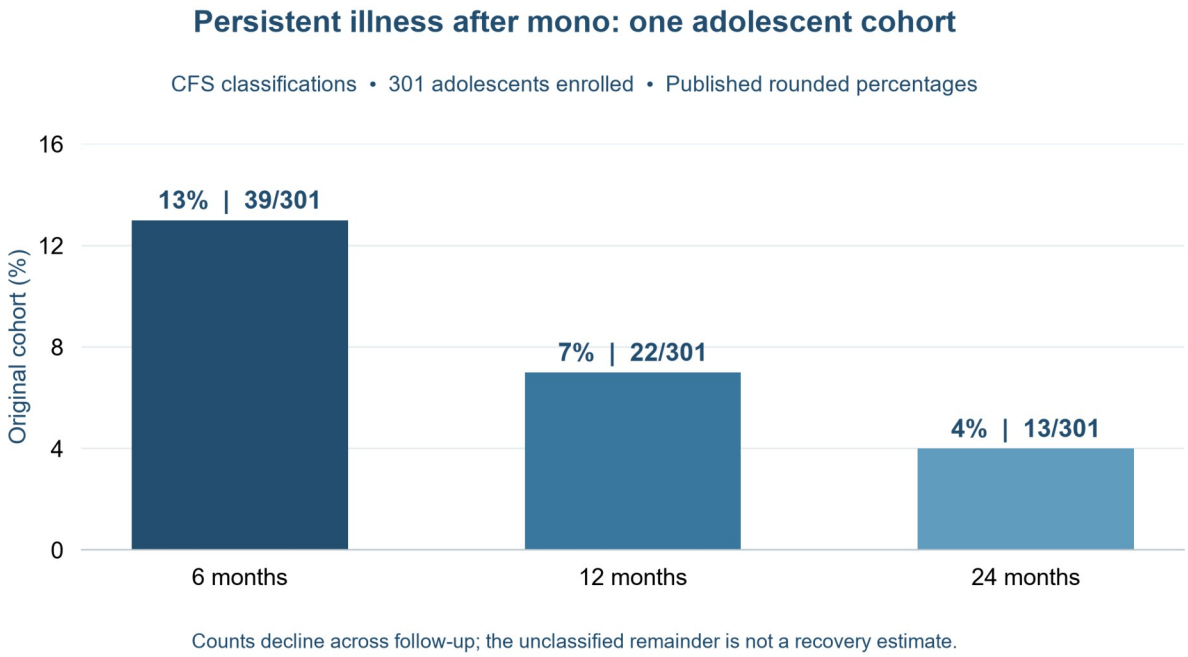


Figure 37. Chronic fatigue syndrome classifications after mono in an adolescent cohort. Labels show identified cases out of the original 301 participants; bar heights use the paper's rounded percentages. These are repeated observations from one cohort, not independent studies. Follow-up losses mean the remaining percentage cannot be labelled recovered. This chart measures clinical classifications, not cognitive impairment or ADHD risk. [160]

The chart answers a narrow question: how many participants were classified with CFS at each follow-up? It does not tell us which mechanism maintained illness, whether an individual's course was steady, or whether ADHD changed their recovery. The different college-cohort definitions and the selected seven-year follow-up belong in the text, not pooled into these bars.

Appendix H. Mono outcomes and follow-up

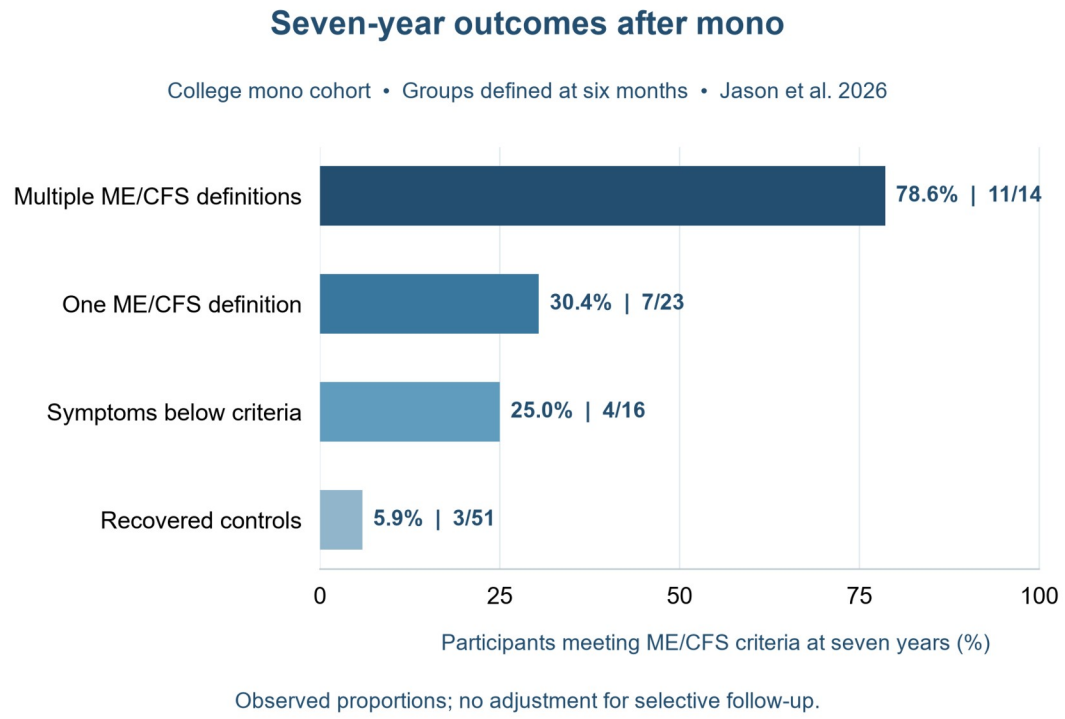


Figure 38. ME/CFS at seven years among returning participants, grouped by six-month status. Percentages are calculated from the labelled counts. These selected groups do not estimate risk for everyone with mono; group names replace the paper's definition-based "severe" and "moderate" labels. [172]

Follow-up behind the outcome estimates

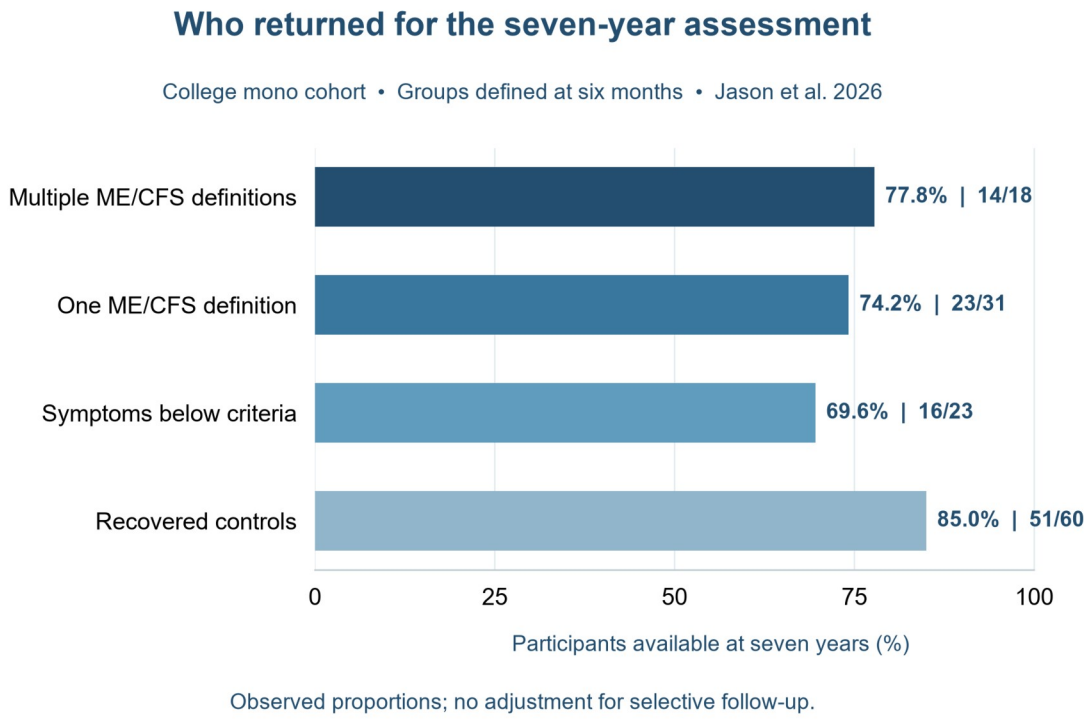


Figure 39. Follow-up availability in the same study, by six-month status. Denominators are the eligible groups with baseline data in the follow-up report, not all 238 original mono cases. Missing participants could have different outcomes. These bars show retention, not recovery. [172]

Appendix I. Fact audit of edition v20

The audit dated 6 September 2026 covered the manuscript's 177 reference entries, internal citation targets, 12 tables and 30 figure captions, with selected original-source checks. Europe PMC matched 161 entries to publication records. That confirms source identity, not the accuracy of every claim or method. Previous primary-source checks were retained where a fresh full-text check was not completed.

The changes are specific: the auricular VNS confidence interval now follows the paper's main text; the famotidine account states its reported registry limitation; and the EBV–MS estimate identifies its comparator, analysis sample and wide interval. Bibliographic details and the chart workbook's outdated guide were corrected. The displayed chart values did not change.

Eleven cited papers had indexed correction notices. Nine notices were checked against original correction text or publisher-indexed text; none changed a result reproduced here. Full correction text for REWIND [72] and the 2026 ADHD dose-effect synthesis [78] remains an access gap. Their notices are linked with the references. No retraction notice was flagged in the 161 matched records; this is not an exhaustive retraction search.

Three substantive limits remain: the adult waking-EEG methods [15], exact regional spectroscopy estimates [45], and independent comparison with the famotidine registry [49]. These are incomplete checks, not evidence that the papers are wrong. The audit does not turn this perspective into a systematic review, establish the proposed shared causal pathways, or replace independent scientific review.

Later source checks

The v28 checks covered gut interventions, hookworms, brake wear, microplastics and cancer, including negative primary outcomes and corrections. Several findings were checked through original indexed abstracts. The full microplastics correspondence was inaccessible; its discussion is limited to the stated analytical dispute. Earlier EEG, spectroscopy and registry-access limits remain.

The v29 developmental-profile addition checked five original reports: dyslexia genetics and the autism register study in full text; twin, melatonin and autism microbiome findings through indexed original abstracts. Earlier references were not fully re-audited.

The v34 checks covered semaglutide in Alzheimer's disease, vagal stimulation in rheumatoid arthritis, EBV–ADHD and endometriosis–ADHD associations, GLP-1 safety, tirzepatide contraception advice, MCAS, meningeal mast cells and dyslexia–ADHD co-reporting. Accessible primary text, original abstracts and official notices were used; this was not a full methods audit. The 2023 Doust correction [234] restores a genomic-control Methods paragraph without changing the chart's 24% and 9% values. Earlier unresolved checks remain unresolved.

Appendix J. A provisional resilience score

J1. What the measurement research adds

There is no validated NRI for this framework in the evidence reviewed here. The studies below offer design lessons. Their populations, disturbances and outcomes differ from the proposed ADHD study; their sample sizes do not determine how many participants that study needs.

Table J1. Original studies informing the proposed measurement

Study and sample	What was measured	What it tells us
Whitson 2026 [178] 170 older adults	Pain interference before knee replacement and at 1, 2, 4 and 6 months.	Trajectory groups and outcomes relative to predictions were not interchangeable. The reference defines the question.
SPRING 2025 [179] 79 older adults	Multisystem responses before major clinical stressors; cardiac and cortisol measures.	Feasibility and associations with resilience surrogates support further testing, not an ADHD classification.
Liang 2026 [180] 248 adults	Cortisol, blood pressure and heart-rate responses to the Trier Social Stress Test.	A physiological index and distress associations in a selected sample; internal modeling is not external validation.
Malinen 2026 [181] 26 trained runners	Repeated prolonged-running tests. Peak-speed change ICC 0.81; VO_2max and ventilatory-threshold changes 0.07–0.36.	Reliability depends on the endpoint. Reliable absolute values do not guarantee reliable changes. ICC (intraclass correlation) measures reliability, not treatment benefit.
Kolk 2024 [183] 109 eligible; 30 enrolled	Thirty days of monitoring after an emergency visit; 28 participants completed follow-up.	Completion among volunteers does not establish representative coverage or complete daily observations.

Source notes: SPRING is summarized from its original indexed abstract. The other four empirical reports were checked in accessible full text. COSMIN provides useful distinctions among reliability, measurement error, validity and responsiveness. Its checklist was developed for health-status measurement instruments; citing it does not certify this proposed score. [178, 179, 180, 181, 182, 183]

J2. Define the score before collecting outcomes

Start with one task outcome, one defined disturbance and one observation window. Orient the task so higher values mean better performance. Estimate the reference from repeated observations under the comparison condition, accounting for practice, time of day and the prespecified medication context. A person's single best day is not a reliable baseline.

$$B = \int_0^T \max(0, C_{ref}(t) - C_{obs}(t))\, dt / (T \times \Delta)$$

$$NRI = 100 \times \max(0, 1 - B)$$

Cref(t) is the estimated reference performance at time t. Cobs(t) is the observed-response trajectory estimated using a prespecified measurement model. T is the fixed follow-up duration. Δ is a positive scaling change in the task's own units. B is the average deficit expressed relative to Δ; the integral adds up deficit across time. Better-than-reference performance earns no extra credit.

Use an independently justified functional anchor for Δ if one exists. Otherwise call it a provisional scaling choice, report its value and test sensitivity to alternatives. Do not present an arbitrary 20-point change as a validated meaningful-change threshold. Comparisons require the same outcome scale, T and Δ, and sufficiently comparable disturbances and reference conditions.

An NRI of 100 means no estimated deficit over the observed window. It does not mean 100% brain health, and it should not be assigned when exposure or observations were inadequate. An NRI of 0 means B is at least 1; retain B and the raw area because the displayed score cannot distinguish larger burdens. Extending T with additional recovered hours can raise the displayed score even if the original disturbance is unchanged. The score stays at its boundary if B is zero or remains at least 1.

This reference asks how performance differs from the specified comparison condition. It is not the expected recovery of a clinical prediction model, and it is not automatically a causal estimate of what would have happened without the disturbance. Sleep, pain and medication may be confounders, components of the state or mediators. Their roles must be specified before adjustment; adjusting away a post-disturbance pathway can change the question. [178]

J3. A worked example with invented values

All three examples start from an expected task score of 80. Set T to 24 hours and Δ to 20 points solely to show the arithmetic. Assume an immediate decline followed by straight-line recovery, with complete, noiseless observation. These are not patient data, clinical cutoffs or expected ADHD responses.

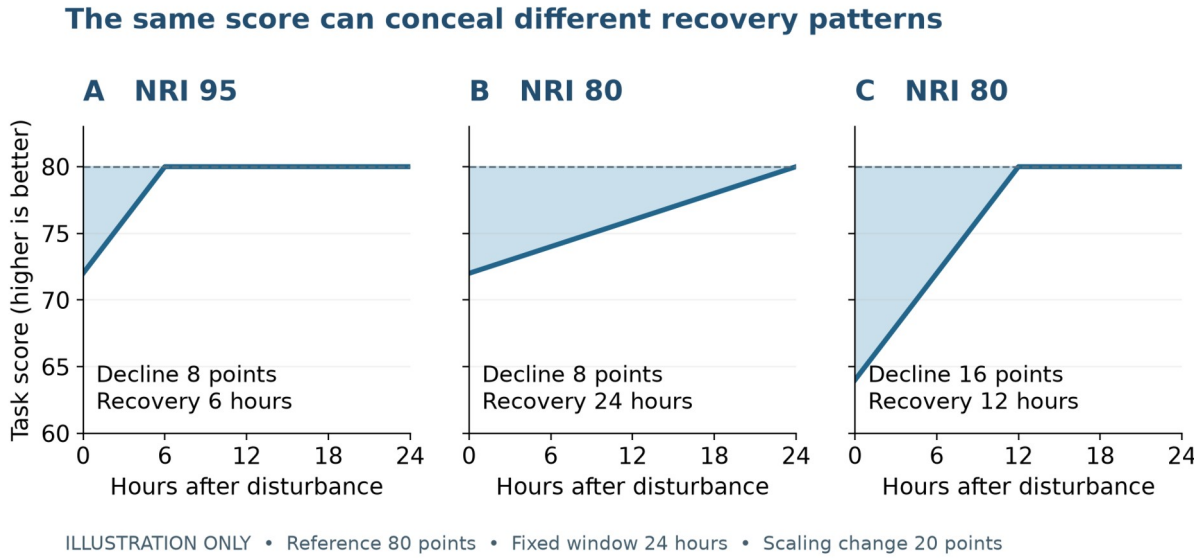


Figure 40. Depth and duration both affect the proposed index. Shading is the accumulated deficit below the reference line. B and C have equal areas and equal scores, despite different maximum declines and recovery times. All trajectories and scaling choices are hypothetical.

Example	Deficit area	B	NRI
A: 8 points; 6 hours	½ × 8 × 6 = 24 point-hours	24 / 480 = 0.05	95
B: 8 points; 24 hours	½ × 8 × 24 = 96 point-hours	96 / 480 = 0.20	80
C: 16 points; 12 hours	½ × 16 × 12 = 96 point-hours	96 / 480 = 0.20	80

Table J2. The denominator is 24 hours × 20 points = 480 point-hours in every row.

The score cannot tell us which person had more pain, a larger physiological change, less practical margin or a harder time completing ordinary tasks. Those outcomes need their own measurements. This is why the response curve and profile remain part of the result.

J4. Rules that keep the number interpretable

Measurement noise. Taking only positive deficits creates apparent burden even when a stable person fluctuates randomly around the reference. Prespecify the trajectory model and test the complete scoring procedure on stable-condition repeats and simulated null data. Report uncertainty from both reference and response estimation. If false burden is too large, revise or reject the score.

Sampling and missingness. Specify assessment timing and the longest acceptable gap. Numerical integration can use trapezoids on the estimated curve where coverage is adequate; it cannot establish what happened during an unobserved flare. Missing a substantial part of the window should produce an unscorable result or explicit bounds, not an assumption of recovery. Report scheduled and completed observations and reasons for missingness.

Recovery. Define return to reference using a prespecified tolerance and sustained observations, rather than one favorable reading. With no detectable decline, recovery time is undefined. If recovery is not seen by the end, report “not observed within follow-up” and treat the time as censored. A later relapse or delayed worsening needs enough follow-up to be observable.

Exposure and context. A smaller disturbance can produce a better score without greater resilience. Compare defined, reasonably comparable conditions or model exposure separately with held-out evaluation. Do not divide by an improvised physiological-load score. A low heart-rate or cortisol response is not inherently healthier.

The accompanying profile. Report baseline function and practical margin; the measured bodily response; the relationship of cognition to that response; and recovery. Also retain the peak deficit, raw deficit area, B, T, Δ, medication context, effort and independent daily function. Cognitive sensitivity requires repeated joint physiological and cognitive data, not just this area calculation.

The enrollment and monitoring losses in the feasibility study illustrate why missingness belongs in the design. A neat response curve from the people who can complete monitoring may leave out those with the hardest days. [183]

J5. What would make the proposal earn its place?

First, establish repeatability under reasonably stable and comparable conditions. Estimate measurement error and uncertainty at the individual level. In the runner study, some changes were much less reliable than absolute performance. That lesson transfers to measurement design; the exercise protocol and its thresholds do not transfer to ADHD or post-infectious illness. [181]

Second, lock the scoring rules and test later observations and new participants. Evaluate whether the score predicts an independently collected functional outcome beyond baseline ability, peak decline, recovery time alone and a simpler sleep–pain–medication model. Keep those outcomes separate from the measurements used to construct the index. Report calibration and useful predictive improvement, not just a statistically significant association.

Third, test generalization rather than assuming it. Start with a defined state and population. Compare ADHD and non-ADHD participants directly, distinguish an interaction from a lower starting level, and examine measurement behavior across relevant treatment and demographic groups. A score from one infection or hormonal state is not yet a stable personal trait. Performance in a new physiological state is a separate test.

Finally, ask whether a change in the score corresponds to a meaningful change in life. Reliability, validity and responsiveness are separate questions. The four-part profile is not a set of interchangeable questionnaire items; a high internal-consistency coefficient would not validate it. Nor does a component explaining considerable physiological variance prove clinical usefulness. [179, 180, 182]

Use naturally occurring variation or clinically justified observations. Do not induce illness, withdraw medication, restrict sleep or provoke exertional worsening just to obtain a score. No cutoffs, treatment choices or labels such as normal, fragile or highly resilient are justified by this proposal.

What would count against it? Apparent deficits generated by noise; poor repeatability; dependence on arbitrary timing or scaling; systematic failure on the worst days; or no useful prediction beyond simpler measures. These are reasons to change or abandon the metric, not reasons to keep adding biomarkers. The broader state hypothesis must remain testable even if this particular score fails.

Appendix K. Two scales of evidence

K1. A long-term association is not a daily mechanism

The paper asks about changes within a person, but it also needs to account for health across decades. This cohort result belongs to that longer timescale. It cannot tell us why cognition changes from one morning to the next.

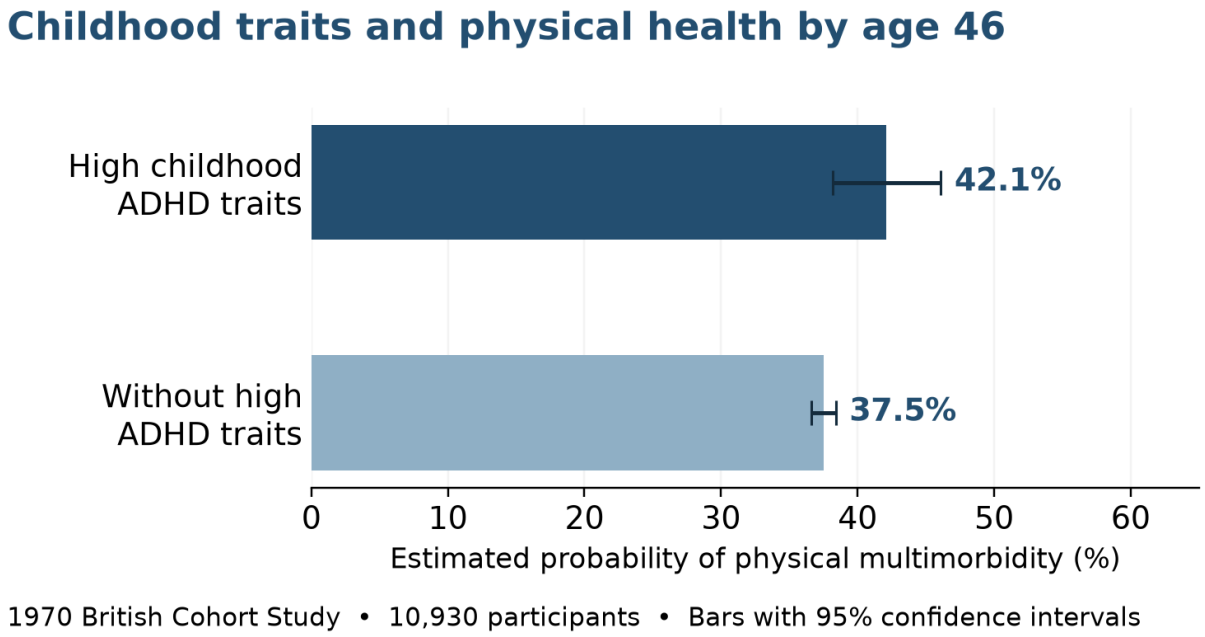


Figure 41. Model-estimated probabilities and 95% CIs for two or more reported physical conditions by age 46: 42.1% (38.2–46.1) with high childhood ADHD traits and 37.5% (36.6–38.4) without. The difference is 4.6 percentage points. These are not raw case counts or a causal treatment contrast. [184]

How to use this result

Keep the levels separate. A population association can identify a group whose health deserves attention. A repeated-measures study can test whether a particular change predicts a later lapse. An intervention can test whether changing a target helps a defined outcome. Evidence from one level can motivate the next test, but cannot substitute for it.

A biological explanation should also leave room for the conditions in which people live. Treatment access, workload and practical support belong in the research design. Recording these factors does not diminish bodily symptoms; it helps avoid assigning every difficulty to an unmeasured molecule.

K2. What a null cognitive result leaves open

The trial estimated differences between treatment groups, rather than counting anyone who felt better as evidence that the drug worked. The intervals show why “not statistically significant” is different from proof of exactly zero effect.

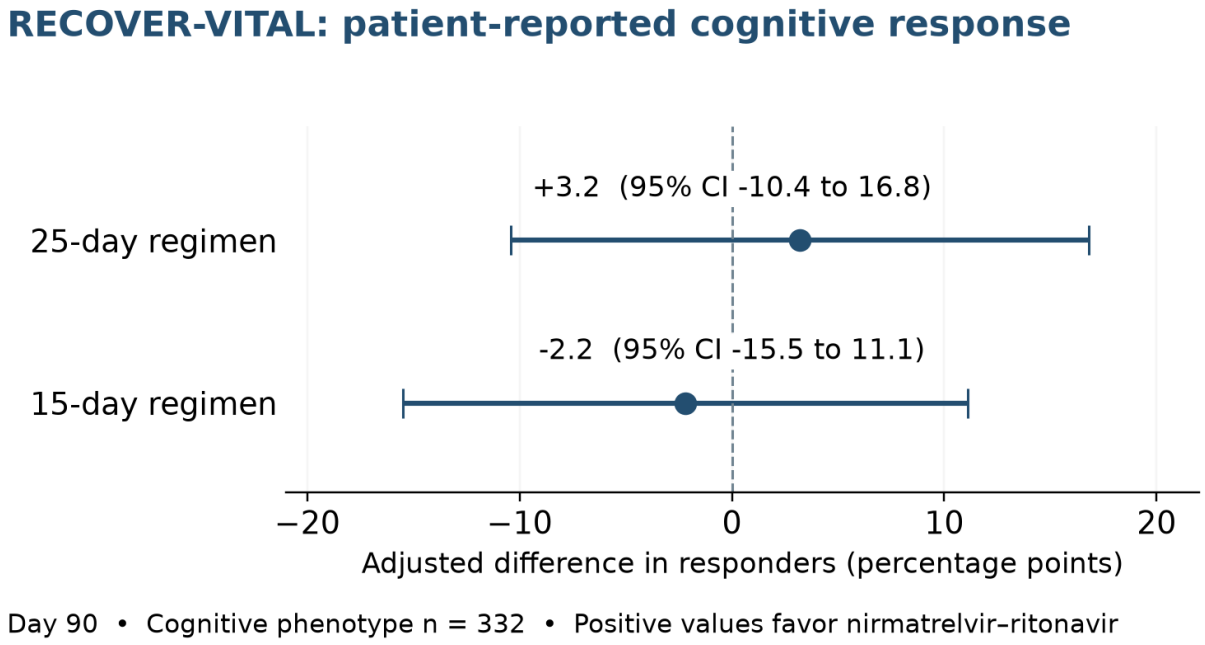


Figure 42. Adjusted responder differences versus ritonavir plus nirmatrelvir-matching placebo, with 95% CIs. Response meant at least a five-point improvement in the PROMIS Cognitive Function 8a T-score at day 90. Both intervals cross zero. This is a self-reported outcome, not a percentage cured. [190]

The implication for our hypothesis

A plausible target is not enough to predict benefit. A negative result should narrow the claim to the drug, population, regimen and endpoint actually tested. A remaining mechanism needs a specific explanation and another test, not an automatic appeal to a hidden subgroup.

Functional adaptation matters too. A person may meet a chosen goal more reliably by changing how the task is done. That benefit can be meaningful without demonstrating restoration of every cognitive process. For the proposed resilience index, an outcome directly trained during treatment should not also serve as independent proof of predictive validity.

Source-checking note

The v25 pass added nine original reports. RECOVER-VITAL was checked through its original abstract, investigator record and trial registry; the adult sleep study through its original abstract and publisher appendix. The sleep abstract lists 33 in its third arm, whereas the appendix lists 23, consistent with the total of 70; this paper uses the total and primary-comparison counts. Other additions used accessible original full text. The historical audit gaps in Appendix I remain unresolved. No new full audit of all references is claimed.

References

The 238 references link to original reports, evidence syntheses, official records or prescribing information. Published correction notices are linked alongside the relevant entries. Appendix I records the source checks and unresolved limits; Appendix J develops the provisional score; Appendix K records the v25 additions. Section 3.1 describes the review method. Matching a citation to a publication does not verify every claim made from it.

1. Willcutt EG, Doyle AE, Nigg JT, Faraone SV, Pennington BF. Validity of the executive function theory of attention-deficit/hyperactivity disorder: a meta-analytic review. Biol Psychiatry. 2005;57(11):1336-1346. doi:10.1016/j.biopsych.2005.02.006

2. Sergeant JA. The cognitive-energetic model: an empirical approach to attention-deficit hyperactivity disorder. Neurosci Biobehav Rev. 2000;24(1):7-12. doi:10.1016/S0149-7634(99)00060-3

3. Sergeant JA. Modeling attention-deficit/hyperactivity disorder: a critical appraisal of the cognitive-energetic model. Biol Psychiatry. 2005;57(11):1248-1255. doi:10.1016/j.biopsych.2004.09.010

4. Ben-Dor Cohen M, Maeir A, Eldar E, Nahum M. Everyday cognitive control and emotion dysregulation in young adults with and without ADHD: an ecological momentary assessment study. J Atten Disord. 2023;27(5):539-553. doi:10.1177/10870547231153934

5. Stern Y, Arenaza-Urquijo EM, Bartres-Faz D, et al. Whitepaper: defining and investigating cognitive reserve, brain reserve, and brain maintenance. Alzheimers Dement. 2020;16(9):1305-1311. doi:10.1016/j.jalz.2018.07.219

6. Toplak ME, West RF, Stanovich KE. Practitioner review: do performance-based measures and ratings of executive function assess the same construct? J Child Psychol Psychiatry. 2013;54(2):131-143. doi:10.1111/jcpp.12001

7. Becker SP, Willcutt EG, Leopold DR, et al. Report of a work group on sluggish cognitive tempo: key research directions and a consensus change in terminology to cognitive disengagement syndrome. J Am Acad Child Adolesc Psychiatry. 2023;62(6):629-645. doi:10.1016/j.jaac.2022.07.821

8. Sibley MH, Kennedy TM, Swanson JM, et al. Characteristics and predictors of fluctuating attention-deficit/hyperactivity disorder in the Multimodal Treatment of ADHD (MTA) study. J Clin Psychiatry. 2024;85(4):24m15395. doi:10.4088/JCP.24m15395

9. Becker SP, Epstein JN, Tamm L, et al. Shortened sleep duration causes sleepiness, inattention, and oppositionality in adolescents with attention-deficit/hyperactivity disorder: findings from a crossover sleep restriction/extension study. J Am Acad Child Adolesc Psychiatry. 2019;58(4):433-442. doi:10.1016/j.jaac.2018.09.439

10. Dan O, Cohen A, Asraf K, Saveliev I, Haimov I. The impact of sleep deprivation on continuous performance task among young men with ADHD. J Atten Disord. 2021;25(9):1284-1294. doi:10.1177/1087054719897811

11. Lunsford-Avery JR, Carskadon MA, Kollins SH, Krystal AD. Sleep physiology and neurocognition among adolescents with attention-deficit/hyperactivity disorder. J Am Acad Child Adolesc Psychiatry. 2025;64(2):276-289. doi:10.1016/j.jaac.2024.03.005

12. Hiscock H, Sciberras E, Mensah F, Gerner B, Efron D, Khano S, Oberklaid F. Impact of a behavioural sleep intervention on symptoms and sleep in children with attention deficit hyperactivity disorder, and parental mental health: randomised controlled trial. BMJ. 2015;350:h68. doi:10.1136/bmj.h68

13. Doran SM, Van Dongen HPA, Dinges DF. Sustained attention performance during sleep deprivation: evidence of state instability. Arch Ital Biol. 2001;139(3):253–267. PMID:11330205. PubMed record

14. Andrillon T, Burns A, Mackay T, Windt J, Tsuchiya N. Predicting lapses of attention with sleep-like slow waves. Nat Commun. 2021;12:3657. doi:10.1038/s41467-021-23890-7

15. Pinggal E, Jackson J, Kusztor A, et al. Sleep-like slow waves during wakefulness mediate attention and vigilance difficulties in adult attention-deficit/hyperactivity disorder. J Neurosci. 2026;46(15):e1694252025. doi:10.1523/JNEUROSCI.1694-25.2025

16. Gong S, et al. Sleep-like slow waves in ADHD: regional specificity in combined type. Int J Psychophysiol. 2026;222:113349. doi:10.1016/j.ijpsycho.2026.113349

17. Gabay L, Miller P, Alia-Klein N, Lewin MP. Circadian effects on attention and working memory in college students with attention deficit and hyperactivity symptoms. Front Psychol. 2022;13:851502. doi:10.3389/fpsyg.2022.851502

18. Becker SP, Brown A, Langberg JM, Beebe DW. Later (“evening”) circadian preference is associated with poorer executive, academic, and attentional functioning in adolescents with and without ADHD. J Child Psychol Psychiatry. 2025;66(1):53-63. doi:10.1111/jcpp.14030

19. van Andel E, Bijlenga D, Vogel SWN, Beekman ATF, Kooij JJS. Effects of chronotherapy on circadian rhythm and ADHD symptoms in adults with attention-deficit/hyperactivity disorder and delayed sleep phase syndrome: a randomized clinical trial. Chronobiol Int. 2021;38(2):260–269. doi:10.1080/07420528.2020.1835943

20. van Andel E, Bijlenga D, Vogel SWN, Beekman ATF, Kooij JJS. Attention-deficit/hyperactivity disorder and delayed sleep phase syndrome in adults: a randomized clinical trial on the effects of chronotherapy on sleep. J Biol Rhythms. 2022;37(6):673–689. doi:10.1177/07487304221124659

21. Chong TTJ, Fortunato E, Bellgrove MA. Amphetamines improve the motivation to invest effort in attention-deficit/hyperactivity disorder. J Neurosci. 2023;43(41):6898-6908. doi:10.1523/JNEUROSCI.0982-23.2023

22. Chat IKY, Hansson LS, Lekander M, et al. Obesity, low-grade inflammation, and inflammatory response to immune challenge modulate willingness to expend effort for reward. Brain Behav Immun. 2026;136:106548. doi:10.1016/j.bbi.2026.106548

23. Faherty T, Raymond JE, McFiggans G, Pope FD. Acute particulate matter exposure diminishes executive cognitive functioning after four hours regardless of inhalation pathway. Nat Commun. 2025;16:1339. doi:10.1038/s41467-025-56508-3 Published correction: 10.1038/s41467-025-57017-z

24. An D, You Y, Ma Q, et al. Deficiency of histamine H2 receptors in parvalbumin-positive neurons leads to hyperactivity, impulsivity, and impaired attention. Neuron. 2025;113(4):572-589.e6. doi:10.1016/j.neuron.2024.12.002

25. Colwell MJ, van Uum FJE, Cowen PJ, et al. Histamine shapes the neurocomputational dynamics of human learning. Nat Commun. 2026;17:7124. doi:10.1038/s41467-026-73865-9

26. Mao X, Wang J, Ding S, et al. MRGPRX2 mediates mast cell-induced endometriosis pain through the sensitization of sensory neurons via histamine/HRH1/TRPV1 signaling pathway. FASEB J. 2025;39(13):e70778. doi:10.1096/fj.202501493R

27. Herring WJ, Wilens TE, Adler LA, et al. Randomized controlled study of the histamine H3 inverse agonist MK-0249 in adult attention-deficit/hyperactivity disorder. J Clin Psychiatry. 2012;73(7):e891-e898. doi:10.4088/JCP.11m07178

28. Kang J, Lee H, Kim S, et al. Comorbid health conditions in people with attention-deficit/hyperactivity disorders: an umbrella review of systematic reviews and meta-analyses. Asian J Psychiatr. 2024;99:104135. doi:10.1016/j.ajp.2024.104135

29. van der Schans J, Cao Q, Bos EH, et al. The temporal order of fluctuations in atopic disease symptoms and attention-deficit/hyperactivity disorder symptoms: a time-series study in ADHD patients. Eur Child Adolesc Psychiatry. 2020;29(2):137-144. doi:10.1007/s00787-019-01336-2

30. Chen Q, Sun Y, Pan N, et al. Inflammatory cytokine alterations in autism spectrum disorder and attention-deficit/hyperactivity disorder: a systematic review and network meta-analysis. Eur Child Adolesc Psychiatry. 2026. doi:10.1007/s00787-026-03068-6

31. Jin H, Li M, Jeong E, Castro-Martinez F, Zuker CS. A body-brain circuit that regulates body inflammatory responses. Nature. 2024;630:695-703. doi:10.1038/s41586-024-07469-y

32. Roberts B, Eisenlohr-Moul T, Martel MM. Reproductive steroids and ADHD symptoms across the menstrual cycle. Psychoneuroendocrinology. 2018;88:105-114. doi:10.1016/j.psyneuen.2017.11.015

33. Zaritsky R, Reed SC, Evans SM. Changes in ADHD symptoms and mood across the menstrual cycle in females treated with stimulants: a pilot study. J Atten Disord. 2026;30(3):329-341. doi:10.1177/10870547251400038

34. Gao M, Koupil I, Sjoqvist H, et al. Psychiatric comorbidity among women with endometriosis: nationwide cohort study in Sweden. Am J Obstet Gynecol. 2020;223(3):415.e1-415.e16. doi:10.1016/j.ajog.2020.02.033

35. Li N, Chen Y, Zhao W, et al. Deciphering the shared genetic architecture between female reproductive disorders and psychiatric disorders. J Ovarian Res. 2026;19(1):62. doi:10.1186/s13048-026-01970-w

36. Voltolini Velho R, Freitag JH, Brueckner AM, et al. The histamine-associated inflammatory landscape of endometriosis: molecular profiling of HDC, HRH1-HRH4, and cytokines across lesion subtypes. Int J Mol Sci. 2025;27(1):212. doi:10.3390/ijms27010212

37. Gil-Gouveia R, Oliveira AG, Martins IP. Cognitive dysfunction during migraine attacks: a study on migraine without aura. Cephalalgia. 2015;35(8):662-674. doi:10.1177/0333102414553823

38. Goadsby PJ, Reuter U, Hallström Y, et al. A controlled trial of erenumab for episodic migraine. N Engl J Med. 2017;377(22):2123-2132. doi:10.1056/NEJMoa1705848

39. Geiss L, Stemmler M, Beck B, Hillemacher T, Widder M, Hosl KM. Dysregulation of the autonomic nervous system in adult attention deficit hyperactivity disorder: a systematic review. Cogn Neuropsychiatry. 2023;28(4):285-306. doi:10.1080/13546805.2023.2255336

40. Bruton AM, Levy L, Rai NK, Colgan DD, Johnstone JM. Diminished interoceptive accuracy in attention-deficit/hyperactivity disorder: a systematic review. Psychophysiology. 2025;62(2):e14750. doi:10.1111/psyp.14750

41. Maier A, Schopen L, Thiel JC, et al. Cognitive functioning in postural orthostatic tachycardia syndrome among different body positions: a prospective pilot study (POTSKog study). Clin Auton Res. 2023;33:459-468. doi:10.1007/s10286-023-00950-0

42. Verbree J, Bronzwaer AS, Ghariq E, et al. Assessment of middle cerebral artery diameter during hypocapnia and hypercapnia in humans using ultra-high-field MRI. J Appl Physiol (1985). 2014;117(10):1084-1089. doi:10.1152/japplphysiol.00651.2014

43. Boban M, Crnac P, Junakovic A, Malojcic B. Hemodynamic monitoring of middle cerebral arteries during cognitive tasks performance. Psychiatry Clin Neurosci. 2014;68(11):795-803. doi:10.1111/pcn.12191

44. Greene C, Connolly R, Brennan D, et al. Blood-brain barrier disruption and sustained systemic inflammation in individuals with long COVID-associated cognitive impairment. Nat Neurosci. 2024;27(3):421-432. doi:10.1038/s41593-024-01576-9 Published correction: 10.1038/s41593-024-01644-0

45. Weber-Fahr W, Dommke S, Sack M, et al. Reduced adenosine triphosphate-to-phosphocreatine ratios in neuropsychiatric post-COVID condition: evidence from 31P magnetic resonance spectroscopy. Biol Psychiatry. 2026;100(5):546-555. doi:10.1016/j.biopsych.2026.01.004

46. Centers for Disease Control and Prevention. Long COVID clinical guidance. Updated March 9, 2026. Accessed September 5, 2026. CDC guidance

47. Wechsler JB, Butuci M, Wong A, Kamboj AP, Youngblood BA. Mast cell activation is associated with post-acute COVID-19 syndrome. Allergy. 2022;77(4):1288-1291. doi:10.1111/all.15188

48. Lenning OB, Jonsson G, Grimstad T, et al. No signs of mast cell involvement in long-COVID: a case-control study. Scand J Immunol. 2024;100(5):e13407. doi:10.1111/sji.13407

49. Momtazmanesh S, Ansari S, Izadi Z, et al. Effect of famotidine on cognitive and behavioral dysfunctions induced in post-COVID-19 infection: a randomized, double-blind, and placebo-controlled study. J Psychosom Res. 2023;172:111389. doi:10.1016/j.jpsychores.2023.111389

50. STIMULATE-ICP consortium. Efficacy and safety of rivaroxaban, colchicine, and famotidine-loratadine with specialist supportive clinical care for fatigue in patients with post-COVID-19 condition in the UK: a multisite, open-label, randomised controlled trial. Lancet Infect Dis. 2026. doi:10.1016/S1473-3099(26)00242-2

51. Yang H, George SJ, Thompson DA, et al. Famotidine activates the vagus nerve inflammatory reflex to attenuate cytokine storm. Mol Med. 2022;28:57. doi:10.1186/s10020-022-00483-8

52. Wong CK, McLean BA, Baggio LL, et al. Central glucagon-like peptide 1 receptor activation inhibits Toll-like receptor agonist-induced inflammation. Cell Metab. 2024;36(1):130-143.e5. doi:10.1016/j.cmet.2023.11.009

53. Afrin LB, Weinstock LB, Dempsey TT, Aschenbrenner K, Blitshteyn S, Schofield JR. Utility of glucagon-like peptide-1 receptor agonists in mast cell activation syndrome. Am J Med Sci. 2025;370(4):377-382. doi:10.1016/j.amjms.2025.07.006

54. Hendershot CS, et al. Once-weekly semaglutide in adults with alcohol use disorder: a randomized clinical trial. JAMA Psychiatry. 2025;82:395–405. doi:10.1001/jamapsychiatry.2024.4789

55. Ulland TK, Song WM, Huang SC-C, et al. TREM2 maintains microglial metabolic fitness in Alzheimer's disease. Cell. 2017;170(4):649-663.e13. doi:10.1016/j.cell.2017.07.023

56. Cummings JL, Atri A, Sano M, et al. Efficacy and safety of oral semaglutide 14 mg (flexible dose) in early-stage symptomatic Alzheimer's disease (evoke and evoke+): two phase 3, randomised, placebo-controlled trials. Lancet. 2026;407(10544):2167-2179. doi:10.1016/S0140-6736(26)00459-9

57. Tesser JRP, Crowley AR, Box EJ, et al. Vagus nerve-mediated neuroimmune modulation for rheumatoid arthritis: a pivotal randomized controlled trial. Nat Med. 2026;32:369-378. doi:10.1038/s41591-025-04114-7

58. Liu F, Li Y. Sorting the mind: cognitive enhancement through transcutaneous auricular vagus nerve stimulation: a systematic review and meta-analysis. Psychol Med. 2026;56:e207. doi:10.1017/S0033291726105017

59. Wolf V, et al. Does transcutaneous auricular vagus nerve stimulation affect vagally mediated heart rate variability? A living and interactive Bayesian meta-analysis. Psychophysiology. 2021;58(11):e13933. doi:10.1111/psyp.13933

60. Lakens D. Equivalence tests: a practical primer for t tests, correlations, and meta-analyses. Soc Psychol Personal Sci. 2017;8(4):355-362. doi:10.1177/1948550617697177

61. Curran PJ, Bauer DJ. The disaggregation of within-person and between-person effects in longitudinal models of change. Annu Rev Psychol. 2011;62:583-619. doi:10.1146/annurev.psych.093008.100356

62. Fisher AJ, Medaglia JD, Jeronimus BF. Lack of group-to-individual generalizability is a threat to human subjects research. Proc Natl Acad Sci USA. 2018;115(27):E6106-E6115. doi:10.1073/pnas.1711978115

63. Hedge C, Powell G, Sumner P. The reliability paradox: why robust cognitive tasks do not produce reliable individual differences. Behav Res Methods. 2018;50(3):1166-1186. doi:10.3758/s13428-017-0935-1

64. Kucina T, Wells L, Lewis I, et al. Calibration of cognitive tests to address the reliability paradox for decision-conflict tasks. Nat Commun. 2023;14:2234. doi:10.1038/s41467-023-37777-2

65. Van der Heijden KB, Smits MG, Van Someren EJW, Ridderinkhof KR, Gunning WB. Effect of melatonin on sleep, behavior, and cognition in ADHD and chronic sleep-onset insomnia. J Am Acad Child Adolesc Psychiatry. 2007;46(2):233–241. doi:10.1097/01.chi.0000246055.76167.0d

66. Jang D, Zhang J, Elfenbein HA. Menstrual cycle effects on cognitive performance: a meta-analysis. PLoS One. 2025;20:e0318576. doi:10.1371/journal.pone.0318576

67. Hsu TW, Chen MH, Chu CS, et al. Attention deficit hyperactivity disorder and risk of migraine: a nationwide longitudinal study. Headache. 2022;62(5):634–641. doi:10.1111/head.14306

68. Strom MA, Fishbein AB, Paller AS, Silverberg JI. Association between atopic dermatitis and attention deficit hyperactivity disorder in U.S. children and adults. Br J Dermatol. 2016;175:920–929. doi:10.1111/bjd.14697

69. Cheng Y, et al. Associations of atopic dermatitis with attention deficit/hyperactivity disorder and autism spectrum disorder: a systematic review and meta-analysis. Dermatology. 2024;240:13–25. doi:10.1159/000533366

70. Hampshire A, Azor A, Atchison C, et al. Cognition and memory after Covid-19 in a large community sample. N Engl J Med. 2024;390:806–818. doi:10.1056/NEJMoa2311330

71. Jamal SM, Landers DB, Hollenberg SM, et al. Prospective evaluation of autonomic dysfunction in post-acute sequela of COVID-19. J Am Coll Cardiol. 2022;79:2325–2330. doi:10.1016/j.jacc.2022.03.357

72. Cukierman-Yaffe T, Gerstein HC, Colhoun HM, et al. Effect of dulaglutide on cognitive impairment in type 2 diabetes: an exploratory analysis of the REWIND trial. Lancet Neurol. 2020;19:582–590. doi:10.1016/S1474-4422(20)30173-3 Published correction: 10.1016/S1474-4422(20)30303-3

73. Lincoff AM, Brown-Frandsen K, Colhoun HM, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. N Engl J Med. 2023;389:2221–2232. doi:10.1056/NEJMoa2307563

74. Appelman B, Charlton BT, Goulding RP, et al. Muscle abnormalities worsen after post-exertional malaise in long COVID. Nat Commun. 2024;15:17. doi:10.1038/s41467-023-44432-3

75. Dan O, Haimov I, Harel A, et al. Sleep deprivation alters early event-related potentials during emotional face processing in adults with ADHD. Sci Rep. 2026;16:6956. doi:10.1038/s41598-026-38376-z

76. Cortese S, et al. Comparative efficacy and tolerability of medications for attention-deficit hyperactivity disorder in children, adolescents, and adults: a systematic review and network meta-analysis. Lancet Psychiatry. 2018;5:727–738. Original source

77. Ostinelli EG, et al. Comparative efficacy and acceptability of pharmacological, psychological, and neurostimulatory interventions for ADHD in adults: a systematic review and component network meta-analysis. Lancet Psychiatry. 2025;12:32–43. Original source

78. Nourredine M, et al. Pharmacological interventions for ADHD: a systematic review and dose-effect network meta-analysis. Lancet Psychiatry. 2026;13:485–495. Original source Published correction: 10.1016/S2215-0366(26)00175-6

79. Vertessen K, et al. Meta-analysis: Dose-Dependent Effects of Methylphenidate on Neurocognitive Functioning in Children With Attention-Deficit/Hyperactivity Disorder. J Am Acad Child Adolesc Psychiatry. 2022;61:626–646. Original source

80. Pelham WE, et al. The effect of stimulant medication on the learning of academic curricula in children with ADHD: a randomized crossover study. J Consult Clin Psychol. 2022;90:367–380. Original source

81. Zhang L, et al. ADHD drug treatment and risk of suicidal behaviours, substance misuse, accidental injuries, transport accidents, and criminality: emulation of target trials. BMJ. 2025;390:e083658. Figure-color correction: BMJ 2025;390:r1827. Original source Published correction: 10.1136/bmj.r1827

82. Kidwell KM, et al. Stimulant medications and sleep for youth with ADHD: a meta-analysis. Pediatrics. 2015;136:1144–1153. Original source

83. Farhat LC, et al. Comparative cardiovascular safety of medications for attention-deficit hyperactivity disorder in children, adolescents, and adults: a systematic review and network meta-analysis. Lancet Psychiatry. 2025;12:355–365. Original source Published correction: 10.1016/S2215-0366(25)00171-3

84. Green EA, et al. Effects of norepinephrine reuptake inhibition on postural tachycardia syndrome. J Am Heart Assoc. 2013;2:e000395. Original source

85. Zhang L, et al. Attention-deficit/hyperactivity disorder medications and long-term risk of cardiovascular diseases. JAMA Psychiatry. 2024;81:178–187. Original source

86. DailyMed. Atomoxetine capsules: US prescribing information, especially section 7.2 (CYP2D6 inhibitors). Accessed 5 September 2026. Original source

87. DailyMed. Guanfacine extended-release tablets: US prescribing information, warnings and drug interactions. Accessed 5 September 2026. Original source

88. DailyMed. Topiramate: US prescribing information, cognitive/neuropsychiatric adverse reactions in migraine trials. Accessed 5 September 2026. Original source

89. Weiler JM, et al. Effects of fexofenadine, diphenhydramine, and alcohol on driving performance: a randomized, placebo-controlled trial in the Iowa driving simulator. Ann Intern Med. 2000;132:354–363. Original source

90. US Food and Drug Administration. Wegovy (semaglutide): prescribing information, drug interactions. 2026. Accessed 5 September 2026. Original source

91. DailyMed. Lisdexamfetamine dimesylate: US prescribing information, serotonin syndrome and drug interactions. Accessed 5 September 2026. Original source

92. Redline S, Cook K, Chervin RD, et al. Adenotonsillectomy for snoring and mild sleep apnea in children: a randomized clinical trial. JAMA. 2023;330:2084–2095. doi:10.1001/jama.2023.22114

93. Ramin-Wright A, et al. Fatigue—a symptom in endometriosis. Human Reproduction. 2018;33:1459–1465. doi:10.1093/humrep/dey115

94. Goldberg HR, et al. High prevalence of undiagnosed iron deficiency in endometriosis patients: a cross-sectional study. International Journal of Gynecology & Obstetrics. 2025;168:1321–1327. doi:10.1002/ijgo.15994

95. Vaucher P, Druais PL, Waldvogel S, Favrat B. Effect of iron supplementation on fatigue in nonanemic menstruating women with low ferritin: a randomized controlled trial. CMAJ. 2012;184:1247–1254. doi:10.1503/cmaj.110950

96. Henderson VW, et al. Cognitive effects of estradiol after menopause: a randomized trial of the timing hypothesis. Neurology. 2016;87:699–708. doi:10.1212/WNL.0000000000002980

97. Baker LD, Espeland MA, Whitmer RA, et al. Structured vs self-guided multidomain lifestyle interventions for global cognitive function: the US POINTER randomized clinical trial. JAMA. 2025;334:681–691. doi:10.1001/jama.2025.12923

98. Reboussin DM, Gaussoin SA, Pajewski NM, et al. Long-term effect of intensive vs standard blood pressure control on mild cognitive impairment and probable dementia in SPRINT. Neurology. 2025;104:e213334. doi:10.1212/WNL.0000000000213334 Published correction: 10.1212/WNL.0000000000213686

99. Scott JC, et al. A quantitative meta-analysis of neurocognitive functioning in posttraumatic stress disorder. Psychological Bulletin. 2015;141:105–140. doi:10.1037/a0038039

100. Semkovska M, et al. Cognitive function following a major depressive episode: a systematic review and meta-analysis. Lancet Psychiatry. 2019;6:851–861. doi:10.1016/S2215-0366(19)30291-3

101. Stern A, et al. Associations between abuse/neglect and ADHD from childhood to young adulthood: a prospective nationally-representative twin study. Child Abuse & Neglect. 2018;81:274–285. doi:10.1016/j.chiabu.2018.04.025

102. González RA, et al. Evidence of concurrent and prospective associations between early maltreatment and ADHD through childhood and adolescence. Social Psychiatry and Psychiatric Epidemiology. 2019;54:671–682. doi:10.1007/s00127-019-01659-0

103. Song H, et al. Association of stress-related disorders with subsequent autoimmune disease. JAMA. 2018;319:2388–2400. doi:10.1001/jama.2018.7028

104. Benros ME, et al. Autoimmune diseases and severe infections as risk factors for mood disorders: a nationwide study. JAMA Psychiatry. 2013;70:812–820. doi:10.1001/jamapsychiatry.2013.1111

105. Hopkins CWP, et al. Cognitive impairment in adult inflammatory bowel disease: a systematic review and meta-analysis. Journal of the Academy of Consultation-Liaison Psychiatry. 2021;62:387–403. doi:10.1016/j.psym.2020.10.002

106. Bjornevik K, et al. Longitudinal analysis reveals high prevalence of Epstein–Barr virus associated with multiple sclerosis. Science. 2022;375:296–301. doi:10.1126/science.abj8222

107. Graus F, et al. A clinical approach to diagnosis of autoimmune encephalitis. Lancet Neurology. 2016;15:391–404. doi:10.1016/S1474-4422(15)00401-9

108. Radjabzadeh D, et al. Gut microbiome-wide association study of depressive symptoms. Nature Communications. 2022;13:7128. doi:10.1038/s41467-022-34502-3

109. Nikolova VL, et al. Acceptability, tolerability, and estimates of putative treatment effects of probiotics as adjunctive treatment in patients with depression: a randomized clinical trial. JAMA Psychiatry. 2023;80:842–847. doi:10.1001/jamapsychiatry.2023.1817

110. Rojo-Marticella M, Arija V, Canals-Sans J. Effect of probiotics on the symptomatology of autism spectrum disorder and/or attention deficit/hyperactivity disorder in children and adolescents: pilot study. Research on Child and Adolescent Psychopathology. 2025;53:163–178. doi:10.1007/s10802-024-01278-7

111. Raison CL, et al. A randomized controlled trial of the tumor necrosis factor antagonist infliximab for treatment-resistant depression: the role of baseline inflammatory biomarkers. JAMA Psychiatry. 2013;70:31–41. doi:10.1001/2013.jamapsychiatry.4

112. Faherty T, et al. Neurological and respiratory outcomes of the HIPTox controlled double-blind air pollution exposure trial. npj Clean Air. 2026;2:34. doi:10.1038/s44407-026-00068-3

113. Malhotra A, et al. Tirzepatide for the treatment of obstructive sleep apnea and obesity. N Engl J Med. 2024;391:1193–1205. Original source Published correction: 10.1056/NEJMx240005

114. Kosiborod MN, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. N Engl J Med. 2023;389:1069–1084. Original source

115. Packer M, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. N Engl J Med. 2025;392:427–437. Original source

116. Bliddal H, et al. Once-weekly semaglutide in persons with obesity and knee osteoarthritis. N Engl J Med. 2024;391:1573–1583. Original source

117. Perkovic V, et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. N Engl J Med. 2024;391:109–121. Original source

118. Sanyal AJ, et al. Phase 3 trial of semaglutide in metabolic dysfunction-associated steatohepatitis. N Engl J Med. 2025;392:2089–2099. Original source

119. Loomba R, et al. Tirzepatide for metabolic dysfunction-associated steatohepatitis with liver fibrosis. N Engl J Med. 2024;391:299–310. Baseline-table correction published 2026. Original source Published correction: 10.1056/NEJMx250014

120. Klausen MK, et al. Once-weekly semaglutide versus placebo in patients with alcohol use disorder and comorbid obesity: a randomised, double-blind, placebo-controlled trial. Lancet. 2026;407:1687–1698. Original source

121. Schacht JP, Sakai JT, Raymond K, Shelton R. Oral semaglutide for alcohol use disorder: a randomized clinical trial. Am J Psychiatry. 2026;183:636–645. Original source

122. Braca S, et al. Effectiveness and tolerability of liraglutide as add-on treatment in patients with obesity and high-frequency or chronic migraine: a prospective pilot study. Headache. 2025;65:1831–1838. Original source

123. Mitchell JL, et al. The effect of GLP-1RA exenatide on idiopathic intracranial hypertension: a randomized clinical trial. Brain. 2023;146:1821–1830. Original source

124. Grech O, et al. Effect of glucagon-like peptide-1 receptor agonist exenatide used as an intracranial pressure lowering agent on cognition in idiopathic intracranial hypertension. Eye. 2024;38:1374–1379. Original source

125. Meissner WG, et al. Trial of lixisenatide in early Parkinson's disease. N Engl J Med. 2024;390:1176–1185. Original source

126. Vijiaratnam N, et al. Exenatide once a week versus placebo as a potential disease-modifying treatment for people with Parkinson's disease in the UK: a phase 3, multicentre, double-blind, parallel-group, randomised, placebo-controlled trial. Lancet. 2025;405:627–636. Original source

127. Kellner DA, et al. Effect of glucagon-like peptide 1 receptor agonists on patients with rheumatoid arthritis. ACR Open Rheumatol. 2025;7:e70103. Original source

128. Saadeh M, et al. GLP-1 receptor agonists are associated with improved outcomes in inflammatory bowel disease. GastroHep. 2026;2026:2239686. Original source

129. Yeh KH, et al. Adjunctive GLP1 receptor agonists in patients with inflammatory bowel diseases and obesity and/or diabetes: a target trial emulation. Clin Gastroenterol Hepatol. 2026. Online publication. Original source

130. Gholiof M, Kalani N, Leonardi M. Effects of glucagon-like peptide-1 receptor agonists (GLP-1RAs) on endometriosis symptoms management and quality of life: an international survey study. Authorea. 24 February 2026. Preprint; not peer reviewed. Original source

131. Vogel JM, et al. Insights from a double-blind randomized direct-to-participant intervention trial for Long COVID. medRxiv. 22 August 2026. Operations report; preprint, not peer reviewed; no efficacy results. Original source

132. Bissonnette R, et al. Oral icotrokinra for plaque psoriasis in adults and adolescents. N Engl J Med. 2025;393:1784–1795. Original source

133. US Food and Drug Administration. Drug Trials Snapshots: ICOTYDE. Original approval 17 March 2026. Accessed 5 September 2026. Original source

134. Jeppesen PB, et al. Teduglutide reduces need for parenteral support among patients with short bowel syndrome with intestinal failure. Gastroenterology. 2012;143:1473–1481.e3. Original source

135. Craft S, et al. Safety, efficacy, and feasibility of intranasal insulin for the treatment of mild cognitive impairment and Alzheimer disease dementia: a randomized clinical trial. JAMA Neurol. 2020;77:1099–1109. Original source

136. Sikich L, et al. Intranasal oxytocin in children and adolescents with autism spectrum disorder. N Engl J Med. 2021;385:1462–1473. Original source

137. US Food and Drug Administration. BPC-157 briefing material for the July 2026 Pharmacy Compounding Advisory Committee meeting. Staff evidence review; accessed 5 September 2026. Original source

138. US Food and Drug Administration. TB-500 briefing material for the July 2026 Pharmacy Compounding Advisory Committee meeting. Staff evidence review; accessed 5 September 2026. Original source

139. US Food and Drug Administration. MOTS-c briefing material for the July 2026 Pharmacy Compounding Advisory Committee meeting. Staff evidence review; accessed 5 September 2026. Original source

140. Hornstein MD, Surrey ES, Weisberg GW, Casino LA; Lupron Add-Back Study Group. Leuprolide acetate depot and hormonal add-back in endometriosis: a 12-month study. Obstet Gynecol. 1998;91:16–24. Original source

141. Lebwohl M, et al. Ixekizumab with or without tirzepatide in adults with psoriasis and overweight or obesity: a phase 3b randomized clinical trial. JAMA Dermatol. 2026;162:709–719. Corrected July 2026. Original source Published correction: 10.1001/jamadermatol.2026.2678

142. US Food and Drug Administration. FDA requests removal of suicidal behavior and ideation warning from GLP-1 receptor agonist medications. Drug Safety Communication. 13 January 2026. Accessed 5 September 2026. Original source

143. Eli Lilly and Company. ZEPBOUND (tirzepatide) US prescribing information. DailyMed. Current label accessed 5 September 2026. Original source

144. Abramson J, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature. 2024;630:493–500. Original source

145. Avsec Ž, et al. Advancing regulatory variant effect prediction with AlphaGenome. Nature. 2026;649:1206–1218. Original source

146. Wang Y, et al. A generative artificial intelligence approach for the discovery of antimicrobial peptides against multidrug-resistant bacteria. Nature Microbiology. 2025;10:2997–3012. Original source

147. Xu Z, Ren F, Wang P, et al. A generative AI-discovered TNIK inhibitor for idiopathic pulmonary fibrosis: a randomized phase 2a trial. Nature Medicine. 2025;31:2602–2610. Original source

148. Tozzi L, et al. Personalized brain circuit scores identify clinically distinct biotypes in depression and anxiety. Nature Medicine. 2024;30:2076–2087. Original source

149. Hack LM, et al. A stratified precision medicine trial targeting α2A-adrenergic receptor agonism as a treatment for the cognitive biotype of depression. Nature Mental Health. 2025;3:1363–1373. Original source

150. Demontis D, Walters GB, Athanasiadis G, et al. Genome-wide analyses of ADHD identify 27 risk loci, refine the genetic architecture and implicate several cognitive domains. Nature Genetics. 2023;55:198–208. doi:10.1038/s41588-022-01285-8. Original source Published correction: 10.1038/s41588-023-01350-w

151. Sibley MH, Arnold LE, Swanson JM, et al. Variable Patterns of Remission From ADHD in the Multimodal Treatment Study of ADHD. American Journal of Psychiatry. 2022;179:142–151. doi:10.1176/appi.ajp.2021.21010032. Original source

152. Sibley MH, Rohde LA, Swanson JM, et al. Late-Onset ADHD Reconsidered With Comprehensive Repeated Assessments Between Ages 10 and 25. American Journal of Psychiatry. 2018;175:140–149. doi:10.1176/appi.ajp.2017.17030298. Original source

153. Handke A, Axelsson J, Benson S, et al. Acute inflammation and psychomotor slowing: Experimental assessment using lipopolysaccharide administration in healthy humans. Brain, Behavior, & Immunity – Health. 2020;8:100130. doi:10.1016/j.bbih.2020.100130. Original source

154. Irwin MR, Boyle CC, Cho JH, et al. Inflammatory Exposure and Depression in Older Adults With Insomnia: A Randomized Clinical Trial. JAMA Psychiatry. 2025;82:859–867. doi:10.1001/jamapsychiatry.2025.1327. Original source

155. Lin FR, Pike JR, Albert MS, et al. Hearing intervention versus health education control to reduce cognitive decline in older adults with hearing loss in the USA (ACHIEVE): a multicentre, randomised controlled trial. The Lancet. 2023;402:786–797. doi:10.1016/S0140-6736(23)01406-X. Original source

156. Dangour AD, Allen E, Clarke R, et al. Effects of vitamin B-12 supplementation on neurologic and cognitive function in older people: a randomized controlled trial. American Journal of Clinical Nutrition. 2015;102:639–647. doi:10.3945/ajcn.115.110775. Original source

157. Launer LJ, et al. Effects of intensive glucose lowering on brain structure and function in people with type 2 diabetes (ACCORD MIND): a randomised open-label substudy. The Lancet Neurology. 2011;10:969–977. doi:10.1016/S1474-4422(11)70188-0. Original source

158. Centers for Disease Control and Prevention. About Infectious Mononucleosis (Mono). 9 May 2024. Original source

159. Centers for Disease Control and Prevention. Laboratory Testing for Epstein-Barr Virus (EBV). 10 April 2024. Original source

160. Katz BZ, Shiraishi Y, Mears CJ, Binns HJ, Taylor R. Chronic fatigue syndrome after infectious mononucleosis in adolescents. Pediatrics. 2009;124:189–193. doi:10.1542/peds.2008-1879. Original source

161. Jason LA, Cotler J, Islam MF, Sunnquist M, Katz BZ. Risks for Developing Myalgic Encephalomyelitis/Chronic Fatigue Syndrome in College Students Following Infectious Mononucleosis: A Prospective Cohort Study. Clinical Infectious Diseases. 2021;73:e3740–e3746. doi:10.1093/cid/ciaa1886. Original source

162. Hickie I, Davenport T, Wakefield D, et al. Post-infective and chronic fatigue syndromes precipitated by viral and non-viral pathogens: prospective cohort study. BMJ. 2006;333:575. doi:10.1136/bmj.38933.585764.AE. Original source

163. Cvejic E, Lemon J, Hickie IB, Lloyd AR, Vollmer-Conna U. Neurocognitive disturbances associated with acute infectious mononucleosis, Ross River fever and Q fever: a preliminary investigation of inflammatory and genetic correlates. Brain, Behavior, and Immunity. 2014;36:207–214. doi:10.1016/j.bbi.2013.11.002. Original source

164. Øie MG, Rødø ASB, Bølgen MS, Pedersen M, Asprusten TT, Wyller VBB. Subjective and objective cognitive function in adolescent with chronic fatigue following Epstein-Barr virus infection. Journal of Psychosomatic Research. 2022;163:111063. doi:10.1016/j.jpsychores.2022.111063. Original source

165. Kristiansen MS, Stabursvik J, O'Leary EC, et al. Clinical symptoms and markers of disease mechanisms in adolescent chronic fatigue following Epstein-Barr virus infection: An exploratory cross-sectional study. Brain, Behavior, and Immunity. 2019;80:551–563. doi:10.1016/j.bbi.2019.04.040. Original source

166. Wang J, Li Y, Geng X, Zhang X, Xiao Y, Wang W. Associations of Epstein-Barr Virus Infection with Attention Deficit Hyperactivity Disorder, Learning Disability, and Special Education in US Children. International Journal of General Medicine. 2022;15:2187–2195. doi:10.2147/IJGM.S355263. Original source

167. Peluso MJ, Deveau TM, Munter SE, et al. Chronic viral coinfections differentially affect the likelihood of developing long COVID. Journal of Clinical Investigation. 2023;133:e163669. doi:10.1172/JCI163669. Original source

168. Hoeggerl AD, et al. Epstein-Barr virus reactivation is not causative for post-COVID-19-syndrome in individuals with asymptomatic or mild SARS-CoV-2 disease course. BMC Infectious Diseases. 2023;23:800. doi:10.1186/s12879-023-08820-w. Original source

169. Lanz TV, et al. Clonally expanded B cells in multiple sclerosis bind EBV EBNA1 and GlialCAM. Nature. 2022;603:321–327. doi:10.1038/s41586-022-04432-7. Original source

170. Younis S, et al. Epstein-Barr virus reprograms autoreactive B cells as antigen-presenting cells in systemic lupus erythematosus. Science Translational Medicine. 2025;17:eady0210. doi:10.1126/scitranslmed.ady0210. Original source

171. Kawada J, Ito Y, Ohshima K, et al. Updated guidelines for chronic active Epstein–Barr virus disease. International Journal of Hematology. 2023;118:568–576. doi:10.1007/s12185-023-03660-5. Original source

172. Jason LA, Furst J, Worth R, Katz BZ. Outcomes of ME/CFS following infectious mononucleosis: seven-year follow-up of a prospective study. Frontiers in Medicine. 2026;13:1676628. doi:10.3389/fmed.2026.1676628. Original source

173. Shen L, Nicolazzo J, Sletten TL, et al. Daily fluctuations in adolescents' sleep predict next-day attention, sleepiness, and fatigue: an ecological momentary assessment study over 28 days. Journal of Child Psychology and Psychiatry. 2025;66:686–696. doi:10.1111/jcpp.14076. Original source

174. Douaud G, Lee S, Alfaro-Almagro F, et al. SARS-CoV-2 is associated with changes in brain structure in UK Biobank. Nature. 2022;604:697–707. doi:10.1038/s41586-022-04569-5. Original source

175. Selvakumar J, Havdal LB, Drevvatne M, et al. Prevalence and Characteristics Associated With Post–COVID-19 Condition Among Nonhospitalized Adolescents and Young Adults. JAMA Network Open. 2023;6:e235763. doi:10.1001/jamanetworkopen.2023.5763. Original source

176. Kennedy TM, Molina BSG, Pedersen SL. Change in Adolescents' Perceived ADHD Symptoms Across 17 Days of Ecological Momentary Assessment. Journal of Clinical Child & Adolescent Psychology. 2024;53:397–412. doi:10.1080/15374416.2022.2096043. Original source

177. van Campen CLMC, Verheugt FWA, Rowe PC, Visser FC. Cerebral blood flow is reduced in ME/CFS during head-up tilt testing even in the absence of hypotension or tachycardia: A quantitative, controlled study using Doppler echography. Clinical Neurophysiology Practice. 2020;5:50–58. doi:10.1016/j.cnp.2020.01.003. Original source

178. Whitson HE, Ashner M, Kraus VB, Peskoe S, Huebner JL, Simon CB, Hall K, Colon-Emeric C. Comparing two methods to quantify physical resilience. J Frailty Aging. 2026;15:100179. doi:10.1016/j.tjfa.2026.100179

179. Bandeen-Roche K, Zhu J, Xue QL, Buta B, Laskow T, Walston JD, Varadhan R. Characterization of Dynamic Adaptation to Stressors Using Multisystem Stimulus-Response Data: The Study of Physical Resilience in Aging Pilot. J Gerontol A Biol Sci Med Sci. 2025;80:glaf056. doi:10.1093/gerona/glaf056

180. Liang L, Chen CX, Chan NY, Yau SY, Liu Y, Li SX, Wing YK, Lee TM, Hou WK. Predicting acute stress recovery: A resilience index of physiological responses to Trier Social Stress Test. Int J Clin Health Psychol. 2026;26:100652. doi:10.1016/j.ijchp.2025.100652

181. Malinen T, Nuuttila OP, Matomäki P, Uusitalo A, Kyröläinen H. Test-Retest Reliability of Physiological Resilience During and After Prolonged Moderate-Intensity Running in Well-Trained Runners. Eur J Sport Sci. 2026;26:e70178. doi:10.1002/ejsc.70178

182. Mokkink LB, Terwee CB, Patrick DL, Alonso J, Stratford PW, Knol DL, Bouter LM, de Vet HC. The COSMIN checklist for assessing the methodological quality of studies on measurement properties of health status measurement instruments: an international Delphi study. Qual Life Res. 2010;19:539-549. doi:10.1007/s11136-010-9606-8

183. Kolk D, MacNeil Vroomen JL, Melis RJF, Ridderikhof ML, Buurman BM. Assessing dynamical resilience indicators in older adults: a prospective feasibility cohort study. Eur Geriatr Med. 2024;15:445-451. doi:10.1007/s41999-023-00904-9

184. Stott J, O'Nions E, Corrigan L, et al. Attention-Deficit/Hyperactivity Disorder Traits in Childhood and Physical Health in Midlife. JAMA Netw Open. 2026;9:e2554802. doi:10.1001/jamanetworkopen.2025.54802

185. Luo Y, Blakey R, Gkatzionis A, et al. Investigating the relationship between attention-deficit hyperactivity disorder (ADHD) and C-reactive protein (CRP): observational, polygenic risk score, and Mendelian randomization analyses. Psychol Med. 2025;55:e103. doi:10.1017/S0033291725000480

186. Zarchev M, Gigase FAJ, de Witte L, et al. Maternal infection - but not inflammation - is associated with attention-deficit/hyperactivity disorder symptoms during childhood: a population-based cohort study. Brain Behav Immun. 2026;131:106134. doi:10.1016/j.bbi.2025.106134

187. van der Ham M, Bijlenga D, Molenaar N, et al. The Effects of Sleep Treatment on Symptoms of ADHD, Sleep Quality, Fatigue, and Depressive Symptoms in Adults. J Atten Disord. 2026;30:354-369. doi:10.1177/10870547251379103

188. Edison P, Femminella GD, Ritchie C, et al. Liraglutide in mild to moderate Alzheimer's disease: a phase 2b clinical trial. Nat Med. 2026;32:353-361. doi:10.1038/s41591-025-04106-7

189. Ganeshalingam AA, Uhrenholt N, Arnfred S, et al. Semaglutide Treatment of Antipsychotic-Treated Patients With Schizophrenia, Prediabetes, and Obesity: The HISTORI Randomized Clinical Trial. JAMA Psychiatry. 2025;82:1065-1074. doi:10.1001/jamapsychiatry.2025.2332

190. Baden LR, Shah NS, Liu STH, et al. Nirmatrelvir–ritonavir targeting viral persistence in post-COVID-19 condition (long COVID) in the USA (RECOVER-VITAL): a randomised, double-blind, placebo-controlled, phase 2 trial. Lancet Infect Dis. 2026. Published online August 2026. doi:10.1016/S1473-3099(26)00406-8

191. Knopman DS, Koltai D, Laskowitz DT, et al. Evaluation of Interventions for Cognitive Symptoms in Long COVID: A Randomized Clinical Trial. JAMA Neurol. 2026;83:49-59. doi:10.1001/jamaneurol.2025.4415

192. Vanova M, Patel AMR, Scott I, et al. Cognitive Rehabilitation and Functional Outcomes in Long COVID-Related Cognitive Impairment: A Randomized Clinical Trial. JAMA Netw Open. 2026;9:e2620687. doi:10.1001/jamanetworkopen.2026.20687

193. Hawks ZW, Beck ED, Jung L, et al. Dynamic associations between glucose and ecological momentary cognition in Type 1 Diabetes. npj Digital Medicine. 2024;7:59. doi:10.1038/s41746-024-01036-5

194. Singh S, Strong R, Xu I, et al. Ecological Momentary Assessment of Cognition in Clinical and Community Samples: Reliability and Validity Study. J Med Internet Res. 2023;25:e45028. doi:10.2196/45028

195. Walter A, Martz E, Weiner L, et al. Sleep, rest-activity rhythm, cognitive and emotional symptoms in adult ADHD: unraveling the links with an actimetry-based approach. BMC Psychiatry. 2026;26:317. doi:10.1186/s12888-026-07947-9

196. Pedersen ML, et al. Selective enhancement of cost sensitivity by methylphenidate in adult ADHD: a randomized placebo-controlled trial. Psychological Medicine. 2026;56:e234. doi:10.1017/S0033291726104954

197. Lau RI, et al. A synbiotic preparation (SIM01) for post-acute COVID-19 syndrome in Hong Kong (RECOVERY): a randomised, double-blind, placebo-controlled trial. Lancet Infect Dis. 2024;24:256–265. doi:10.1016/S1473-3099(23)00685-0

198. Moura DS, Sultan S, Georgin-Lavialle S, et al. Evidence for cognitive impairment in mastocytosis: prevalence, features and correlations to depression. PLoS One. 2012;7:e39468. doi:10.1371/journal.pone.0039468

199. Castells M, Giannetti MP, Hamilton MJ, et al. Mast cell activation syndrome: current understanding and research needs. J Allergy Clin Immunol. 2024;154:255–263. doi:10.1016/j.jaci.2024.05.025

200. Mamuladze T, Zaninelli TH, Smyth LCD, et al. Mast cells regulate the brain-dura interface and CSF dynamics. Cell. 2025;188:5487–5498.e16. doi:10.1016/j.cell.2025.06.046

201. Wu ML, Xie C, Li X, Sun J, Zhao J, Wang JH. Mast cell activation triggered by SARS-CoV-2 causes inflammation in brain microvascular endothelial cells and microglia. Front Cell Infect Microbiol. 2024;14:1358873. doi:10.3389/fcimb.2024.1358873

202. Zaitsu M, Narita S, Lambert KC, et al. Estradiol activates mast cells via a non-genomic estrogen receptor-alpha and calcium influx. Mol Immunol. 2007;44:1977–1985. doi:10.1016/j.molimm.2006.09.030

203. Foer D, Amin T, Nagai J, et al. Glucagon-like peptide-1 receptor pathway attenuates platelet activation in aspirin-exacerbated respiratory disease. J Immunol. 2023;211:1806–1813. doi:10.4049/jimmunol.2300102

204. Broughton T, Lambert E, Wertz J, Agnew-Blais J. Increased risk of provisional premenstrual dysphoric disorder (PMDD) among females with attention-deficit hyperactivity disorder (ADHD): cross-sectional survey study. Br J Psychiatry. 2025;226:410–417. doi:10.1192/bjp.2025.104

205. Schmidt PJ, Martinez PE, Nieman LK, et al. Premenstrual dysphoric disorder symptoms following ovarian suppression: triggered by change in ovarian steroid levels but not continuous stable levels. Am J Psychiatry. 2017;174:980–989. doi:10.1176/appi.ajp.2017.16101113

206. Wei SM, Wakim P, Martinez PE, Nieman LK, Rubinow DR, Schmidt PJ. Differential effects of ovarian steroids in women with and without premenstrual dysphoric disorder: a replication and extension of findings. Am J Psychiatry. 2025;182:922–934. doi:10.1176/appi.ajp.20240596

207. Resnick SM, Matsumoto AM, Stephens-Shields AJ, et al. Testosterone treatment and cognitive function in older men with low testosterone and age-associated memory impairment. JAMA. 2017;317:717–727. doi:10.1001/jama.2016.21044

208. Martinez PE, Rubinow DR, Nieman LK, et al. 5α-Reductase inhibition prevents the luteal phase increase in plasma allopregnanolone levels and mitigates symptoms in women with premenstrual dysphoric disorder. Neuropsychopharmacology. 2016;41:1093–1102. doi:10.1038/npp.2015.246

209. Luchsinger JA, Rosin SP, Kazemi EJ, et al. Glucose-lowering medications, glycemia, and cognitive outcomes: the GRADE randomized clinical trial. JAMA Intern Med. 2025;185:778–787. doi:10.1001/jamainternmed.2025.1189

210. Verhulst CEM, Fabricius TW, Nefs G, et al. Consistent effects of hypoglycemia on cognitive function in people with or without diabetes. Diabetes Care. 2022;45:2103–2110. doi:10.2337/dc21-2502

211. Gabery S, Salinas CG, Paulsen SJ, et al. Semaglutide lowers body weight in rodents via distributed neural pathways. JCI Insight. 2020;5:e133429. doi:10.1172/jci.insight.133429

212. Hummel J, Benkendorff C, Fritsche L, et al. Brain insulin action on peripheral insulin sensitivity in women depends on menstrual cycle phase. Nat Metab. 2023;5:1475–1482. doi:10.1038/s42255-023-00869-w

213. Stiernborg M, Yang LL, Skott E, et al. Altered gut microbiome function in ADHD: More Prevotella, less vitamin B12 biosynthesis, and beneficial modulation by synbiotic treatment. Brain Behav Immun. 2026;133:106259. doi:10.1016/j.bbi.2026.106259

214. Skott E, Yang LL, Stiernborg M, et al. Effects of a synbiotic on symptoms, and daily functioning in attention deficit hyperactivity disorder: a double-blind randomized controlled trial. Brain Behav Immun. 2020;89:9–19. doi:10.1016/j.bbi.2020.05.056

215. Ni Lochlainn M, Bowyer RCE, Moll JM, et al. Effect of gut microbiome modulation on muscle function and cognition: the PROMOTe randomised controlled trial. Nat Commun. 2024;15:1859. doi:10.1038/s41467-024-46116-y

216. Ni Lochlainn M, et al. Author Correction: Effect of gut microbiome modulation on muscle function and cognition: the PROMOTe randomised controlled trial. Nat Commun. 2025;16:3393. doi:10.1038/s41467-025-58771-w

217. Pierce D, et al. Effect of experimental hookworm infection on insulin resistance in people at risk of type 2 diabetes. Nat Commun. 2023;14:4503. doi:10.1038/s41467-023-40263-4

218. Tanasescu R, et al. Hookworm treatment for relapsing multiple sclerosis: a randomized double-blinded placebo-controlled trial. JAMA Neurol. 2020;77:1089–1098. doi:10.1001/jamaneurol.2020.1118

219. Croese J, et al. Randomized, placebo controlled trial of experimental hookworm infection for improving gluten tolerance in celiac disease. Clin Transl Gastroenterol. 2020;11:e00274. doi:10.14309/ctg.0000000000000274

220. Wastyk HC, Fragiadakis GK, Perelman D, et al. Gut-microbiota-targeted diets modulate human immune status. Cell. 2021;184:4137–4153.e14. doi:10.1016/j.cell.2021.06.019

221. Parkin JGH, Dean LSN, Bell JA, et al. Copper-enriched automotive brake wear particles perturb human alveolar cellular homeostasis. Part Fibre Toxicol. 2025;22:4. doi:10.1186/s12989-024-00617-2

222. Kim G, Jeon S, Shin TH, et al. Pulmonary toxicity of brake wear particles: effects of composition and surface reactivity. Part Fibre Toxicol. 2026;23:27. doi:10.1186/s12989-026-00684-7

223. Nihart AJ, Garcia MA, El Hayek E, et al. Bioaccumulation of microplastics in decedent human brains. Nat Med. 2025;31:1114–1119. doi:10.1038/s41591-024-03453-1

224. Monikh FA, Materic D, Valsami-Jones E, et al. Challenges in studying microplastics in human brain. Nat Med. 2025;31:4034–4035. doi:10.1038/s41591-025-04045-3

225. Campen MJ, West AB, Garcia M, et al. Reply to: Challenges in studying microplastics in human brain. Nat Med. 2025;31:4036–4037. doi:10.1038/s41591-025-04046-2

226. Nihart AJ, et al. Author Correction: Bioaccumulation of microplastics in decedent human brains. Nat Med. 2025;31:1367. doi:10.1038/s41591-025-03675-x

227. Evangelou I, Bucci S, Stohl A. Atmospheric microplastic emissions from land and ocean. Nature. 2026;649:1186–1189. doi:10.1038/s41586-025-09998-6

228. Marfella R, Prattichizzo F, Sardu C, et al. Microplastics and nanoplastics in atheromas and cardiovascular events. N Engl J Med. 2024;390:900–910. doi:10.1056/NEJMoa2309822

229. Janelsins MC, Heckler CE, Peppone LJ, et al. Cognitive complaints in survivors of breast cancer after chemotherapy compared with age-matched controls: an analysis from a nationwide, multicenter, prospective longitudinal study. J Clin Oncol. 2017;35:506–514. doi:10.1200/JCO.2016.68.5826

230. Janelsins MC, Heckler CE, Peppone LJ, et al. Longitudinal trajectory and characterization of cancer-related cognitive impairment in a nationwide cohort study. J Clin Oncol. 2018;36:3231–3239. doi:10.1200/JCO.2018.78.6624

231. Otto-Dobos LD, Grant CV, Lahoud AA, et al. Chemotherapy-induced gut microbiome disruption, inflammation, and cognitive decline in female patients with breast cancer. Brain Behav Immun. 2024;120:208–220. doi:10.1016/j.bbi.2024.05.039

232. Porcari S, Ciccarese C, Heidrich V, et al. Fecal microbiota transplantation plus pembrolizumab and axitinib in metastatic renal cell carcinoma: the randomized phase 2 TACITO trial. Nat Med. 2026;32:1316–1324. doi:10.1038/s41591-025-04189-2

233. Choi IJ, Kim CG, Lee JY, et al. Family history of gastric cancer and Helicobacter pylori treatment. N Engl J Med. 2020;382:427–436. doi:10.1056/NEJMoa1909666

234. Doust C, Fontanillas P, Eising E, et al. Discovery of 42 genome-wide significant loci associated with dyslexia. Nat Genet. 2022;54:1621–1629. doi:10.1038/s41588-022-01192-y | Author correction (2023)

235. van Bergen E, et al. Co-Occurrence and Causality Among ADHD, Dyslexia, and Dyscalculia. Psychol Sci. 2025;36:204–217. doi:10.1177/09567976241293999

236. Ghirardi L, Brikell I, Kuja-Halkola R, et al. The familial co-aggregation of ASD and ADHD: a register-based cohort study. Mol Psychiatry. 2018;23:257–262. doi:10.1038/mp.2017.17

237. Gringras P, Nir T, Breddy J, et al. Efficacy and Safety of Pediatric Prolonged-Release Melatonin for Insomnia in Children With Autism Spectrum Disorder. J Am Acad Child Adolesc Psychiatry. 2017;56:948–957.e4. doi:10.1016/j.jaac.2017.09.414

238. Yap CX, Henders AK, Alvares GA, et al. Autism-related dietary preferences mediate autism-gut microbiome associations. Cell. 2021;184:5916–5931.e17. doi:10.1016/j.cell.2021.10.015

Author and research transparency

Author: Alex Yesilcimen. Article type: hypothesis/perspective with targeted narrative synthesis. No original participant dataset, new meta-analysis or conducted intervention is reported. Proposed studies are not registered protocols. The NRI is an unvalidated research proposal; its worked example contains invented values. Figures 1, 2, 7, 13, 30, 31 and 40 and Visual guides A, C and D are hypothetical or conceptual. Other figures show attributed study results or labeled calculations. Figures 32 and 33 describe sample inclusion and completion, not clinical outcomes. AI assistance supported literature organization, source checking, editorial development and figure production. This is not independent peer review.

The source audit distinguishes verified numerical claims from retained narrative material and remaining access limits. The adult waking-EEG methods and exact regional spectroscopy effects still require fuller review. Funding, conflicts of interest, affiliation and correspondence require author confirmation before submission. No clinical or institutional endorsement is implied.